

Chapter 9 Custom Power Devices for Distribution Power Quality: DSTATCOM, DVR, UPQC and Advanced Compensators

Learning Objectives

By the end of this chapter, you should be able to:

- distinguish custom power devices from the transmission FACTS controllers of Chapters 6 to 8, and explain why the same converter principles serve a different purpose at distribution voltage;
- classify voltage-quality and current-quality problems, read a voltage-tolerance curve and compute THD, TDD, displacement and true power factor and voltage unbalance;
- explain the operating principle, classification, topologies, control algorithms and design of a distribution static compensator (DSTATCOM) for power-factor correction, voltage regulation, load balancing, harmonic elimination and neutral-current compensation;
- explain the components, energy-source options, compensation strategies, control, protection and design of a dynamic voltage restorer (DVR), and state the exact condition under which a DVR can restore a sag without real power;
- derive the voltage, current and VA ratings of a unified power quality conditioner (UPQC) in its UPQC-P, UPQC-Q and UPQC-S forms, for right-shunt and left-shunt connections, and distinguish an operating duty from an installed rating;
- size DC-link storage and capacitance, interfacing inductors, ripple filters, injection transformers and semiconductor ratings;
- derive the instantaneous pq theory and the synchronous-reference-frame method used to generate compensator references;
- analyse shunt, series and hybrid active filters from the harmonic-source model and the network impedance, and size a hybrid filter;
- apply IEEE Std 519-2022 current and voltage criteria at a point of common coupling, including the statistical assessment, and size a compensator for compliance;
- select a custom power device from an audited disturbance profile using rating, energy, protection and life-cycle cost arguments rather than device capability alone.

9.1 From Transmission FACTS to Distribution Custom Power

Chapters 6 to 8 used voltage-source converters to control power flow, voltage and stability on transmission networks. When the same converter technology is applied inside a distribution system, at a customer bus or a feeder, the goal changes. The converter now protects a load from a bad supply, or protects the supply from a bad load. Hingorani called this family custom power [1]. Three members dominate practice and give this chapter its structure: the distribution static compensator (DSTATCOM), the dynamic voltage restorer (DVR) and the unified power quality conditioner (UPQC). Active power filters (APFs), solid-state transfer switches and fault-current limiters belong to the same family.

The three main devices differ by how they connect (Figure 9.1). A DSTATCOM is connected in shunt and injects current, so it naturally corrects problems that originate in the load current: poor power factor, unbalance, neutral current and harmonics. A DVR is connected in series through an injection transformer and inserts voltage, so it corrects problems in the supply voltage: sags, swells, unbalance, notches and voltage harmonics. A UPQC places a series converter and a shunt converter on a common DC link and can do both jobs, subject to the ratings of each converter and the energy available on the shared link.

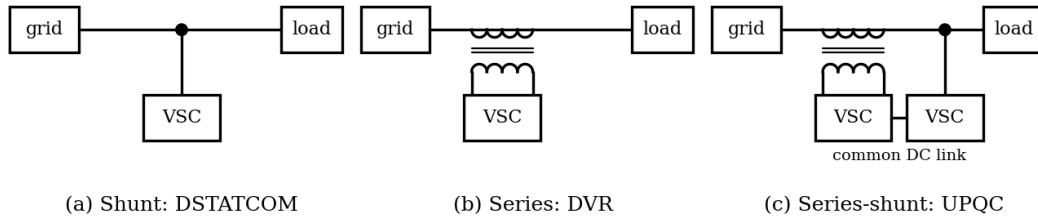


Figure 9.1. Connection families of custom power devices: (a) shunt, (b) series and (c) series-shunt.

Typical installations range from a few kVA at 230 V to several MVA at medium voltage. Neither voltage nor size defines the family; the purpose and control objective do. The series-voltage, shunt-current and DC-energy relationships of Chapters 6 to 8 remain valid, but distribution adds its own constraints: neutral conductors and zero-sequence current, single-phase and nonlinear loads, injection-transformer saturation, fault bypass, very short disturbance-detection times and the tolerance of the specific equipment being protected. A DVR controls protected-load voltage, while an SSSC mainly controls a line's effective series compensation; shared converter mathematics does not mean identical duties.

Figure 9.2 summarizes how the three devices are classified in the literature [2], [3], [4]: by converter type (current-source or voltage-source), by supply system (single-phase two-wire, three-phase three-wire and three-phase four-wire), by topology, by the source of DC energy and, for the UPQC, by shunt location and by how the series converter shares real and reactive power.

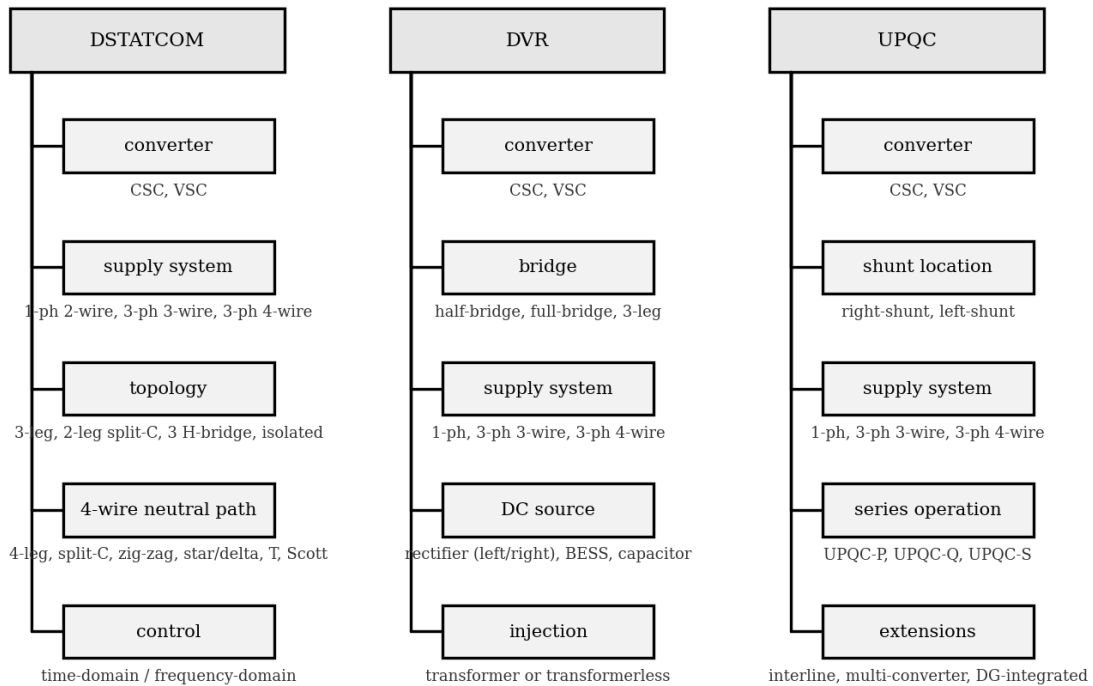


Figure 9.2. Classification of DSTATCOM, DVR and UPQC.

Voltage-source converters (VSCs) have largely displaced current-source converters (CSCs) for custom power. A CSC needs a large DC inductor and sizeable AC capacitors, has higher losses and is not easily extended to multilevel forms. A PWM VSC with a DC capacitor is lighter, cheaper, has lower losses and scales to higher ratings with multilevel structures. Unless stated otherwise, every device in this chapter uses a PWM VSC.

Table 9.1. Transmission FACTS and distribution custom power compared.

Attribute	Transmission FACTS	Distribution custom power
Purpose	Network power flow, voltage and stability	Voltage and current quality at a load, bus or feeder

Attribute	Transmission FACTS	Distribution custom power
Typical scale	Tens to hundreds of MVA	kVA to a few MVA; voltage alone is not the definition
Typical disturbance	Contingencies, oscillations, transfer limits	Sags, swells, harmonics, unbalance, flicker, poor power factor
Response	Cycles to seconds for most functions	Often sub-cycle for sag correction and current shaping
Special constraints	Insulation level, protection coordination, SSR	Neutral current, single-phase loads, transformer saturation, bypass, equipment tolerance

9.1.1 A physical view of custom power compensation

The natural entry point is to ask where the disturbance originates and which electrical quantity must be synthesized. If the load current is undesirable while the supply voltage is acceptable, the compensator should inject current in shunt. If the supply voltage is undesirable while the load current must continue, it should inject voltage in series. If both are unacceptable, the two actions can be coordinated through a common DC link. A shunt converter cannot directly create a missing series voltage, although its current can move the PCC voltage through the feeder impedance; a series converter cannot directly cancel a nonlinear load current at the source, although its voltage can shape that current indirectly. Network impedance, converter rating, available DC energy and control bandwidth therefore decide what can actually be corrected (Table 9.2).

Table 9.2. Compensation location and its principal physical limit.

Location	Controlled quantity	Primary disturbance	Typical device	Principal physical limit
Shunt	Injected current	Reactive, harmonic, unbalanced or neutral load current	DSTATCO M / shunt APF	Converter current and PCC voltage
Series	Injected voltage	Sag, swell, unbalance, voltage harmonics	DVR / series APF	Injection voltage, line current and stored energy
Series + shunt	Voltage and current together	Simultaneous supply and load problems	UPQC	Both converter ratings and common DC-link balance
Multi-feeder	Coordinated series voltages	Sag support across several feeders	IDVR / MC-UPQC	Cross-feeder real-power balance and converter ratings

9.1.2 From ideal compensator to practical converter

Each device is developed in four layers. The first is the ideal compensation law obtained from Kirchhoff's laws and phasors. The second is the converter realization: VSC topology, transformer or coupling inductor, DC link and filters. The third is the control system that estimates the disturbance and forces the converter to reproduce the required voltage or current. The fourth is the practical envelope set by semiconductor current, modulation voltage, stored energy, thermal limits, protection and the upstream network. This layering guards against a common error in custom-power studies, in which a controller computes a mathematically perfect reference that the converter cannot produce. Every derivation in the chapter therefore ends with a feasibility check.

9.2 Power-Quality Problems in Distribution Systems

9.2.1 Voltage-quality and current-quality problems

It is useful to sort distribution power-quality problems by where they originate. A voltage-quality problem is already present in the supply voltage before the protected load sees it: a sag caused by a remote fault, a swell on the healthy phases during a single-line-to-ground fault on an ungrounded or impedance-grounded system, voltage unbalance, notching caused by a neighbour's converter, flicker or voltage harmonics. A current-quality problem is created by the load and pushed back into the network: reactive current, unbalanced phase currents, excessive neutral current and harmonic current. Current-quality problems become voltage-quality problems for other customers once the current flows through the network impedance (Table 9.3).

Table 9.3. Voltage-quality and current-quality problems and the natural compensator.

Voltage-quality problems (supply side)	Current-quality problems (load side)
Sag, swell, interruption	Poor displacement power factor (reactive current)
Voltage unbalance	Unbalanced phase currents
Notching, flicker, fluctuation	Excessive neutral current (triplen harmonics, single-phase loads)
Voltage harmonics, transients	Harmonic currents, load fluctuation
Natural compensator: series (DVR)	Natural compensator: shunt (DSTATCOM, shunt APF)

9.2.2 Disturbance categories

Figure 9.3 shows eight commonly encountered disturbances; it is not an exhaustive classification. Under IEEE Std 1159-2019 [5], a short-duration voltage sag has an RMS magnitude of 0.1 to 0.9 pu lasting from half a cycle to one minute. A swell starts at about 1.1 pu, and its typical upper magnitude depends on duration. An interruption has an RMS voltage below 0.1 pu; a sustained interruption lasts longer than one minute. Harmonics are integer multiples of the fundamental, usually generated by nonlinear loads. Notches are short, periodic voltage dips produced when a line-commutated converter momentarily shorts two phases during commutation. Voltage fluctuations at a few hertz produce visible lamp flicker. Transients may be impulsive (lightning) or oscillatory (capacitor switching). Unbalance, interharmonics, DC offset, noise and frequency variation also matter.

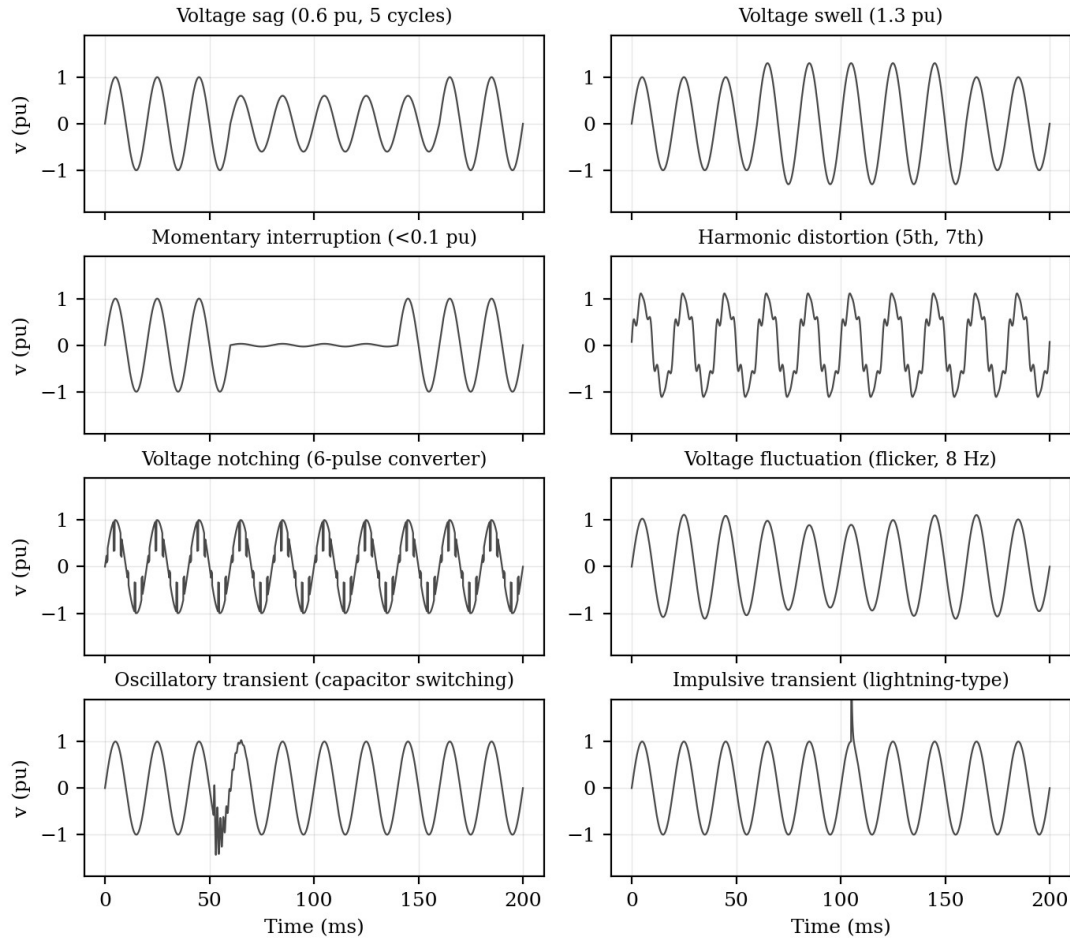


Figure 9.3. Illustrative power-quality disturbances (simulated, 50 Hz, per unit).

Table 9.4. Selected categories and typical characteristics, after IEEE Std 1159-2019.

Category	Typical duration	Typical magnitude
Instantaneous sag / swell	0.5 to 30 cycles	0.1 to 0.9 pu / 1.1 to 1.8 pu
Momentary sag / swell	30 cycles to 3 s	0.1 to 0.9 pu / 1.1 to 1.4 pu
Temporary sag / swell	3 s to 1 min	0.1 to 0.9 pu / 1.1 to 1.2 pu
Momentary / temporary interruption	0.5 cycle to 3 s / 3 s to 1 min	below 0.1 pu
Sustained interruption	longer than 1 min	0 pu
Undervoltage / overvoltage	longer than 1 min	0.8 to 0.9 pu / 1.1 to 1.2 pu
Voltage imbalance	steady state	0.5 to 2 % (negative sequence)
Harmonics	steady state	0 to 20 %
Voltage fluctuation (flicker)	intermittent	0.1 to 7 %, below 25 Hz

9.2.3 Reading a voltage-tolerance curve

Equipment sensitivity to sags and swells is commonly summarized by a tolerance curve such as the one published by the Information Technology Industry Council (ITI), historically the CBEMA curve [6]. Figure 9.4 reproduces only its general shape. The ITI curve is a specific reference for 120 V, 60 Hz information-

technology equipment, and its application note should be consulted for exact coordinates. For a real installation, use the protected equipment's own ride-through specification at its actual supply voltage and frequency.

The curve plots acceptable voltage against disturbance duration. A brief, deep sag may be tolerated, while the same depth held longer causes a trip. Duration, and not only depth, is therefore a first-class design parameter throughout the chapter: it sets the DVR's stored energy and the response time its controller must achieve.

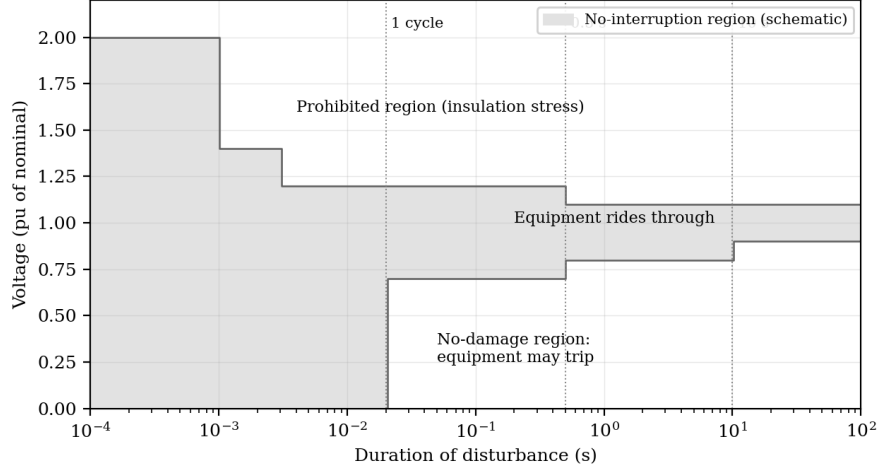


Figure 9.4. Schematic equipment voltage-tolerance regions (shape only, not a compliance curve).

9.2.4 Power-quality indices

Several indices recur in this chapter. For a periodic quantity with RMS harmonic components X_h , total harmonic distortion is

$$THD = \frac{\sqrt{\sum_{h \geq 2} X_h^2}}{X_1} \times 100\% \quad (9.1)$$

where X_1 is the RMS fundamental. Total demand distortion (TDD) has the same numerator but uses the maximum demand load current as denominator (Section 9.7). For a load with RMS fundamental current I_1 , displacement angle ϕ_1 and total RMS current I , the true power factor separates into a displacement factor and a distortion factor:

$$PF = \frac{P}{VI} = \frac{I_1}{I} \cos \phi_1 = \frac{\cos \phi_1}{\sqrt{1 + THD_I^2}} \quad (9.2)$$

with THD_I as a fraction, assuming a sinusoidal voltage. A capacitor bank can only correct $\cos \phi_1$; a DSTATCOM can correct both factors. Voltage unbalance is most rigorously expressed by the voltage unbalance factor,

$$VUF = \frac{|V_n|}{|V_p|} \times 100\% \quad (9.3)$$

where V_p and V_n are the positive- and negative-sequence components of Chapter 2.

9.2.5 Matching problems to devices

A DVR corrects load-voltage sags and swells within its voltage, current, energy and control limits. A DSTATCOM or shunt APF compensates load-generated reactive, unbalanced, neutral and harmonic currents, and through the feeder impedance can also regulate the bus voltage. A UPQC combines these services. A

UPQC can replenish the series converter's real power from the grid during a sag only while adequate source voltage remains; with the source absent, it needs usable stored energy and a topology that can safely supply an isolated load (Section 9.8.2). Flicker mitigation needs enough bandwidth and current or voltage margin. Very fast impulsive transients usually need surge protection, not a converter (Table 9.5).

Table 9.5. Power-quality problems and candidate devices.

Dominant problem	Candidate device and required checks
Low power factor, reactive demand	DSTATCOM in UPF mode; capacitor bank if the load is steady and linear
Feeder voltage regulation	DSTATCOM in ZVR mode; voltage effect depends on feeder impedance
Unbalanced load, neutral current	Three-phase four-wire DSTATCOM (four-leg, split-capacitor or transformer-assisted)
Load-current harmonics	DSTATCOM / shunt APF; hybrid filter after a resonance study
Voltage sag or swell	DVR with enough injection voltage, current, energy and phase-transition capability
Sags on several feeders	Interline DVR, if joint operating points and shared-link balance are feasible
Supply-voltage harmonics, notches	DVR or series APF designed for harmonic voltage injection
Both voltage and current problems	UPQC, subject to both converter duties, source capacity and DC balance
Sustained interruption	UPS, standby source or island-capable storage scheme with isolation

9.3 Distribution Static Compensator (DSTATCOM)

The DSTATCOM is a shunt-connected VSC whose job is to make the current drawn from the supply look like the current of an ideal load: sinusoidal, balanced and, if required, in phase with the voltage. It is the distribution counterpart of the STATCOM of Chapter 6 [7], but its duty is wider. A transmission STATCOM mainly exchanges positive-sequence fundamental reactive power; a DSTATCOM is commonly asked to cancel harmonics, balance single-phase loads and remove neutral current as well. Commercial units range from a few hundred kVA to several MVA at medium voltage, often coordinating switched capacitor or reactor banks.

9.3.1 Operating principle

Figure 9.5 shows the single-line arrangement. The VSC is connected at the point of common coupling (PCC) through an interfacing inductor L_r , and a small R_f - C_f ripple filter at the PCC absorbs switching-frequency current. The DC side has only a capacitor; no external energy source is needed because, ideally, the DSTATCOM exchanges no average real power. Voltage and current sensors feed a DSP or rapid-prototyping controller that generates the gate pulses.

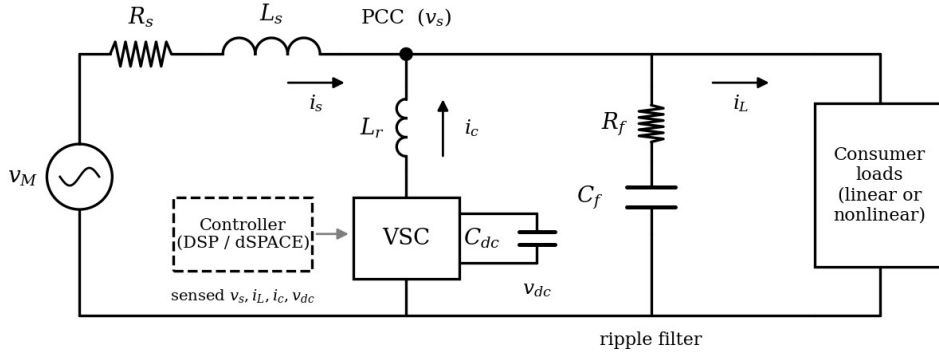


Figure 9.5. DSTATCOM single-line diagram with interfacing inductor, ripple filter and DC-link capacitor.

With the compensator current i_c defined positive from the VSC into the PCC, Kirchhoff's current law gives

$$i_s = i_L - i_c \quad (9.4)$$

so whatever part of the load current the DSTATCOM supplies locally no longer flows from the source. Decompose the load current of phase k into four parts:

$$i_{Lk} = i_{pk} + i_{qk} + i_{uk} + i_{hk} \quad (9.5)$$

where i_{pk} is the fundamental positive-sequence current in phase with the voltage (it carries the load's average real power), i_{qk} the fundamental positive-sequence current in quadrature, i_{uk} the negative- and zero-sequence fundamental components and i_{hk} the harmonics. In full compensation the source reference is $i_{sk}^* = i_{pk} + i_{loss,k}$, where i_{loss} is a small in-phase current that supplies converter losses and holds the DC-link voltage, and the DSTATCOM must inject $i_{ck}^* = i_{qk} + i_{uk} + i_{hk} - i_{loss,k}$. Which components remain in the source current is a design choice: in unity-power-factor operation only i_p remains, while in zero-voltage-regulation operation a deliberate fundamental reactive component is retained so that the feeder drop yields the required PCC magnitude.

Treating the VSC fundamental output as a voltage $V_c \angle \delta$ behind a reactance X_r , with PCC voltage $V \angle 0$, the power delivered to the PCC per phase is

$$P_c = \frac{V V_c \sin \delta}{X_r}, Q_c = \frac{V (V_c \cos \delta - V)}{X_r} \quad (9.6)$$

When $V_c \cos \delta > V$ the converter delivers reactive power (capacitive operation); when $V_c \cos \delta < V$ it absorbs it (inductive operation). A small angle δ draws real power to charge the DC capacitor and cover losses. In practice the converter runs in current-controlled PWM, so these relations describe what the current controller achieves rather than how it is commanded.

The same hardware supports several objectives, often simultaneously:

- *Power-factor correction or unity power factor (UPF) mode.* Source current is forced into phase with the PCC voltage, so the source supplies only real power. The PCC voltage settles wherever the feeder impedance puts it.
- *Zero voltage regulation (ZVR) mode.* The DSTATCOM injects enough reactive current to hold the PCC voltage magnitude at its reference. The source current then contains a reactive component, so the source power factor is no longer unity.
- *Load balancing.* Negative-sequence and, in four-wire systems, zero-sequence load currents are supplied locally, so the source sees a balanced set.
- *Harmonic elimination.* Load harmonic currents are supplied locally; this is shunt active filtering.

- *Neutral-current compensation.* In a four-wire system the zero-sequence and triplen currents return through the compensator's own neutral path, so the source neutral carries almost nothing.

Figure 9.6 compares the three cases that matter most for fundamental quantities. Without compensation, the lagging load current produces a drop across the feeder and the load voltage falls. In UPF mode the source current is aligned with the PCC voltage; the drop $jX_s I_s$ is now in quadrature with V_L , so the magnitude falls less but still falls. In ZVR mode the source current is advanced until $|V_L| = |V_s|$.

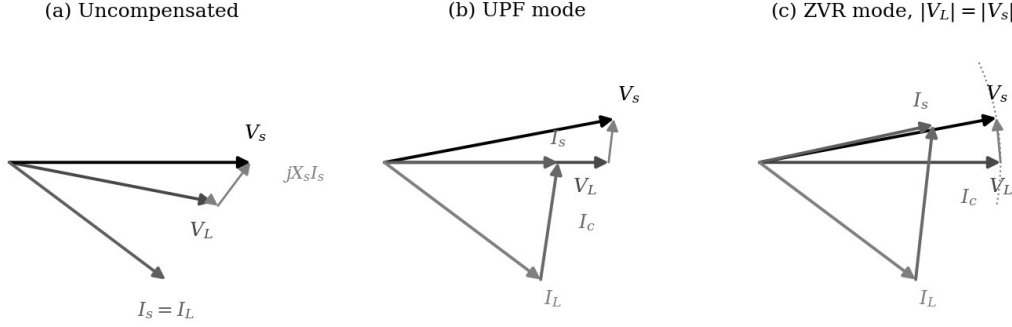


Figure 9.6. Phasor diagrams: (a) uncompensated lagging load, (b) DSTATCOM in UPF mode, (c) DSTATCOM in ZVR mode.

Reactive current needed for ZVR. Take the PCC voltage V_L as reference and the source EMF magnitude $|V_s|$ as given. With per-phase load real power P and a lossless compensator, the source current is $I_s = I_p + jI_q$ with $I_p = P/V_L$. KVL across the feeder impedance $Z_s = R_s + jX_s$ gives $V_s = V_L + Z_s I_s$, and requiring $|V_s|$ to equal the specified source magnitude leads to a quadratic in I_q :

$$(R_s^2 + X_s^2)I_q^2 + 2(BR_s - AX_s)I_q + A^2 + B^2 - |V_s|^2 = 0, A = V_L + R_s I_p, B = X_s I_p \quad (9.7)$$

The smaller root, normally positive (leading source current), is used. The compensator current then follows from $I_c = I_L - I_s$ in each phase. ZVR is only as good as the feeder impedance allows: on a stiff bus the reactive current needed to move the voltage becomes large, while on a weak bus a small current has a large effect. A controller that ignores R_s may work on an inductive medium-voltage feeder but decouples poorly on low-voltage feeders where R/X is appreciable. Example 9.4 works through a complete case.

DSTATCOM compared with an SVC. Chapter 6 showed that an SVC behaves as a controlled susceptance, so its maximum capacitive current falls in proportion to voltage. A DSTATCOM is current-limited: it can hold rated capacitive current down to very low PCC voltage (Figure 9.7). During the voltage dips that distribution customers care most about, this matters. The DSTATCOM also responds within a fraction of a cycle, injects harmonics and negative sequence on command, and does not create the low-order resonances that capacitor banks can.

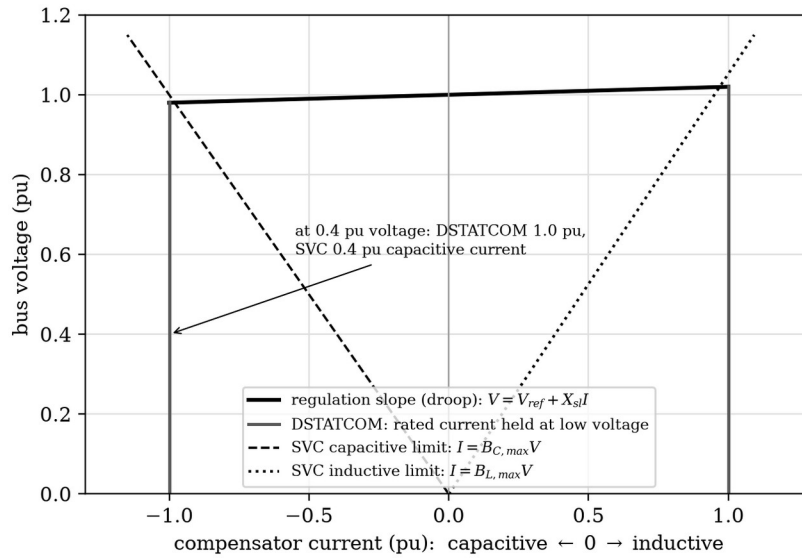


Figure 9.7. V-I characteristic of a DSTATCOM compared with an SVC (schematic). Along the droop line the bus voltage rises as the compensator moves from capacitive to inductive current.

9.3.2 Classification and topologies

Converter type. A current-fed DSTATCOM uses a CSC with a DC inductor; a voltage-fed DSTATCOM uses a VSC with a DC capacitor. For the reasons given in Section 9.1, VSCs dominate.

Three-phase three-wire topologies. Figure 9.8 shows the four common arrangements. The three-leg VSC (a) uses six switches and one DC capacitor and is the standard choice. The two-leg VSC with split capacitors (b) saves two switches by tying the third phase to the capacitor midpoint, at the cost of a higher DC voltage and capacitor-voltage balancing. Three single-phase H-bridges (c) each need an isolation transformer but give independent phase control and redundancy. An isolating star/delta transformer (d) matches a low-voltage converter to a medium-voltage bus and provides galvanic isolation.

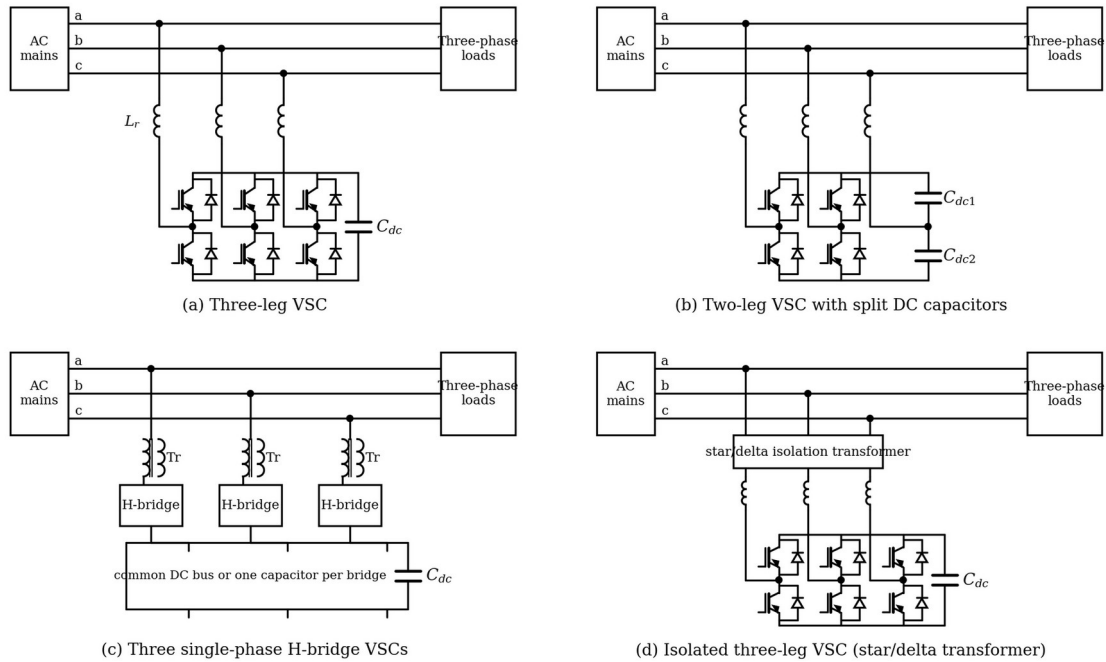


Figure 9.8. Three-phase three-wire DSTATCOM topologies.

Three-phase four-wire topologies. Four-wire distribution feeds single-phase loads between line and neutral, so zero-sequence current flows and triplen harmonics add in the neutral. A six-switch three-leg bridge cannot

synthesize zero-sequence current, so a neutral path must be added (Figure 9.9). The four-leg VSC (a) adds a fourth leg with its own inductor L_{rn} and controls the neutral current directly. The split-capacitor three-leg VSC (b) returns the neutral current through the capacitor midpoint; it needs no extra switches but large, balanced capacitors because the neutral current flows through them. Transformer-assisted schemes (c) and (d) let a three-leg VSC handle positive and negative sequence while a passive transformer provides a low-impedance zero-sequence path. Zig-zag, star/delta, T-connected, star/hexagon, star/polygon and Scott-connected transformers have all been used [8]. Three single-phase H-bridges with isolation transformers are also used. Table 9.6 summarizes the trade-offs.

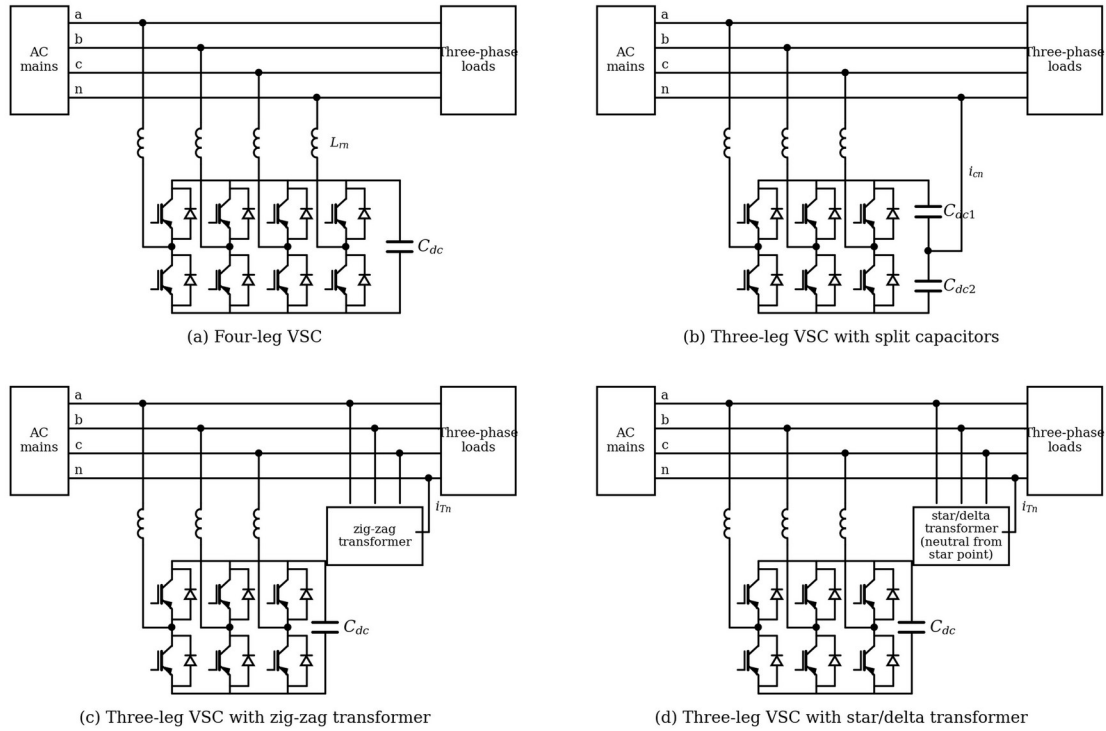


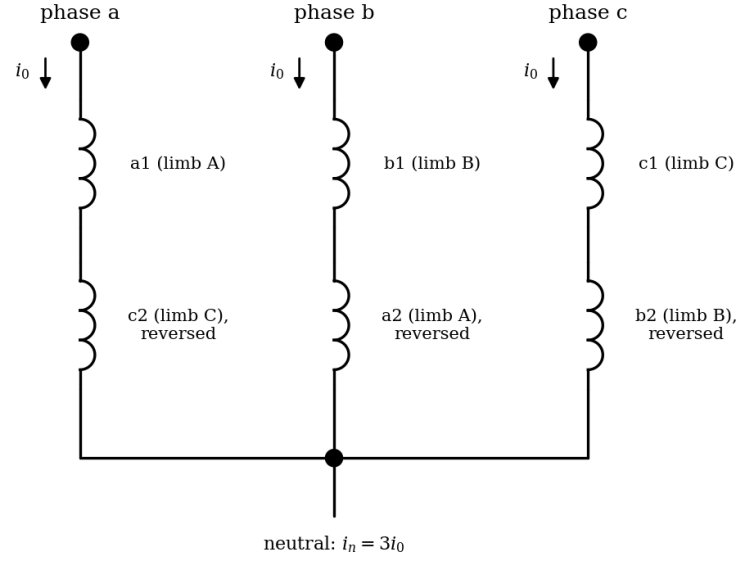
Figure 9.9. Three-phase four-wire DSTATCOM topologies.

Table 9.6. Four-wire DSTATCOM topologies compared.

Topology	Zero-sequence capability	Main advantage	Main design concern
Three-leg VSC	No direct neutral-current path	Lowest switch count for three-wire systems	Cannot compensate feeder neutral current
Four-leg VSC	Full controlled path	Independent neutral-current control	Extra leg, three-dimensional modulation and current limit
Split DC link	Neutral tied to midpoint	No fourth leg	Capacitor-voltage imbalance and ripple
Zig-zag assisted	Passive zero-sequence path	Reduces converter neutral duty	Transformer size, losses and harmonic heating

The zig-zag transformer works because each limb carries two windings belonging to different phases, connected in opposition (Figure 9.10). Zero-sequence currents are equal in all three phases, so the two mmfs on each limb cancel and the zero-sequence current sees only leakage impedance. Positive- and negative-sequence currents do not cancel and see the full magnetizing impedance, so the transformer draws almost nothing from the balanced system. Connected at the load, it diverts most of the neutral current away from the source. Its

effectiveness depends on its zero-sequence impedance relative to the source neutral path, so it is usually combined with active control rather than relied on alone.



Limb A carries a1 (current i_a) and a2 (current i_b , reversed): net mmf $\propto i_a - i_b$.
Zero-sequence currents ($i_a = i_b = i_c = i_0$) give zero mmf on every limb,
so i_0 meets only leakage impedance and returns through the neutral.

Figure 9.10. Zig-zag transformer as a passive zero-sequence (neutral-current) path.

9.3.3 Control algorithms

A DSTATCOM controller has three layers. The reference generator extracts from measured voltages and currents the source-current reference $\dot{i}_s^*$ (indirect current control) or the compensator-current reference $\dot{i}_c^* = \dot{i}_L - \dot{i}_s^*$ (direct current control). A DC-link voltage regulator, usually a PI controller, adds the small active current $\dot{i}_{loss}$ that keeps v_{dc} at its reference. A current controller, hysteresis or PWM, makes the VSC track the reference. The current loop must be substantially faster than the outer voltage loops: an outer DC-voltage loop that is too fast tries to correct switching-scale and twice-frequency energy pulsations and injects distortion into the reference, while a slow current loop cannot reproduce harmonic commands. Bandwidth separation is therefore a physical requirement, not a tuning preference.

Many reference-generation algorithms have been published [2], [9]. In the time domain they include unit-template (PI-based) control, the power balance theory, the $I \cos \phi$ algorithm, current synchronous detection, instantaneous reactive power (pq) theory, synchronous reference frame (SRF or dq) theory, instantaneous symmetrical component theory, single-phase pq and dq theories, Adaline (LMS) estimators and enhanced-PLL methods. In the frequency domain they include Fourier, DFT and recursive-DFT extraction, Kalman filtering and wavelet methods. The four most widely taught are developed below; the others are summarized in Table 9.7.

Unit-template (PI-controller-based) control

This is the simplest scheme (Figure 9.11). The amplitude of the PCC voltage is estimated from the three sampled phase voltages,

$$V_t = \sqrt{2/3 (v_{sa}^2 + v_{sb}^2 + v_{sc}^2)} \quad (9.8)$$

and in-phase unit templates are formed as $u_a = v_{sa}/V_t$, $u_b = v_{sb}/V_t$, $u_c = v_{sc}/V_t$. For balanced sinusoidal voltages these are unit-amplitude sinusoids locked to the supply. Quadrature unit templates, each leading its in-phase template by 90° , follow algebraically:

$$w_a = \frac{-u_b + u_c}{\sqrt{3}}, w_b = \frac{\sqrt{3}u_a}{2} + \frac{u_b - u_c}{2\sqrt{3}}, w_c = -\frac{\sqrt{3}u_a}{2} + \frac{u_b - u_c}{2\sqrt{3}} \quad (9.9)$$

where i_{loss} is the output of the DC-voltage PI and i_{qr} that of an AC-voltage PI. The inverse transformation $i_{sa} = i_d \cos \theta - i_q \sin \theta$ (and likewise for b and c with $\theta \mp 2\pi/3$) returns balanced, sinusoidal source-current references. A positive i_q is a leading source current, which raises the PCC voltage through the feeder inductance. Because the reference is built from the PLL angle rather than from the measured voltage waveform, SRF control gives clean sinusoidal source currents even when the PCC voltage is distorted. Its accuracy depends on the PLL and on the low-pass filter, which trades dynamic response against ripple rejection.

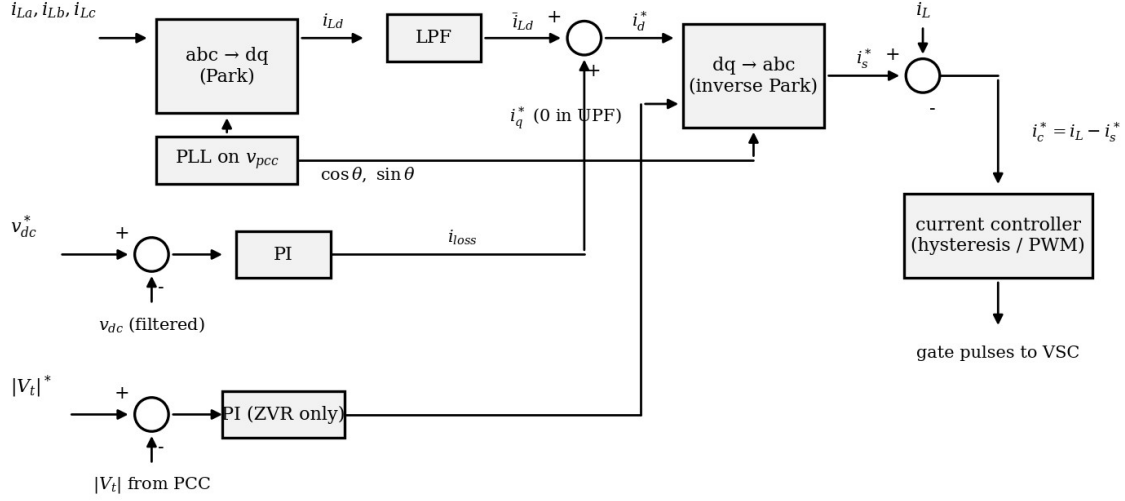


Figure 9.12. SRF (dq) control of a three-leg DSTATCOM.

Instantaneous reactive power (pq) theory

Akagi, Kanazawa and Nabae's instantaneous power theory [11], [12] works in the stationary $\alpha\beta$ frame. The first step applies the Clarke transformation to voltages and currents (Figure 9.13):

$$\begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} = \frac{2}{3} \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} \quad (9.13)$$

The $2/3$ factor is amplitude-invariant: balanced currents of peak I_m trace an $\alpha\beta$ circle of radius I_m . With zero sequence absent, true instantaneous three-phase power is $P_{3\phi} = \frac{3}{2} p$; a power-invariant transform uses a different scaling.

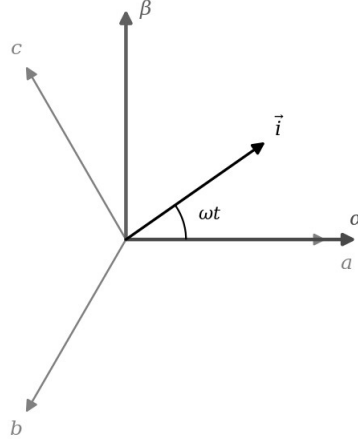


Figure 9.13. Amplitude-invariant Clarke axes.

The instantaneous real and imaginary powers are

$$p = v_\alpha i_\alpha + v_\beta i_\beta, q = v_\alpha i_\beta - v_\beta i_\alpha \quad (9.14)$$

With the positive- β axis and cosine phase sequence used here, a lagging balanced load has $q < 0$ and its conventional reactive power is $Q = -\frac{3}{2}\dot{q}$; this sign must be carried into the inverse and into the controller. For

balanced sinusoidal voltages, a linear balanced load gives constant p and q ; harmonics and unbalance in the current produce oscillating parts. Distorted or unbalanced voltage also produces oscillating power, so practical controllers often build the current reference from the extracted positive-sequence fundamental voltage.

Split each power into an average and an oscillating part, $p = \bar{p} + \tilde{p}$ and $q = \bar{q} + \tilde{q}$. To cancel harmonics only, the compensator powers are $p_C = \tilde{p}$ and $q_C = \tilde{q}$. To cancel harmonics and fundamental reactive current, $p_C = \tilde{p}$ and $q_C = q$. A DC-voltage regulator adds a small term to p_C to cover losses. Inverting Equation (9.14) gives

$$i_{\alpha,C} = \frac{v_\alpha p_C - v_\beta q_C}{v_\alpha^2 + v_\beta^2}, i_{\beta,C} = \frac{v_\beta p_C + v_\alpha q_C}{v_\alpha^2 + v_\beta^2} \quad (9.15)$$

and the phase references follow from the inverse amplitude-invariant transformation,

$$\begin{bmatrix} i_{C,a} \\ i_{C,b} \\ i_{C,c} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} & 1 \\ -\frac{1}{2} & -\frac{\sqrt{3}}{2} & 1 \end{bmatrix} \begin{bmatrix} i_{\alpha,C} \\ i_{\beta,C} \\ i_{0,C} \end{bmatrix} \quad (9.16)$$

with $i_{0,C} = 0$ in a three-wire installation. The denominator in Equation (9.15) must be kept away from zero during deep sags; a practical controller saturates the current, gives priority to the most important duty and has a declared fallback. Figure 9.14 shows the complete signal chain.

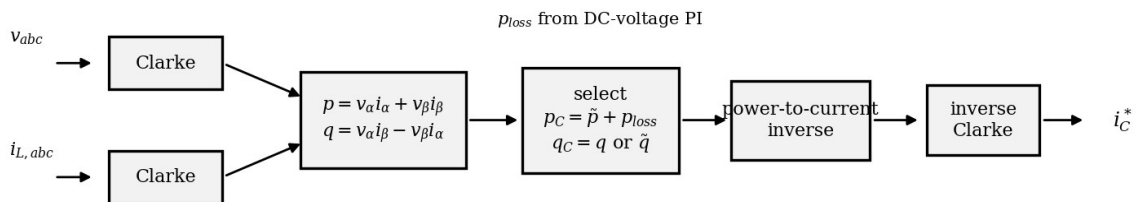


Figure 9.14. pq-based compensating-current reference.

Four-wire extension. A four-wire load can have a nonzero phase-current sum. Extend the transformation with

$$i_0 = \frac{1}{3}(i_a + i_b + i_c), v_0 = \frac{1}{3}(v_a + v_b + v_c) \quad (9.17)$$

For neutral current defined as returning from the load, $i_n = i_a + i_b + i_c = 3i_0$, and the complete real-power identity is

$$P_{3\phi} = \frac{3}{2}(v_\alpha i_\alpha + v_\beta i_\beta) + 3v_0 i_0 \quad (9.18)$$

A four-wire DSTATCOM commands $i_{0,C} = i_{0,L}$ so that the source carries no zero sequence. When $v_0 = 0$ this term must be controlled directly as a current; it cannot be reconstructed by dividing a zero-sequence power by v_0 , because nonzero i_0 can exist with zero p_0 . A three-wire bridge without a neutral path cannot deliver zero-sequence current at all.

Other algorithms in brief

Power balance theory computes the instantaneous load power $p_L = \sum v_{sk} i_{Lk}$, low-pass filters it and divides by the voltage amplitude to obtain the in-phase source-current amplitude, and does the same with an instantaneous reactive power for ZVR. $I \cos \phi$ control extracts, at the zero crossings of each phase voltage, the in-phase and quadrature magnitudes of that phase's load current, averages them over the three phases and multiplies by the unit templates. Current synchronous detection is closely related and works with filtered positive-sequence voltages. Instantaneous symmetrical component theory forces balanced source currents with a chosen power-factor angle ϕ_p directly:

$$i_{sa}^* = \frac{v_{sa} + \beta(v_{sb} - v_{sc})}{\sum_k v_{sk}^2} \dot{p}_L, \beta = \frac{\tan \phi_p}{\sqrt{3}} \quad (9.19)$$

with b and c obtained cyclically and $\beta = 0$ for unity power factor. Adaline (LMS) control treats the fundamental active component of each load current as a weight W_p multiplying the unit template and updates it every sample:

$$W_{pa}(k+1) = W_{pa}(k) + \eta [i_{La}(k) - W_{pa}(k)u_a(k)]u_a(k) \quad (9.20)$$

where η is a convergence factor, typically 0.001 to 0.1. Single-phase pq and dq theories construct the missing orthogonal signal by delaying each phase voltage and current by a quarter cycle, so each phase is compensated independently, which suits single-phase loads and three single-phase VSCs. Proportional-resonant controllers work in the stationary frame and track sinusoidal references without a dq transformation, while repetitive and selective-harmonic controllers provide high gain at periodic harmonic frequencies at the cost of one-cycle memory and robustness margin.

Table 9.7. Comparison of DSTATCOM reference-generation and regulation methods (qualitative; after [9]).

Method	Main idea	Strengths	Limitations
Unit template / PI	DC-link PI sets in-phase amplitude	Very simple, few sensors	Slow (DC-loop limited); sensitive to distorted voltage
SRF (dq)	Park transform, LPF on i_{Ld}	Clean sinusoidal reference with PLL; widely used	Needs PLL; LPF delay; sequence extraction for unbalance
pq (IRPT)	Instantaneous p, q in $\alpha\beta$	Fast; clear harmonic and reactive separation	Needs ideal voltage or positive-sequence extraction
Power	Average active/reactive load	Simple averaging	Response tied to averaging

Method	Main idea	Strengths	Limitations
balance, $I \cos \phi$, CSD	power or current		window
ISC theory	Symmetrical components, forced balanced current	Direct balancing, chosen pf	Sensitive to voltage distortion
Adaline (LMS)	Adaptive weight estimation	Fast tracking, no PLL	Step-size trade-off
DFT / recursive DFT	Fourier extraction per cycle	Selective harmonic control	One-cycle window; computation
PR / repetitive	Resonant or periodic internal model	Zero steady-state error at selected harmonics	Tuning burden; delay and robustness

Current control, limiting and anti-windup

A hysteresis controller switches a phase leg whenever the actual current leaves a band around the reference. It is fast and independent of load parameters but has a variable switching frequency. A PWM current controller compares the current error, passed through a PI or resonant regulator, with a fixed-frequency carrier, which gives a known switching spectrum and easier filter design. Deadbeat, predictive and sliding-mode current controllers are also used. Whatever the method, the current-loop bandwidth must exceed the highest harmonic the DSTATCOM is expected to cancel, and the DC voltage must be high enough to drive di_c/dt through L_r at the steepest part of the reference.

During a severe load step or a depressed PCC voltage, the requested current can exceed the converter rating, and the reference must be limited before modulation. A useful priority is semiconductor safety first, DC-link stability second and the chosen power-quality objective third, although the exact order is application dependent. Anti-windup is needed in every PI loop because a saturated converter cannot realize the controller output. Harmonic cancellation, power-factor correction and voltage regulation compete for the same current, so a controller that claims all three must state how the current limit is shared when their references conflict.

9.3.4 Design of a DSTATCOM

The design of a three-leg VSC-based DSTATCOM covers the DC-bus voltage and capacitance, the interfacing inductor, the ripple filter and the switch ratings [2].

DC-bus voltage. With sinusoidal PWM, the peak phase voltage a three-leg VSC can produce is $m V_{dc}/2$. The DC voltage must therefore exceed twice the peak phase voltage of the PCC:

$$V_{dc} = \frac{2\sqrt{2} V_{LL}}{\sqrt{3} m} \quad (9.21)$$

where m is the modulation index, normally taken as 1. At 415 V this gives 677.7 V, and 700 V is a common choice. A slightly higher value gives headroom for ZVR operation and for the voltage across L_r .

DC-bus capacitance. The capacitor must supply the energy demanded during a load change until the DC-voltage loop responds. Equating the energy released by a permitted voltage drop from V_{dc} to V_{dc1} to the converter's energy over a recovery time t ,

$$\frac{1}{2} C_{dc} (V_{dc}^2 - V_{dc1}^2) = k 3 V_{ph} (a I) t \quad (9.22)$$

where V_{ph} is the phase voltage, I the rated DSTATCOM phase current, a an overload factor (about 1.2) and k the fraction of rated power that must actually be buffered (1 for a conservative estimate). In a split-capacitor converter the neutral current flows into the capacitor midpoint and adds a fundamental-frequency ripple. With neutral-current amplitude $\hat{I}_n$ and two capacitors C in series, the midpoint current divides between them, so the ripple on each capacitor is $\Delta v_{pp} = \hat{I}_n / (\omega C)$ and

$$C \geq \frac{\hat{I}_n}{\omega \Delta v_{pp}} \quad (9.23)$$

A four-leg converter does not pass neutral current through its capacitor, but single-phase loading still produces a twice-frequency power pulsation that must be checked.

Interfacing inductance. The inductor limits the switching-frequency ripple in i_c . For a three-leg VSC with switching frequency f_s and permitted peak-to-peak ripple Δi_{cr} ,

$$L_r = \frac{\sqrt{3} m V_{dc}}{12 a f_s \Delta i_{cr}} \quad (9.24)$$

Ripple is usually allowed to be 5 to 15 % of rated current. A larger L_r reduces ripple but slows the current response and needs more DC voltage. For a single-phase H-bridge, or each bridge of a three-single-phase converter, the corresponding expression is $L_r = m V_{dc} / (4 a f_s \Delta i_{cr})$.

Ripple filter. A series R_f - C_f branch at the PCC presents a low impedance at the switching frequency and a high impedance at 50 Hz. Its corner frequency is placed near half the switching frequency:

$$f_r = \frac{1}{2\pi R_f C_f} = \frac{f_s}{2} \quad (9.25)$$

This is the RC corner frequency, not a resonance; the branch is not tuned. With $R_f = 5 \Omega$ and $f_s = 10$ kHz, $C_f = 6.4 \mu\text{F}$, and 10 μF is usually selected.

Semiconductor ratings. The device voltage rating covers the DC bus plus a dynamic overshoot of about 10 %, $V_{sw} = 1.1 V_{dc}$, and the next standard class above this (1200 V for a 700 V bus) is chosen. The current rating covers the peak of rated current plus the ripple, with a 25 % margin:

$$I_{sw} = 1.25 (\Delta i_{cr} + \sqrt{2} I_c) \quad (9.26)$$

Adding the full peak-to-peak ripple, as here, is slightly conservative; some texts add only half of it. A design should state its convention.

9.3.5 Simulated performance

The figures in this subsection come from a switched simulation written for this chapter: a 415 V, 50 Hz three-wire system feeding a six-pulse rectifier in parallel with a 0.8 power factor linear load, compensated by a three-leg IGBT VSC (700 V DC bus, 3 mF, $L_r = 2.5$ mH, hysteresis band 1 A) under SRF control with a PLL on the PCC voltage. At $t = 0.25$ s the rectifier current rises from 40 A to 60 A and the linear load rises by 50 %.

Figure 9.15 shows UPF operation. The source currents become balanced sinusoids in phase with the PCC voltages within about two cycles of the load step. The DSTATCOM current carries the rectifier's harmonics and the linear load's reactive current. The DC-link voltage dips by about 16 V at the step, because for a few milliseconds the source reference lags the new load demand and the capacitor supplies the difference, and then recovers under the PI loop. Figure 9.16 overlays one phase, and the spectra in Figure 9.17 show the load-current THD of 13.1 % reduced to 1.3 % in the source.

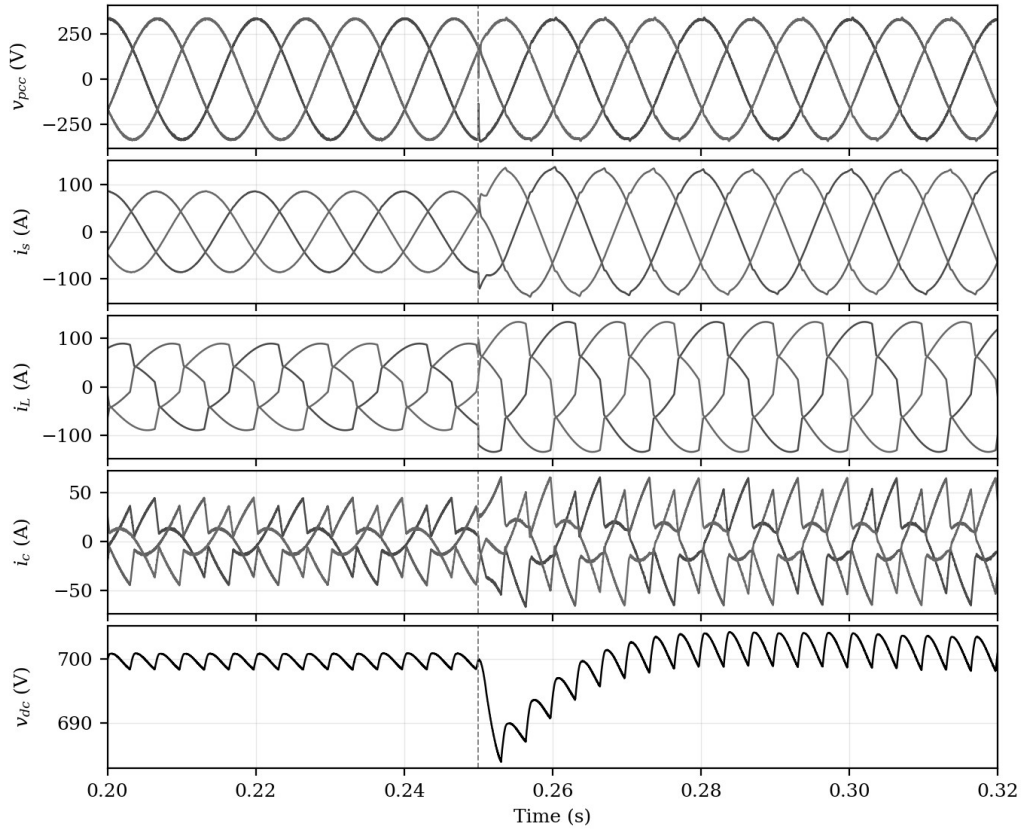


Figure 9.15. Simulated SRF-controlled three-leg DSTATCOM in UPF mode: PCC voltages, source currents, load currents, compensator currents and DC-link voltage, with a load increase at 0.25 s.

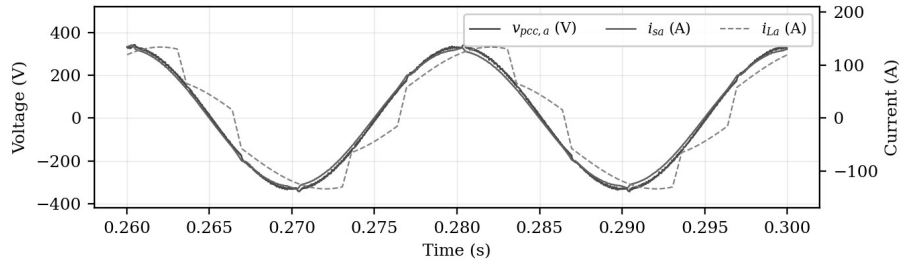


Figure 9.16. Phase-a detail in UPF mode: source current in phase with PCC voltage; load current distorted.

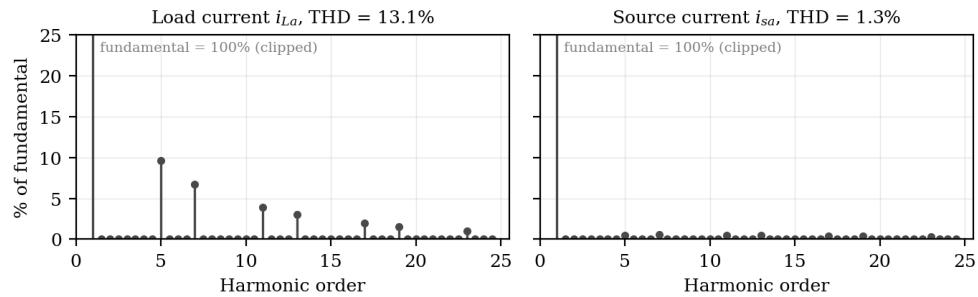


Figure 9.17. Harmonic spectra of load and source current (UPF mode, after the load step).

Figure 9.18 repeats the case on a weaker feeder ($L_s = 2$ mH) and compares UPF and ZVR modes. In UPF mode the PCC fundamental amplitude settles at 333 V before the step and 325 V after it, against a rated 338.8 V. In ZVR mode the AC-voltage loop holds 338 to 339 V in both conditions; the source current is a little larger and

leads slightly, giving a displacement factor of about 0.98. ZVR buys voltage at the price of extra source and compensator current.

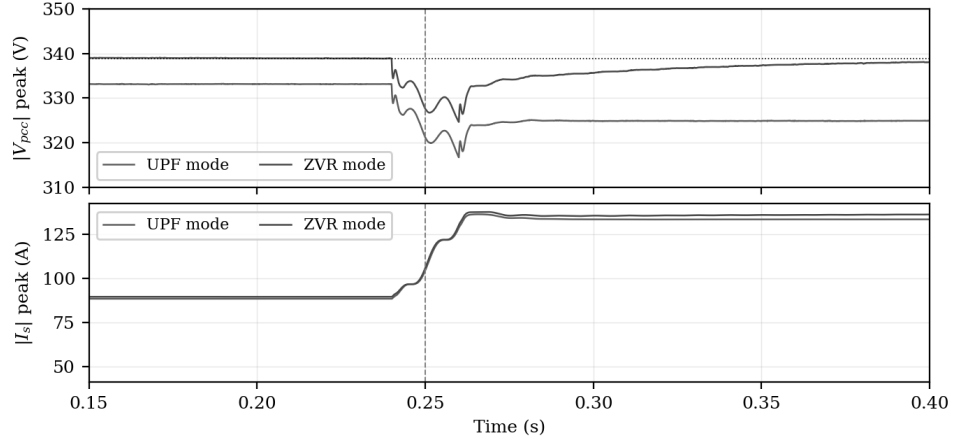


Figure 9.18. PCC fundamental voltage and source current amplitude for UPF and ZVR modes on a weak feeder.

Figure 9.19 shows a three-phase four-wire DSTATCOM (average-value model) balancing a 10 kW single-phase load on phase a, a 5 kW, 0.8 pf load on phase b (removed at 0.2 s) and a single-phase rectifier-type load on phase c. The load neutral current, which includes third-harmonic current, reaches about 55 A peak. After compensation the source currents are balanced sinusoids and the source neutral current is essentially zero apart from a brief transient at the load change; the compensator neutral leg carries the full neutral current.

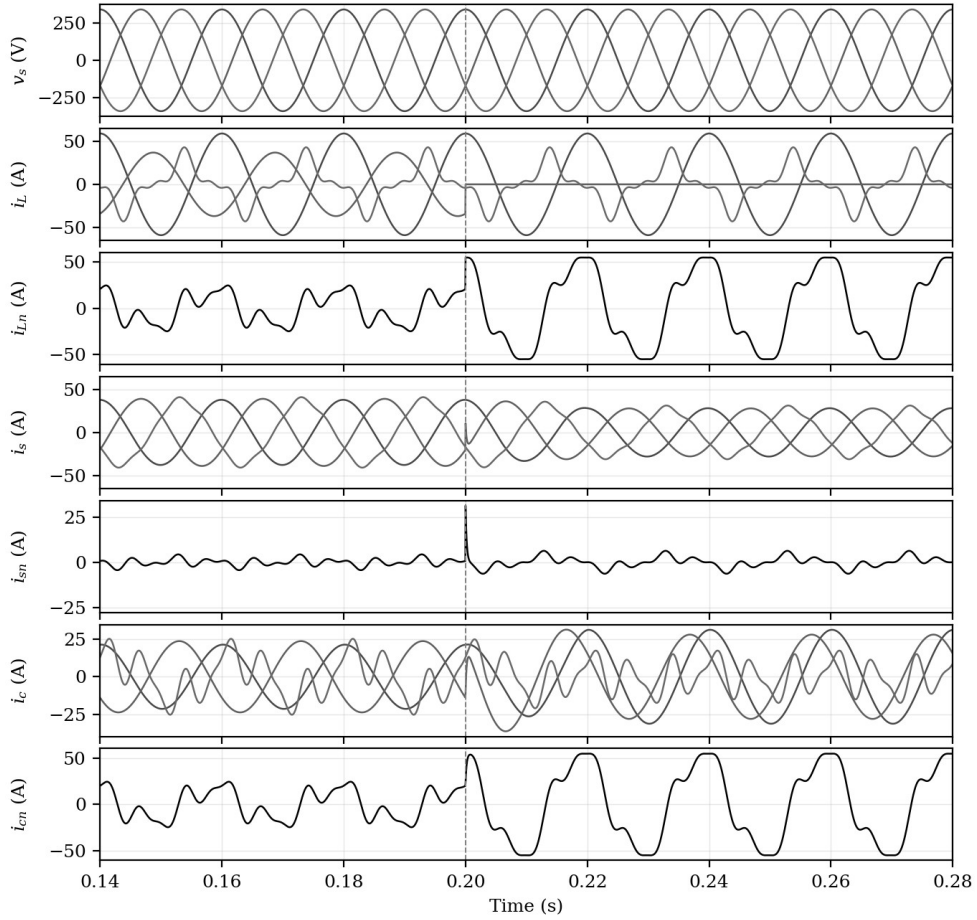


Figure 9.19. Four-wire DSTATCOM for load balancing and neutral-current compensation (average-value model).

9.3.6 Worked examples: DSTATCOM

Example 9.1. UPF compensation and complete design for an induction motor load

A three-phase three-wire DSTATCOM at 415 V, 50 Hz provides reactive power compensation for unity power factor to a delta-connected 25 kW induction motor running at 0.8 lagging power factor. Calculate (a) the supply line current before and after compensation, (b) the equivalent per-phase resistance of the compensated load, (c) the DSTATCOM current, (d) its kVA rating, (e) the interfacing inductance and (f) the DC-bus capacitance. Use $f_s = 2.5$ kHz, $V_{dc} = 700$ V held within 8 %, ripple current 15 % of the DSTATCOM current, $m = 1$, $a = 1.2$, $k = 1$ and a recovery time of 10 ms.

Solution. $V_{ph} = 415/\sqrt{3} = 239.6$ V. (a) Before compensation $I_s = 25\,000/(\sqrt{3} \times 415 \times 0.8) = 43.48$ A. After compensation only the active component remains: $I_s = 43.48 \times 0.8 = 34.78$ A. (b) $R_{eq} = 239.6/34.78 = 6.89 \Omega$ per phase (star equivalent). (c) $I_{DST} = 43.48 \times 0.6 = 26.09$ A. (d) $S_{DST} = 3 \times 239.6 \times 26.09 = 18.75$ kVA, equal to the motor's reactive power $25 \tan 36.87^\circ = 18.75$ kvar. (e) $\Delta i_{cr} = 0.15 \times 26.09 = 3.91$ A, so from Equation (9.24), $L_r = \sqrt{3} \times 700 / (12 \times 1.2 \times 2500 \times 3.91) = 8.61$ mH. (f) With an 8 % band, $V_{dc1} = 644$ V. From Equation (9.22) with $a = 1.2$, $\Delta E = 3 \times 239.6 \times (1.2 \times 26.09) \times 0.01 = 225.0$ J and $C_{dc} = 2 \times 225.0 / (700^2 - 644^2) = 5.98$ mF. Without the overload factor ($a = 1$) the value would be 4.98 mF, so a design must state which convention it uses. The switch ratings from Equation (9.26) are $I_{sw} = 1.25(3.91 + \sqrt{2} \times 26.09) = 51.0$ A and $V_{sw} = 770$ V, so 1200 V, 75 A devices are a reasonable choice. A ripple filter with $R_f = 5 \Omega$ and corner frequency $f_s/2 = 1.25$ kHz needs $C_f = 25.5 \mu\text{F}$.

Example 9.2. Balancing a single-phase load connected across two lines

A three-phase three-wire DSTATCOM at 415 V, 50 Hz is used for load balancing and power-factor correction of a single-phase 50 kW unity-power-factor load connected between lines a and b. Calculate (a) the load line currents, (b) the supply line currents after compensation, (c) the DSTATCOM currents and (d) its kVA rating.

Solution. (a) $V_{ab} = 415 \angle 30^\circ$ V and $|Z_L| = 415^2/50\,000 = 3.44 \Omega$, so $I_{La} = 120.48 \angle 30^\circ$ A, $I_{Lb} = -I_{La} = 120.48 \angle -150^\circ$ A and $I_{Lc} = 0$. (b) The balanced UPF source currents carry 50 kW: $I_s = 50\,000/(3 \times 239.6) = 69.56$ A, so $I_{sa} = 69.56 \angle 0^\circ$, $I_{sb} = 69.56 \angle -120^\circ$ and $I_{sc} = 69.56 \angle 120^\circ$ A. (c) From $I_c = I_L - I_s$: $I_{ca} = 69.56 \angle 60^\circ$ A, $I_{cb} = 69.56 \angle 180^\circ$ A and $I_{cc} = 69.56 \angle -60^\circ$ A. (d) $S_{DST} = 239.6 \times 3 \times 69.56 = 50.0$ kVA. The DSTATCOM rating equals the load rating. This is a general result for a single-phase resistive load balanced by an ideal compensator: the negative-sequence power to be circulated equals the load power, the same conclusion reached for the passive Steinmetz circuit, now realized with a converter.

Example 9.3. Four-wire unbalanced star load realized as a balanced UPF load

A three-phase four-wire unbalanced star-connected load with $Z_a = 9.0 + j3.0$ pu, $Z_b = 3.0 + j1.5$ pu and $Z_c = 7.5 + j1.5$ pu (base impedance 9.15Ω) is supplied at 415 V, 50 Hz. A four-leg DSTATCOM realizes it as a balanced UPF load. Find (a) the source currents, (b) the DSTATCOM phase currents, (c) its neutral current, (d) its kVA rating and (e) the equivalent per-phase resistance.

Solution. In ohms, $Z_a = 82.35 + j27.45$, $Z_b = 27.45 + j13.73$ and $Z_c = 68.63 + j13.73 \Omega$. The load currents are $I_{La} = 2.760 \angle -18.43^\circ$, $I_{Lb} = 7.807 \angle -146.57^\circ$ and $I_{Lc} = 3.424 \angle 108.69^\circ$ A. Total real power is $\sum |I_{Lk}|^2 R_k = 3104.9$ W, or 1035.0 W per phase. (a) $I_s = 1035.0/239.6 = 4.32$ A in each phase, in phase with its voltage. (b) $I_{ca} = 1.912 \angle -152.84^\circ$, $I_{cb} = 4.391 \angle -172.66^\circ$ and $I_{cc} = 1.173 \angle -25.10^\circ$ A. (c) The load neutral current is $I_{Ln} = \sum I_{Lk} = 5.354 \angle -158.85^\circ$ A; the DSTATCOM neutral leg carries

the same current, so the source neutral carries none. (d) Summing phase-leg VA, $S_{DST} = 239.6 \times (1.912 + 4.391 + 1.173) = 1.79$ kVA, with the neutral leg rated for 5.35 A. (e) $R_{eq} = 239.6/4.32 = 55.5 \Omega$. The DSTATCOM supplies the load's reactive power (1.21 kvar in total) and all of its negative- and zero-sequence current, yet its rating is only 1.79 kVA because these duties partly cancel in each leg. A neutral-leg current should not be multiplied by a phase voltage when rating the converter, because the neutral leg's AC terminal is at neutral potential.

Example 9.4. Load balancing and zero voltage regulation on a weak feeder

A 415 V, 50 Hz supply has a feeder impedance of $1 + j3 \Omega$ per phase. A single-phase load $Z_a = 10 + j5 \Omega$ is connected between phase a and neutral; phases b and c are unloaded. A four-leg DSTATCOM at the load bus balances the load and regulates the load-bus voltage to 239.6 V per phase. Find (a) the load-bus voltage without compensation, (b) the source currents, (c) the DSTATCOM currents and (d) its rating.

Solution. (a) Without compensation, phase a sees $V_{La} = 239.6 Z_a / (Z_a + Z_s) = 196.9 \angle -9.46^\circ$ V, an 18 % sag in that phase. (b) With $V_L = 239.6$ V, $I_{La} = 239.6 / (10 + j5) = 21.43 \angle -26.57^\circ$ A and $P_L = 21.43^2 \times 10 = 4592.7$ W, or 1530.9 W per phase, so $I_p = 6.389$ A. In Equation (9.7), $A = 245.99$ and $B = 19.17$, giving $10 I_q^2 - 1437.6 I_q + 3470.0 = 0$ and $I_q = 2.456$ A. Thus $I_{sa} = 6.389 + j2.456 = 6.845 \angle 21.02^\circ$ A, and I_{sb}, I_{sc} have the same magnitude displaced by $\mp 120^\circ$. The source EMF then leads the load bus by 5.18° . Because the source current leads the bus voltage, 588.4 var per phase flows from the load bus back toward the source, and this reactive flow through X_s is what lifts the bus voltage. (c) $I_{ca} = I_{La} - I_{sa} = 17.56 \angle -43.29^\circ$ A, $I_{cb} = -I_{sb} = 6.845 \angle 81.02^\circ$ A and $I_{cc} = -I_{sc} = 6.845 \angle -38.98^\circ$ A. The neutral leg carries $21.43 \angle -26.57^\circ$ A, the whole load current returning through the compensator. (d) $S_{DST} = 239.6 \times (17.56 + 6.845 + 6.845) = 7.49$ kVA, plus a 21.4 A neutral leg. The feeder drop is $|Z_s I_s| = 21.6$ V, but because the source current leads, the drop is almost perpendicular to the load voltage and $|V_s| = |V_L|$: the ZVR principle of Figure 9.6(c).

Example 9.5. Power-factor correction rating

A 415 V, 50 Hz system feeds a 45 kW star-connected load at 0.72 lagging power factor. Find the source current before and after UPF compensation, the load reactive power, the DSTATCOM current and kVA, and the equivalent resistance after compensation.

Solution. Before: $I_s = 45\,000 / (\sqrt{3} \times 415 \times 0.72) = 86.95$ A. After: $I_s = 45\,000 / (\sqrt{3} \times 415) = 62.60$ A, a 28 % reduction. $Q_L = 45 \tan(\cos^{-1} 0.72) = 43.37$ kvar. $I_{DST} = 43\,373 / (\sqrt{3} \times 415) = 60.34$ A and $S_{DST} = 43.37$ kVA. $R_{eq} = 239.6/62.60 = 3.83 \Omega$ per phase.

Example 9.6. Interfacing inductor

A DSTATCOM operates with $V_{dc} = 900$ V, $f_s = 4$ kHz, $m = 1$, $a = 1.2$ and a permitted ripple of 10 % of its 50 A rated current. Find L_r .

Solution. $\Delta i_{cr} = 5$ A, and Equation (9.24) gives $L_r = \sqrt{3} \times 900 / (12 \times 1.2 \times 4000 \times 5) = 5.41$ mH.

Example 9.7. Harmonic compensation current and rating

A 415 V nonlinear load draws 40 A fundamental (in phase with the voltage) with 5th and 7th harmonics of 20 % and 15 % of the fundamental. Find the RMS load current, the harmonic current, the DSTATCOM current for sinusoidal source current and its rating.

Solution. $I_L = 40 \sqrt{1 + 0.2^2 + 0.15^2} = 41.23$ A and $I_h = 40 \sqrt{0.2^2 + 0.15^2} = 10.0$ A, so the current THD is 25 %. With the fundamental already at unity displacement, $I_{DST} = 10.0$ A and $S_{DST} = 3 \times 239.6 \times 10.0 = 7.19$

kVA. The true power factor of the load is $1/\sqrt{1+0.25^2}=0.970$ from Equation (9.2), even though its displacement factor is 1; a capacitor bank could not improve it.

Example 9.8. Rating for several loads

A 415 V system supplies 30 kW at 0.8 lagging and 20 kW at 0.9 lagging. Find the DSTATCOM rating for UPF and the source current after compensation.

Solution. $P=50$ kW and $Q=30 \times 0.75+20 \times 0.4843=32.19$ kvar, so $S_{DST}=32.19$ kVA. After compensation $I_s=50\,000/(\sqrt{3} \times 415)=69.56$ A.

Example 9.9. Clarke transform verification

With balanced voltages $v_a=V_m \cos \omega t$, $v_b=V_m \cos(\omega t-120^\circ)$, $v_c=V_m \cos(\omega t+120^\circ)$, a balanced linear load draws $i_a=10 \cos \omega t$, $i_b=10 \cos(\omega t-120^\circ)$, $i_c=10 \cos(\omega t+120^\circ)$ A. Using Equation (9.13), find i_α and i_β at $\omega t=0$ and the instantaneous powers.

Solution. At $\omega t=0$: $i_a=10$ and $i_b=i_c=-5$ A, so $i_\alpha=(2/3)(10+2.5+2.5)=10$ A and $i_\beta=0$. Throughout the cycle $i_\alpha=10 \cos \omega t$ and $i_\beta=10 \sin \omega t$. With the stated in-phase voltages, $p=10 V_m$, $q=0$ and $P_{3\phi}=15 V_m$. Current data alone, without a voltage reference, could not establish zero reactive power: a circular current locus shows balanced sinusoidal current, not unity power factor.

Example 9.10. Complete three-phase pq compensation calculation

An ideal balanced three-wire 50 Hz supply has $v_k=100 \cos \theta_k$ V with $\theta_a=\theta$, $\theta_b=\theta-2\pi/3$, $\theta_c=\theta+2\pi/3$. The load currents are $i_{L,k}=10 \cos(\theta_k-\pi/6)+2 \cos 5\theta_k+\cos 7\theta_k$ A (peak values). Derive the shunt current reference that cancels harmonics and fundamental reactive current, evaluate it at $\theta=\pi/6$ and check the source current and powers.

Solution. The fifth harmonic is negative sequence and the seventh positive sequence for these phase shifts. The Clarke transform gives $v_\alpha=100 \cos \theta$, $v_\beta=100 \sin \theta$, $i_{L,\alpha}=10 \cos(\theta-\pi/6)+2 \cos 5\theta+\cos 7\theta$ and $i_{L,\beta}=10 \sin(\theta-\pi/6)-2 \sin 5\theta+\sin 7\theta$. From Equation (9.14), $p=866.025+300 \cos 6\theta$ and $q=-500-100 \sin 6\theta$, so $\dot{p}=866.025$, $\dot{q}=-500$, $\tilde{p}=300 \cos 6\theta$ and $\tilde{q}=-100 \sin 6\theta$.

Set $p_C=\tilde{p}$ and $q_C=q$. At $\theta=\pi/6$: $(v_\alpha, v_\beta)=(86.603, 50)$ V, $(p_C, q_C)=(-300, -500)$ and $v_\alpha^2+v_\beta^2=10\,000$. Equation (9.15) gives $i_{C,\alpha}=[86.603(-300)-50(-500)]/10\,000=-0.098$ A and $i_{C,\beta}=[50(-300)+86.603(-500)]/10\,000=-5.830$ A. The inverse Clarke transform gives $(i_{C,a}, i_{C,b}, i_{C,c})=(-0.098, -5.000, 5.098)$ A. The load currents at the same instant are $(7.402, -5.000, -2.402)$ A, so $i_s=i_L-i_C=(7.5, 0, -7.5)$ A. Over the whole cycle $i_{s,k}=8.660 \cos \theta_k$ A: balanced sinusoids in phase with their voltages, 6.124 A RMS.

At this instant the physical powers are $P_L=1.5 \times 566.025=849.04$ W, $P_s=1.5 \times 866.025=1299.04$ W and $P_C=1.5 \times (-300)=-450$ W, so $P_s+P_C=P_L$. The filter absorbs 450 W at this instant and returns it at others, with zero average exchange. The load's fundamental reactive power is $Q_L=-1.5 \dot{q}=750$ var. The RMS compensating phase current is $\sqrt{(5^2+2^2+1^2)}/2=3.873$ A, including the reactive fundamental and both harmonics; harmonic-only compensation would use $q_C=\tilde{q}$ and leave the reactive current in the source.

9.4 Dynamic Voltage Restorer (DVR)

A DVR is a series-connected VSC that inserts a controlled voltage between the supply and a sensitive load, so that the load voltage stays at its reference while the supply voltage sags, swells, becomes unbalanced or distorted. The first utility DVR entered service in 1996 on the Duke Power system at Anderson, South Carolina (12.47 kV), and the device has since become a common way to protect a large sensitive load, such as a

semiconductor fab or a paper machine, against sags caused by faults elsewhere on the network [13]. The series connection is what makes it economical: a DVR supplies only the missing voltage, so its rating is a fraction of the load rating for sags of practical depth.

9.4.1 Components and operating principle

Figure 9.20 shows a complete DVR. Its main parts are:

- a voltage-source converter, usually a three-leg or three H-bridge IGBT inverter switched at several kHz;
- a DC link and energy source, which supplies the real power the injection requires (Section 9.4.2);
- an injection (series) transformer in each phase, which isolates the converter from the line, matches its voltage and current to the line, and blocks zero sequence where required;
- a passive filter, normally an L_f - C_f low-pass filter on the converter side of the transformer (line-side filters avoid transformer harmonic heating but expose the filter to fault current);
- a bypass switch (antiparallel thyristors or a mechanical switch) that short-circuits the transformer's line winding during downstream faults, overloads or maintenance; and
- a controller with voltage and current sensors, a PLL and disturbance detection.

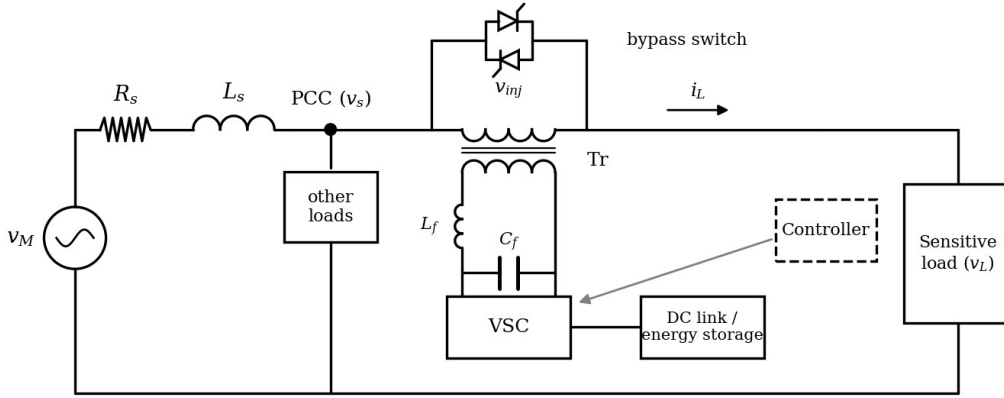


Figure 9.20. Complete DVR arrangement: injection transformer, LC filter, VSC, DC link or energy storage, bypass switch and controller.

With transformer and filter quantities referred to the line side, the governing ideal phasor relationship is

$$V_L = V_s + V_{inj} \quad (9.27)$$

where V_L , V_s and V_{inj} are RMS voltage phasors at the same terminals and positive injection is a rise from source to load, the voltage-rise convention of Chapter 8. Sag depth is $X = 1 - |V_s|/V_{LC}$, where V_{LC} is the rated (target) load voltage. The real and reactive power the DVR exchanges with the line are, per phase,

$$P_{DVR} + jQ_{DVR} = V_{inj} I_{DVR}^* \quad (9.28)$$

and for a standalone DVR the series current equals the load current, $I_{DVR} = I_L$.

A DVR has three operating modes. In standby the supply is healthy and the DVR injects nothing; either the converter applies zero voltage (shorting the transformer's converter winding through its switches) or the bypass switch carries the load current, which keeps losses low. In boost (injection) mode a disturbance is detected and the DVR injects the missing voltage within a few milliseconds. In protection mode a downstream fault drives current far above rating, so the bypass switch closes and the converter is blocked.

9.4.2 Classification and energy-source topologies

DVRs are classified, like the DSTATCOM, by converter type (CSC or VSC), by supply system (single-phase, three-phase three-wire and three-phase four-wire) and by bridge topology. A single-phase DVR may use a half bridge or a full bridge (Figure 9.21). For the same DC voltage the full bridge produces twice the AC output voltage; the half bridge needs two series capacitors and a doubled DC voltage, but only two switches. A three-phase DVR uses either a three-leg VSC feeding three injection transformers with isolated windings (Figure 9.22) or three H-bridges. Three H-bridges, or a four-leg converter, allow independent phase voltages including zero sequence, which a three-leg bridge with delta-connected windings cannot produce. Multilevel and cascaded structures reduce device voltage stress and dv/dt at higher distribution voltages but need more devices, isolated supplies or capacitor balancing. Transformerless DVRs remove the injection transformer but transfer isolation, common-mode voltage and fault-withstand requirements to the converter and its insulation.

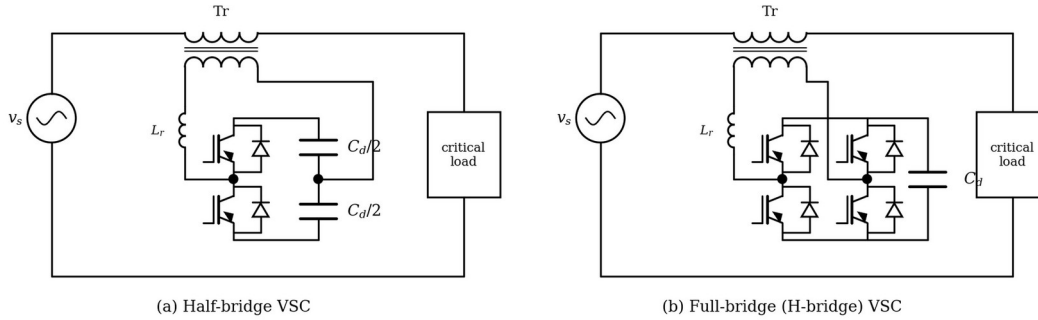


Figure 9.21. Single-phase DVR: (a) half-bridge VSC, (b) full-bridge VSC.

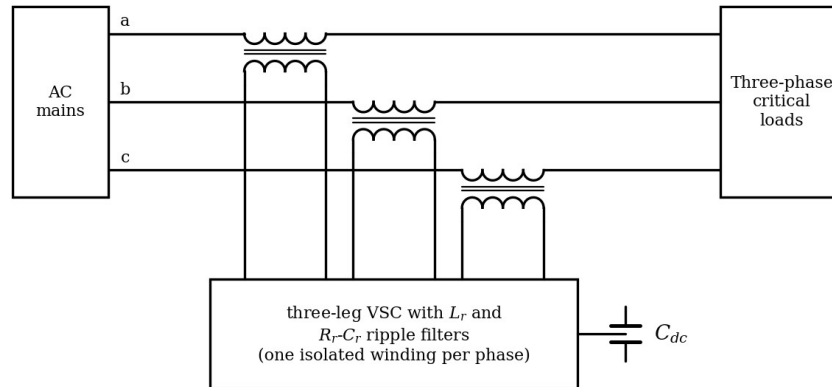


Figure 9.22. Three-phase three-wire DVR: three injection transformers driven by a three-leg VSC with ripple filters.

The most important classification for design is by the source of the injected energy (Figure 9.23) [14]:

- *No storage, supply-side (left-shunt) rectifier.* A rectifier fed from the supply side charges the DC link. The DVR can ride through sags of any duration, but the rectifier's input voltage falls with the sag, so the available DC voltage and power fall exactly when they are needed. Deep sags cannot be corrected unless the DC link is oversized.
- *No storage, load-side (right-shunt) rectifier.* The rectifier is fed from the restored load voltage, so its input stays at rated value. The extra rectifier current flows through the injection transformer, and the scheme forms a positive-feedback loop that must be controlled carefully.
- *Stored energy (BESS, supercapacitor, flywheel).* A battery or other store, usually through a DC/DC converter, supplies real power. The DVR can use in-phase injection (minimum voltage rating) and can also preserve phase, but the store must be sized for the worst sag depth and duration (Section 9.4.5).
- *Capacitor-supported (self-supported).* Only a DC capacitor is present. Ideally the DVR then exchanges no average real power with the line, so it must inject in quadrature with the load current. As shown in

Section 9.4.3, this restores a sag only if the sag is shallow enough for the load power factor, and a small real-power exchange is still needed to cover losses [15].

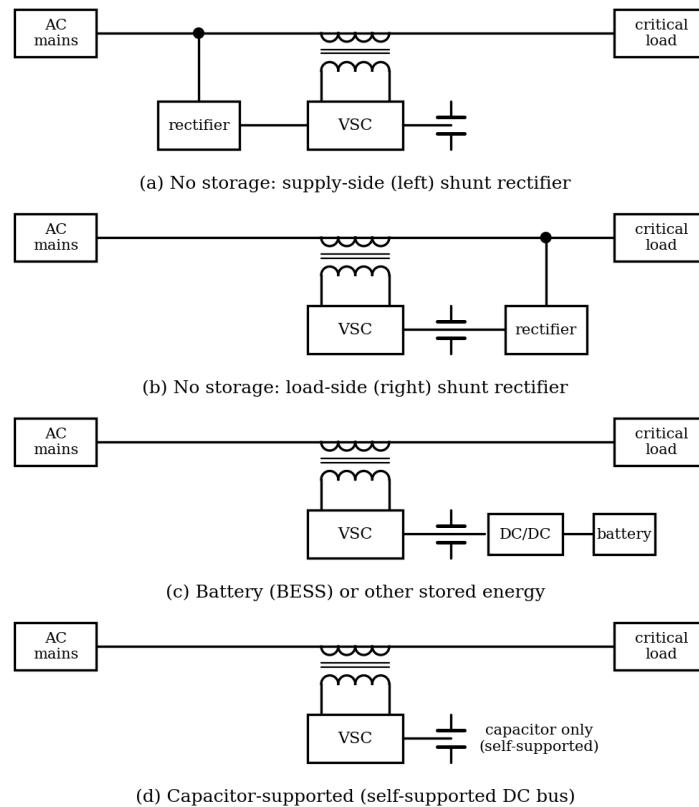


Figure 9.23. DVR energy-source options: (a) supply-side rectifier, (b) load-side rectifier, (c) battery (BESS) with DC/DC converter, (d) capacitor-supported.

Nielsen and Blaabjerg's comparison [14] concluded that, for the same power rating, a supply-side rectifier scheme without storage is simple and cheap and handles long shallow sags well, while a stored-energy DVR is the better choice when deep sags must be corrected and the stored energy needed is modest. Two DVRs with the same series-converter kVA can therefore have very different ride-through durations (Table 9.8).

Table 9.8. DVR energy-source and topology choices.

Choice	Advantage	Limitation that must be checked
Capacitor-supported	Simple DC side	Short real-power duration and DC-voltage droop
BESS-supported	Longer real-power support	Battery power capability, state of charge and converter losses
Supercapacitor-supported	High pulse power and cycle life	Lower energy density and wide voltage variation
Supply-side rectifier	No dedicated storage for moderate sags	Source must retain enough voltage and power
Multilevel VSC	Lower device stress and better waveform	Balancing, device count and protection complexity
Transformerless DVR	Reduced magnetic hardware	Isolation, common-mode and fault-duty challenges

9.4.3 Compensation strategies

The DVR can reach the target load-voltage magnitude with many different injections, because only $|V_L|$ is specified. Figure 9.24 shows the strategies in common use for a sag that also carries a phase jump.

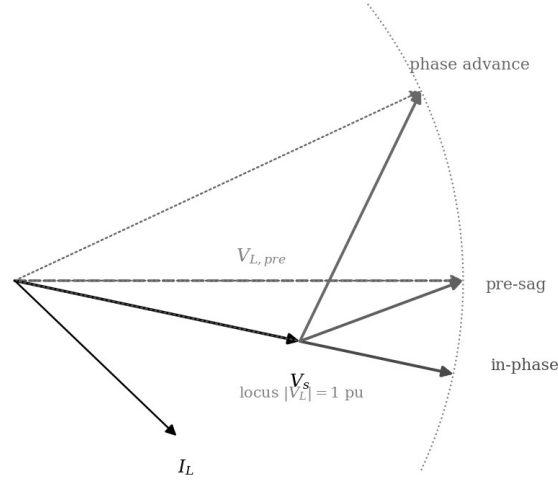


Figure 9.24. DVR compensation strategies for a sag with a phase jump: in-phase, pre-sag and phase-advance injection (the quadrature strategy is treated separately below).

In-phase injection. The injected voltage is in phase with the disturbed supply voltage. With V_s as reference, the injection is a signed scalar,

$$V_{inj} = V_{Lc} - |V_s| = V_{Lc} X \quad (9.29)$$

which is the minimum injection magnitude of any strategy, so it gives the smallest voltage rating and the smallest transformer. For a standalone DVR carrying I_L at load power-factor angle ϕ ,

$$P_{DVR} = (V_{Lc} - |V_s|) I_L \cos \phi, Q_{DVR} = (V_{Lc} - |V_s|) I_L \sin \phi \quad (9.30)$$

For a balanced three-phase sag this means $P_{DVR} = X P_L$: the DVR supplies exactly the fraction of load power that the source can no longer deliver. During a swell ($X < 0$) the signs reverse and the DVR absorbs real power, which needs a regenerative path, a store that can accept charge or a dissipative chopper. In-phase injection does not preserve the pre-sag phase.

Pre-sag compensation. Phase-sensitive loads, such as line-commutated converters and some drives, are disturbed by a phase jump as well as by a magnitude drop. The DVR then restores the pre-sag phasor itself,

$$V_{inj} = V_{L,pre} - V_s \quad (9.31)$$

which needs a larger injection, and usually more real energy, than in-phase compensation. The PLL must hold the pre-disturbance angle during the event [16].

Phase-advance (energy-optimized) compensation. If the load tolerates a phase shift, the load voltage can be rotated ahead of the supply so that the injection contains a larger reactive part and less real power, extending ride-through for a given store [17]. The price is a larger injection voltage and a phase transition at the start and end of the event.

Quadrature (zero-real-power) compensation. If the injection is perpendicular to the series current, $P_{DVR} = \text{Re}\{V_{inj} I_{DVR}^*\} = 0$ and, ideally, no stored energy is needed. This is the only steady-state option for a capacitor-supported DVR. The current reference is decisive: a standalone DVR carries the load current, so its injection must be perpendicular to I_L , not to V_s . In the right-shunt UPQC of Section 9.5 the series current is the source current, which is in phase with V_s ; that is a different geometry.

Exact feasibility of zero-real-power restoration

Choose V_s real and let $V_L = V_{Lc} e^{j\theta_L}$ and $I_L = |I_L| e^{j(\theta_L - \phi)}$. Expanding $\text{Re}\{(V_L - V_s)I_L^* \} = 0$ gives

$$\cos(\theta_L - \phi) = \frac{V_{Lc} \cos \phi}{|V_s|} \quad (9.32)$$

A real solution requires the right-hand side not to exceed one, so a standalone DVR can restore a sag without real power only if

$$|V_s| \geq V_{Lc} \cos \phi \Leftrightarrow X \leq X_{max} = 1 - \cos \phi \quad (9.33)$$

Physically, the source must still deliver the full load real power $V_{Lc} I_L \cos \phi$ through the same current magnitude I_L , and it can deliver at most $|V_s| |I_L|$. Figure 9.25 plots this limit. A load at 0.85 power factor can have a 15 % sag corrected by a capacitor-supported DVR, and a unity-power-factor load cannot have any sag corrected without real power. At the limit, $\theta_L = \phi$, the load current comes into phase with V_s , the injection is perpendicular to V_s and its magnitude is $V_{Lc} \sin \phi$. A formula giving the limit as $1 - \sin \phi$ appears in some teaching material; it confuses the limiting injection $V_{Lc} \sin \phi$ with the limiting supply voltage and must not be used.

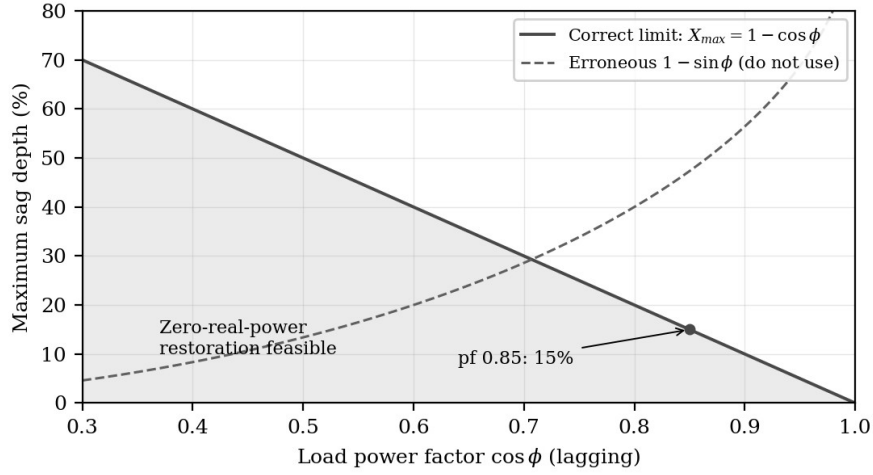


Figure 9.25. Exact zero-real-power feasibility limit of a standalone DVR versus load power factor.

Where the sag is feasible, both roots must be checked:

$$\theta_L = \phi \pm \cos^{-1} \left(\frac{V_{Lc} \cos \phi}{|V_s|} \right), V_{inj} = V_{Lc} e^{j\theta_L} - |V_s| \quad (9.34)$$

For a 230 V, 0.8 lagging-pf load with $|V_s| = 207$ V the roots are 9.604° and 64.136° , with injections of $43.17 \angle 62.73^\circ$ V and $232.83 \angle 117.27^\circ$ V (Figure 9.26). Both give zero real power; the smaller one is preferred if its load-phase shift is acceptable.

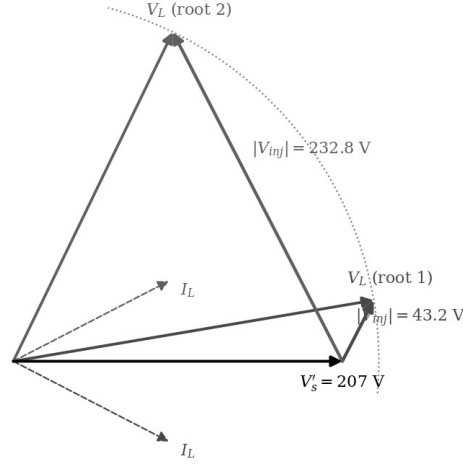


Figure 9.26. The two zero-real-power roots for a 230 V, 0.8 pf load and a 207 V supply.

General-angle injection. For a specified injection magnitude $U = |V_{inj}|$, the law of cosines on the triangle of Equation (9.27) gives the injection angle measured from V_s :

$$\cos \psi = \frac{V_{Lc}^2 - |V_s|^2 - U^2}{2|V_s|U} \quad (9.35)$$

Both signs of ψ are geometrically admissible when $|V_{Lc} - |V_s|| \leq U \leq V_{Lc} + |V_s|$. In-phase injection is the minimum- U boundary. Choosing the angle is a feasibility calculation, not an optimization by itself; an optimization needs an objective such as minimum energy, minimum converter VA or limited phase shift.

Unbalanced sags

A single-line-to-ground or line-to-line fault elsewhere sags the three phases unequally, introduces negative-sequence voltage and can add a phase jump. Restoring each phase magnitude independently, with the fault's phase relationships still in place, would not reconstruct a balanced set. Using the symmetrical components of Chapter 2 with $V_0 = 0$ (an assumption that must be checked, because a grounded-wye winding can support zero sequence),

$$V_a = V_p + V_n, V_b = a^2 V_p + a V_n, V_c = a V_p + a^2 V_n \quad (9.36)$$

and the required injections for the balanced target $V_{Lc} \angle 0^\circ$, $V_{Lc} \angle -120^\circ$, $V_{Lc} \angle 120^\circ$ are

$$V_{inj,k} = V_{k,target} - V_{s,k}, k = a, b, c \quad (9.37)$$

If $V_0 \neq 0$ it must also be subtracted, and the DVR's bridge, transformer connection and neutral path must be able to synthesize zero sequence. Figure 9.27 shows the case $V_p = 0.85$ pu, $V_n = 0.15$ pu, worked in Example 9.38. Unbalanced injection carries a second-harmonic real-power pulsation (visible in Figure 9.32), which adds DC-link ripple even when the average real power is modest.

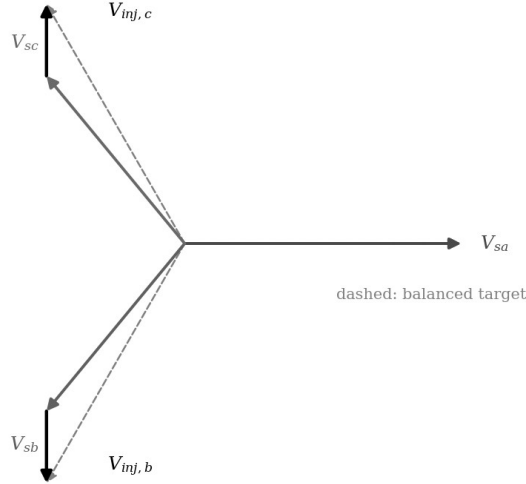


Figure 9.27. Sequence-defined unbalanced sag ($V_p=0.85$ pu, $V_n=0.15$ pu) and the per-phase injections that restore the balanced target.

Voltage harmonics and notches

A DVR with enough bandwidth can also inject the negative of supply-voltage harmonics, $v_{inj,h} = -v_{s,h}$, and fill in commutation notches, so the load sees a clean sinusoid. This is series active filtering (Section 9.6). The injection-transformer leakage inductance and the L_f - C_f filter limit the achievable bandwidth; selective harmonic controllers are used at medium voltage to keep the switching frequency low.

9.4.4 Control of the DVR

Figure 9.28 shows the usual SRF-based controller. The supply voltages are transformed to the dq frame with the angle from a PLL. For pre-sag compensation the PLL is frozen at the pre-disturbance angle when a sag is detected; for in-phase compensation it continues to track the supply. The load-voltage reference in the same frame is $v_{Ld}^i = V_m$ (rated peak phase voltage) and $v_{Lq}^i = 0$, so the feedforward injection reference is

$$v_{inj,d}^i = V_m - v_{sd}, v_{inj,q}^i = -v_{sq} \quad (9.38)$$

Transforming back gives $v_{inj,abc}^i = v_L^i - v_s$, which corrects magnitude, phase, unbalance (if negative sequence is handled, for example by a second frame rotating at $-\omega$) and harmonics in one step. A feedback loop on the measured load voltage removes the drops across the filter and transformer leakage. The injected voltage is produced by a PWM voltage controller, often with an inner loop on filter-capacitor current for active damping of the L_f - C_f resonance.

For a capacitor-supported DVR, a DC-voltage PI adds a small real-power component so that the converter absorbs its own losses:

$$v_{loss}(n) = v_{loss}(n-1) + K_{p1}[v_{de}(n) - v_{de}(n-1)] + K_{i1}v_{de}(n), v_{de} = v_{dc}^i - v_{dc} \quad (9.39)$$

and a second PI on the load-voltage amplitude error sets the quadrature component.

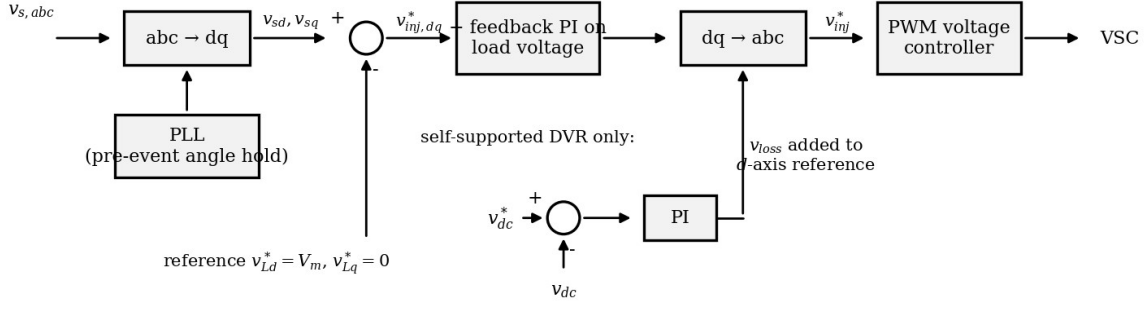


Figure 9.28. SRF-based DVR control with feedforward, load-voltage feedback and (for capacitor-supported DVRs) a DC-voltage loop.

Disturbance detection decides how fast the DVR acts. The useful response time is the sum of detection, reference calculation, PWM delay and the build-up of the output filter, and the load sees the uncompensated disturbance throughout. Half-cycle or full-cycle RMS methods are robust to noise but cannot detect a sag in a few milliseconds; peak detection, the dq-frame magnitude $\sqrt{v_d^2 + v_q^2}$ and missing-voltage methods (comparing the instantaneous voltage with its pre-sag waveform) respond in 1 to 2 ms. A design should therefore state both the restored steady-state voltage and the maximum transient deviation during detection and injection build-up. The controller must also apply converter voltage and current limits, DC-energy limits, injection-transformer flux limits, anti-windup, controlled recovery ramps and bypass logic. A step change of injected voltage at the wrong instant of the cycle can saturate the injection transformer, because flux is the time integral of voltage; flux estimation, soft insertion or a command that limits the first-cycle volt-seconds prevents this.

9.4.5 Energy storage and DC-link requirements

For a storage-supported standalone DVR, injected positive real power and converter losses deplete the store. For arbitrary phase injections the average converter AC real power is $P_{DVR} = \sum_k \text{Re}\{V_{inj,k} \dot{I}_k\}$; for balanced operation it is $3|V_{inj}||I|\cos\gamma$, where γ is the angle between injection and current, and in an in-phase standalone DVR $\gamma = \phi$. For positive, approximately constant delivered power and a stated conversion efficiency η ,

$$E = \frac{P_{DVR}}{\eta} t_{sag} \quad (9.40)$$

The general balance, which also covers zero and negative real power, is

$$E_{drawn} = \int_0^t [P_{DVR}(\tau) + P_{loss}(\tau)] d\tau \quad (9.41)$$

Pure quadrature operation has $P_{DVR} = 0$ but still needs $\int P_{loss} dt$. During absorbing operation the signed power charges the store, and charging limits must be checked. For a capacitor allowed to fall from $V_{dc,max}$ to $V_{dc,min}$, the usable energy is $\frac{1}{2}C(V_{dc,max}^2 - V_{dc,min}^2)$, so the minimum capacitance is

$$C_{dc} = \frac{2E}{V_{dc,max}^2 - V_{dc,min}^2} \quad (9.42)$$

For an in-phase standalone DVR at fixed load power, $P_{DVR} = X P_L$, so $C_{dc} = 2X P_L t / [\eta(V_{dc,max}^2 - V_{dc,min}^2)]$ is linear in both depth and duration (Figure 9.29). Long events quickly make a pure capacitor impractical, which is why batteries, supercapacitors, flywheels or rectifier-supported links are used for longer ride-through.

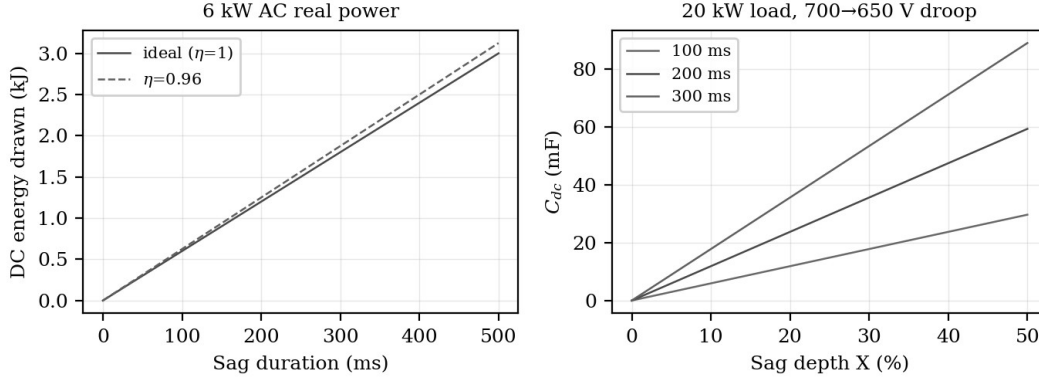


Figure 9.29. Left: DC energy versus sag duration at 6 kW AC real power. Right: energy-based capacitance of an in-phase standalone DVR versus sag depth (20 kW load, 700 to 650 V droop).

Stored energy alone does not size a DVR. The store must provide the event energy, its interface must provide the event power, and the DC-link voltage must stay high enough for the converter to synthesize the commanded AC voltage. A battery with ample watt-hours may still be unsuitable if its allowable discharge current is too small, while a supercapacitor may supply the pulse power but need a wide DC/DC conversion range as its voltage falls. For repeated sags, recharge time and thermal accumulation must also be checked.

9.4.6 Design of a DVR

The design covers the voltage, current and VA ratings of the VSC, the injection transformer, the DC-bus voltage and capacitance, the interfacing inductance and the ripple filter [2].

Voltage rating. The injection voltage is set by the worst disturbance to be corrected and by the strategy. For in-phase injection on a balanced sag, $V_{inj} = X V_{Lc}$, plus an allowance for unbalance (for example 6 % negative sequence) and for phase jumps. Quadrature injection needs $\sqrt{V_{Lc}^2 - |V_s|^2}$ in a UPQC and the larger standalone root in Equation (9.34), so it raises the voltage rating sharply.

Current rating. A standalone DVR carries the full load current, $I_{DVR} = I_L$, and its kVA rating is $S_{DVR} = 3 V_{inj} I_L$ for a three-phase unit.

Injection transformer. Its line winding carries I_L continuously and must withstand downstream fault current until the bypass closes, so it has three distinct duties: continuous current, compensation voltage and short-duration fault withstand. Its rating equals the converter rating. The turns ratio puts the converter at a convenient voltage and current: $n = V_{VSC} / V_{inj}$, with converter-side current I_L / n . The core must not saturate when the injection starts at a voltage zero with full magnitude, so it is commonly designed for about twice the steady flux, or the controller limits the initial flux.

DC-bus voltage. For a three-leg VSC with sinusoidal PWM, $V_{dc} \geq 2\sqrt{2} V_{VSC} / m$, where V_{VSC} is the converter-side phase RMS voltage. For a full bridge, $V_{dc} \geq \sqrt{2} V_{VSC} / m$.

DC-bus capacitance. In a BESS- or rectifier-supported DVR the capacitor only filters ripple and covers the control delay, sized from $\frac{1}{2} C (V_{dc}^2 - V_{dc1}^2) = 3 V_{inj} I_L \Delta t$ for a short support time Δt of a few hundred microseconds. In a capacitor-supported DVR that injects real power, Equation (9.42) with the full event duration applies.

Interfacing inductance. Referred through the transformer ratio n ,

$$L_r = \frac{n(\sqrt{3}/2)mV_{dc}}{6af_s\Delta i_s} \quad (9.43)$$

where Δi_s is the permitted line-side ripple. Without the factor n this is Equation (9.24).

Ripple filter. As for the DSTATCOM, R_r and C_r are chosen with a corner frequency of about $f_s/2$.

9.4.7 Protection

The DVR sits in series with the load, so every downstream fault current passes through its transformer. Protection must close the bypass switch within a fraction of a cycle and block the converter before its switches exceed their current rating; a thyristor crowbar across the converter winding is a common backup. Upstream faults are, by contrast, the reason the DVR exists, and it must not bypass during them. The detection logic must therefore distinguish a supply-side sag (low voltage, normal or low current) from a load-side fault (high current). Coordination with upstream overcurrent relays matters because the DVR's injection and current limit change the fault current those relays see. Transformer inrush at energization, standby losses and the thermal capacity of the bypass for long outages complete the protection study.

9.4.8 Simulated performance

The following waveforms come from an average-value simulation written for this chapter: a 415 V, 20 kVA, 0.8 pf linear load protected by a storage-supported DVR with feedforward SRF control and a second-order filter response (800 Hz natural frequency, damping 0.7).

Figure 9.30 shows a balanced 30 % sag from 0.10 to 0.20 s and a 20 % swell from 0.30 to 0.40 s. The load voltage stays at rated value throughout. During the sag the DVR exports 4.8 kW, which is 30 % of the 16 kW load, exactly the $X P_L$ of Equation (9.30). During the swell it absorbs about 3.2 kW, which must go back into the store. Figure 9.31 expands the sag onset: the load voltage dips for about one millisecond while the injection builds up through the output filter.

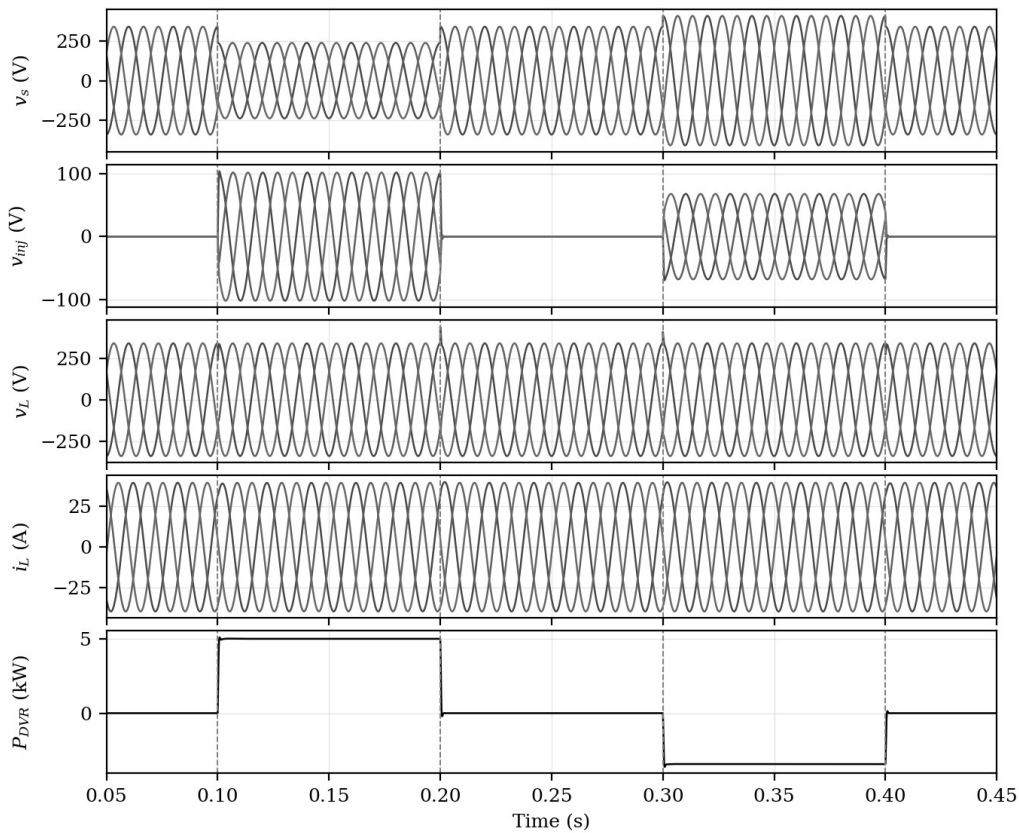


Figure 9.30. Simulated DVR during a 30 % sag and a 20 % swell: supply voltage, injected voltage, load voltage, load current and DVR real power.

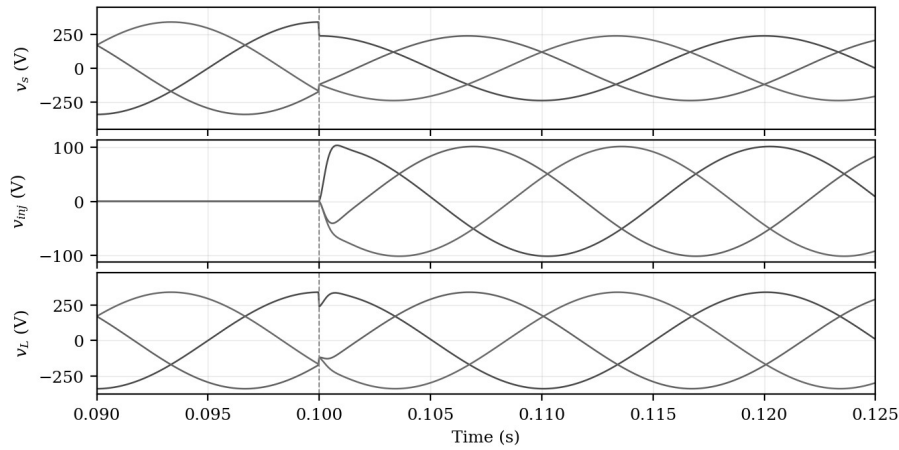


Figure 9.31. Sag onset detail: about 1 ms transient as the injection builds through the output filter.

Figure 9.32 shows an unbalanced sag ($V_p=0.8$ pu, $V_n=0.15 \angle 30^\circ$ pu) from 0.10 to 0.20 s and supply harmonics (10 % fifth, 7 % seventh) from 0.30 to 0.40 s. The phase injections differ in magnitude and angle, and the DVR's real power oscillates at 100 Hz around its mean, which the DC link must buffer. During the harmonic interval the DVR injects only harmonic voltage, the load voltage stays sinusoidal and the real power oscillates with zero mean.

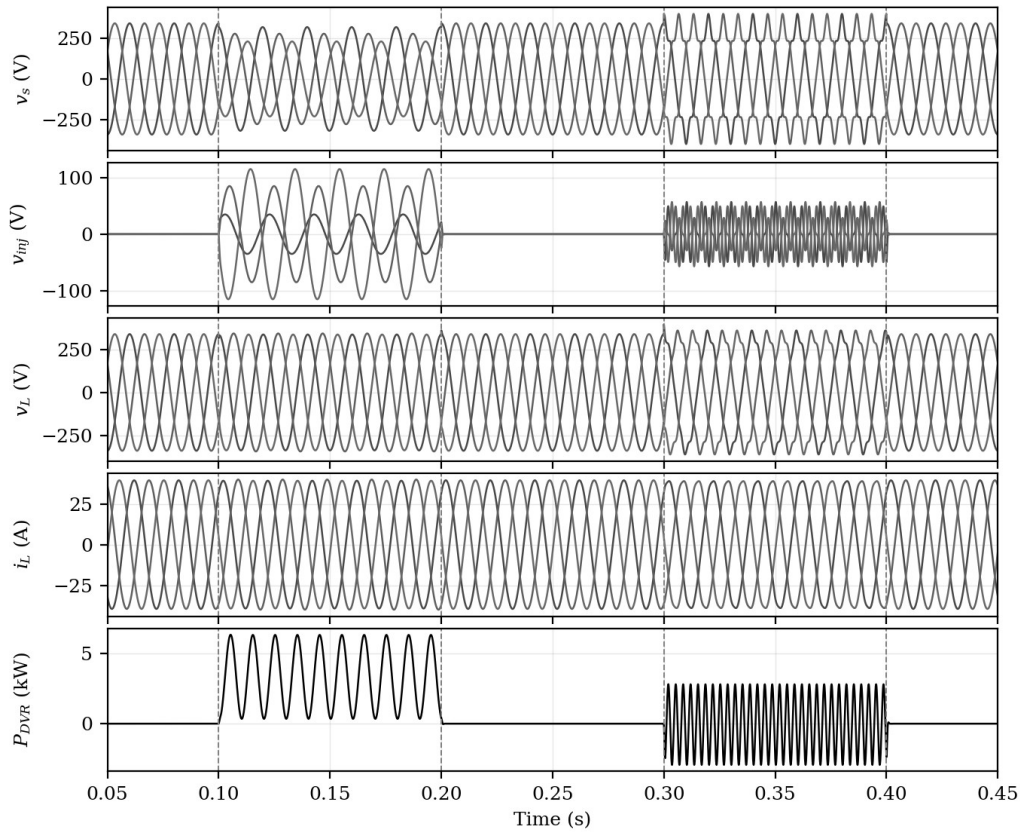


Figure 9.32. Simulated DVR during an unbalanced sag and during supply-voltage harmonics.

9.4.9 The interline DVR (IDVR)

An IDVR shares a DC link between series compensators on different feeders (Figure 9.33), following the interline power-balance principle of Chapter 8. When one feeder sags, the DVR on a healthy feeder absorbs real power from its own line (by injecting a small voltage component in phase opposition to that feeder's current), and the sagged feeder's DVR exports it. The healthy feeder's load voltage must remain acceptable

while this happens, so the healthy feeder is not an unlimited source. The common link still needs capacitance for switching ripple, unbalanced power pulsation and transient mismatch.

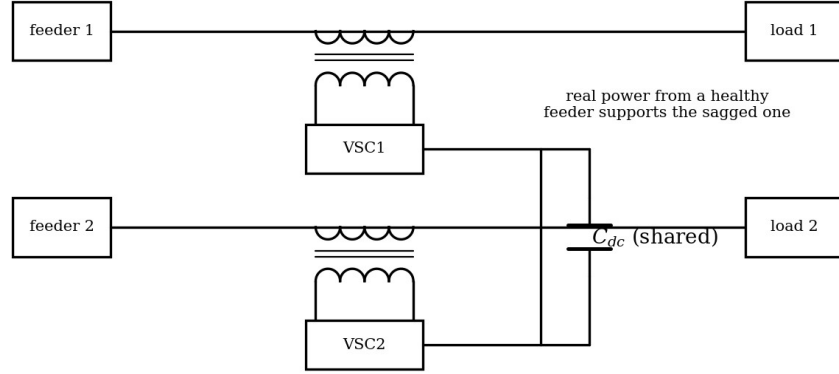


Figure 9.33. Interline DVR: series converters on separate feeders sharing a DC link.

9.4.10 Worked examples: DVR

Example 9.11. Single-phase BESS-supported DVR with in-phase injection

A single-phase 5 kVA, 230 V, 50 Hz, 0.85 lagging pf critical load is protected by a BESS-supported DVR using in-phase injection. The supply sags by 25 %. Find the injected voltage, the DVR current, its kVA rating and the real power drawn from the battery.

Solution. $V_s = 230 \times 0.75 = 172.5$ V and $V_{inj} = 57.5$ V. $I_{DVR} = I_L = 5000/230 = 21.74$ A, so $S_{DVR} = 57.5 \times 21.74 = 1.25$ kVA. From Equation (9.30), $P_{DVR} = 57.5 \times 21.74 \times 0.85 = 1063$ W and $Q_{DVR} = 658$ var.

Example 9.12. Capacitor-supported DVR with a unity-power-factor load

A single-phase capacitor-supported DVR is proposed for a 10 kVA, 230 V, unity-pf load facing a 30 % sag, using an injection of $\sqrt{230^2 - 161^2} = 164.25$ V at right angles to the supply voltage. Check whether this injection really needs no real power.

Solution. With $V_s = 161$ V real and $V_{inj} = j 164.25$ V, the load voltage is $230 \angle 45.57^\circ$ V. A unity-pf load draws its current in phase with its own voltage, $I_L = 43.48 \angle 45.57^\circ$ A. Then $P_{DVR} = \text{Re}\{V_{inj} I_L^*\} = 164.25 \times 43.48 \times \cos(90^\circ - 45.57^\circ) = 5.10$ kW, about half the load power. By Equation (9.33), $X_{max} = 1 - \cos \phi = 0$ for a unity-pf load, so no sag can be corrected without real power, whatever the injection angle. The injection was perpendicular to V_s , but a standalone DVR's zero-power condition is perpendicularity to I_L . A capacitor-supported DVR would discharge within a fraction of a cycle.

Example 9.13. Maximum sag for a capacitor-supported DVR

A capacitor-supported DVR protects a 415 V (239.6 V phase) load at 0.85 lagging pf. Find the maximum sag it can correct without real power, the injection at that limit and the phase shift of the load voltage.

Solution. From Equation (9.33), $X_{max} = 1 - 0.85 = 15\%$, so $|V_{s|min}| = 203.7$ V. At the limit $\theta_L = \phi = 31.79^\circ$, the injection is perpendicular to V_s and $|V_{inj}| = 239.6 \sin 31.79^\circ = 126.2$ V. The load voltage leads the supply by 31.79° . The value 0.5268 pu is $\sin \phi$, the injection magnitude in per unit; mistaking it for the minimum supply voltage gives an incorrect “47 % maximum sag”.

Example 9.14. DVR rating for different injection schemes

A three-phase 415 V, 30 kVA, 0.8 pf critical load experiences a balanced 35 % sag. The phase current is 41.74 A. Compare in-phase injection with zero-real-power injection.

Solution. $|V_s| = 0.65 \times 239.6 = 155.74$ V. In-phase: $V_{inj} = 83.86$ V and $S_{DVR} = 3 \times 83.86 \times 41.74 = 10.50$ kVA (35 % of the load rating), with $P_{DVR} = 0.35 \times 24 = 8.4$ kW. Zero real power: by Equation (9.33) a 0.8 pf load allows at most a 20 % sag, so a 35 % sag is infeasible for a standalone capacitor-supported DVR. The figure of 182.1 V (22.8 kVA) obtained from $\sqrt{V_{Lc}^2 - |V_s|^2}$ is the injection perpendicular to the supply voltage. It is the correct UPQC-Q series voltage (Section 9.5), where the series current is in phase with V_s , but it is not a zero-power solution for a standalone DVR carrying a lagging load current.

Example 9.15. Swell compensation and power direction

A single-phase DVR protects a 2.5 kVA, 230 V, unity-pf load against a 20 % swell. Find the injection, the kVA rating and the power direction.

Solution. $V_s = 276$ V, so $V_{inj} = 230 - 276 = -46$ V (in phase opposition). $I_L = 10.87$ A and $S_{DVR} = 500$ VA. $P_{DVR} = -46 \times 10.87 = -500$ W: the DVR absorbs 500 W, which must charge the store or be dissipated in a chopper.

Example 9.16. BESS energy for an extended sag

A three-phase 415 V, 15 kVA, 0.8 pf critical load is protected by a BESS-supported in-phase DVR against a 30 % sag lasting 500 ms. Find the load power, the battery power and the energy per event.

Solution. $P_L = 12$ kW. $V_{inj} = 71.88$ V and $I = 20.86$ A, so $P_{BESS} = 3 \times 71.88 \times 20.86 \times 0.8 = 3.60$ kW, which is $0.3 P_L$. $E = 3.60 \times 0.5 = 1.80$ kJ, or 0.50 Wh. At 95 % conversion efficiency the battery delivers 1.89 kJ. The energy is tiny, but the battery must deliver 3.6 kW for half a second: DVR storage is sized by power and discharge rate at least as much as by energy, which is why supercapacitors are attractive.

Example 9.17. Standalone DVR energy storage

A balanced three-phase storage-supported DVR in in-phase mode protects a 415 V, 20 kW, 0.9 lagging pf load against a 30 % sag for 200 ms. Find the phase voltage, series current, injected and sagged phase voltages, delivered real power and ideal energy.

Solution. $V_{ph} = 239.60$ V and $I_L = 20000 / (\sqrt{3} \times 415 \times 0.9) = 30.92$ A. $U = 0.30 V_{ph} = 71.88$ V and $|V_s| = 167.72$ V. $P_{DVR} = 3 U I_L \cos \phi = 0.30 P_L = 6000$ W and the VA duty is 6667 VA. Ideal energy: $E = 6000 \times 0.2 = 1200$ J. Source real power is 14 kW and DVR export 6 kW, totalling the 20 kW load demand.

Example 9.18. Loss-inclusive capacitance

For Example 9.17, with the DC bus permitted to droop from 700 V to 650 V and $\eta = 0.96$, find the DC-link capacitance and compare with the ideal value.

Solution. $E = (6000 / 0.96) \times 0.2 = 1250$ J, 4.2 % above 1200 J. From Equation (9.42), $C_{dc} = 2 \times 1250 / (700^2 - 650^2) = 37.04$ mF (ideal 35.56 mF). These are energy-based preliminary values; ESR, ripple current, voltage derating, tolerance and lifetime must also be checked. For a 240 ms event the ideal values become 1440 J and 42.67 mF, and 44.44 mF at 96 % efficiency.

Example 9.19. Complete design of a three-phase BESS-supported DVR

Design a three-phase DVR for a 415 V, 20 kVA, 50 Hz critical load. The supply varies from -30% to $+20\%$ with up to 6% unbalance; the switching frequency is 10 kHz. Use in-phase injection, a converter-side voltage of 50 V, $m=1$, $a=1.2$ and a 2% line-side ripple.

Solution. Voltage rating: $0.30 \times 239.6 = 71.9$ V for the deepest balanced sag, plus about $0.06 \times 239.6 = 14.4$ V for unbalance, giving 86.3 V; select 90 V per phase. Current rating: $I_L = 20\,000 / (3 \times 239.6) = 27.82$ A. kVA: $S = 3 \times 90 \times 27.82 = 7.51$ kVA. Injection transformer: 7.5 kVA, ratio $n = 50/90 = 0.556$, converter-side current $27.82/0.556 = 50.1$ A. DC bus: $V_{dc} \geq 2\sqrt{2} \times 50 = 141.4$ V; select 150 V. DC capacitor for 200 μ s of support with 5% droop: $\frac{1}{2}C(150^2 - 142.5^2) = 3 \times 90 \times 27.82 \times 200 \times 10^{-6} = 1.50$ J, so $C = 1.37$ mF; select 2.2 mF, 200 V.

Inductor from Equation (9.43): $\Delta i_s = 0.556$ A and $L_r = 0.556 \times 0.866 \times 150 / (6 \times 1.2 \times 10^4 \times 0.556) = 1.80$ mH. Ripple filter: $R_r = 5\ \Omega$, $f_r = 5$ kHz, $C_r = 6.4\ \mu$ F; select 10 μ F. A capacitor-supported design using the quadrature voltage $\sqrt{239.6^2 - 167.7^2} = 171.1$ V would need 14.3 kVA, nearly twice as much, and would restore a 30% sag without real power only if the load power factor were 0.7 or lower (Equation (9.33)).

9.5 Unified Power Quality Conditioner (UPQC)

A UPQC combines a series converter (the DVR part) and a shunt converter (the DSTATCOM part) on a common DC link [4], [18]. It is best read as a coordinated two-port power-quality controller rather than a DVR and a DSTATCOM placed in one cabinet. The series converter keeps the load voltage clean and regulated whatever the supply does; the shunt converter keeps the source current sinusoidal, balanced and in phase whatever the load draws, and it also regulates the shared DC voltage. The common DC link couples their real-power duties, and that coupling determines rating, control priority and disturbance capability. The UPQC is to distribution what the UPFC of Chapter 8 is to transmission: the same back-to-back structure with different objectives.

9.5.1 Structure and power flow

The series VSC is voltage-controlled through injection transformers; the shunt VSC is current-controlled through interfacing inductors. Both need ripple filters, and both share the DC capacitor. With $P_{into,k}$ defined as the real power absorbed from the AC system by converter k (positive into the converter), the DC-link energy obeys

$$\frac{dW_{dc}}{dt} = \frac{d}{dt} \left(\frac{1}{2} C_{dc} V_{dc}^2 \right) = \sum_k P_{into,k} - P_{loss} \quad (9.44)$$

At constant average DC energy, $\sum P_{into,k} = P_{loss}$. Equivalently, with AC exports positive, $P_{se} + P_{sh} + P_{loss} = 0$. When the series converter exports real power to boost a sagged supply, the shunt converter must draw the same power from the AC system, so in a feasible grid-fed steady state the real power circulates: source to shunt converter, through the DC link, to series converter and back into the line. This is why a UPQC can correct long sags without a large energy store, provided the sagged source can still supply the load power at a higher current. The capacitor only buffers transient mismatch and ripple; it is not a source of sustained real power.

9.5.2 Classification

UPQCs are classified by converter type (CSC or VSC), by supply system (single-phase two-wire, three-phase three-wire and three-phase four-wire), by the location of the shunt converter and by how the series converter shares real and reactive power (Figure 9.34) [4]. Extensions include the interline UPQC (series and shunt

converters on different feeders), the multi-converter UPQC and UPQCs with distributed generation or storage on the DC link.

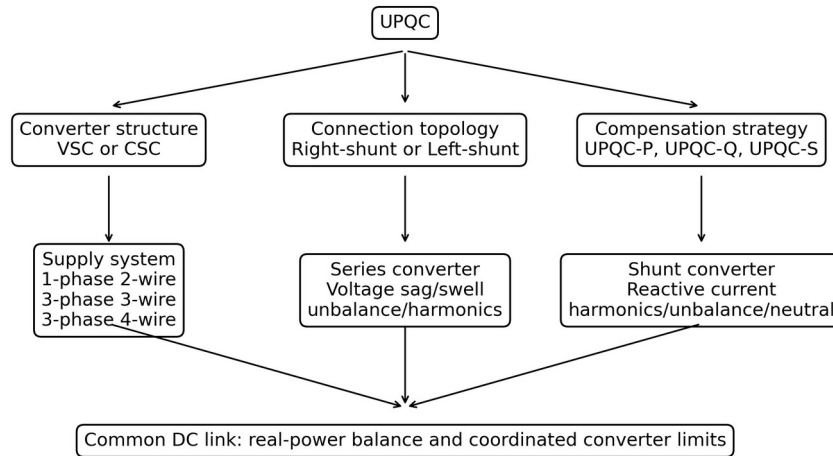


Figure 9.34. Classification of UPQC by converter structure, connection topology, supply system and compensation objective.

Right-shunt and left-shunt UPQC. In the right-shunt UPQC (Figure 9.35(a)) the shunt converter is on the load side of the series converter. The series converter then carries the source current, which the shunt converter has made sinusoidal and in phase with the supply, and the shunt converter sees the regulated load voltage; during a sag it draws the series converter's real power from the restored bus, so the source current rises as the source voltage falls. In the left-shunt UPQC (Figure 9.35(b)) the shunt converter is on the supply side: the series converter carries the load current, with its harmonics and reactive component, and the shunt converter sees the disturbed supply voltage. The right-shunt arrangement is usually preferred because the shunt converter works at a clean voltage, its current harmonics do not flow through the series transformer and the series converter carries a well-defined sinusoidal current. The left-shunt arrangement is used when the shunt converter must also compensate other loads at the PCC. The two arrangements can deliver similar external objectives but do not have identical converter ratings, so comparisons must be made at the same load, sag, source power factor and control objective.

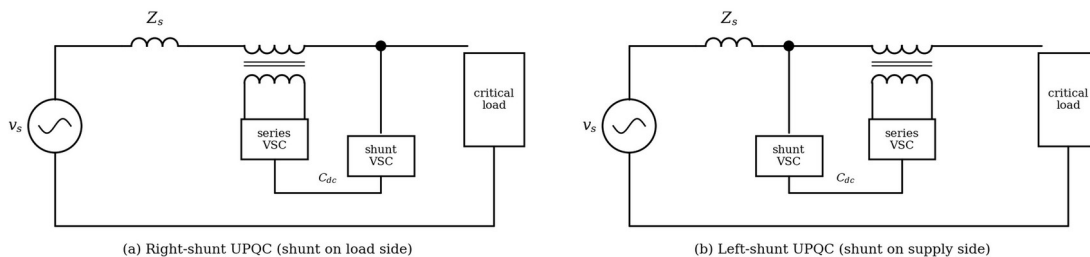


Figure 9.35. (a) Right-shunt UPQC, (b) left-shunt UPQC.

Single-phase and three-phase UPQCs. Balanced three-phase phasor sizing can use a per-phase equivalent, but switching and control implementations are not obtained simply by repeating a single leg. A single-phase full bridge needs a complete return path; a three-wire three-phase UPQC uses three-leg converters; a four-wire UPQC needs a rated neutral path for the shunt converter, such as a fourth leg or a split DC link (Figure 9.36). The neutral leg must be rated from its current, since its terminal voltage can be small while its conduction loss is large. Single-phase units have twice-line-frequency DC-energy ripple, and unbalanced three-phase operation also produces ripple; both enter the capacitor sizing.

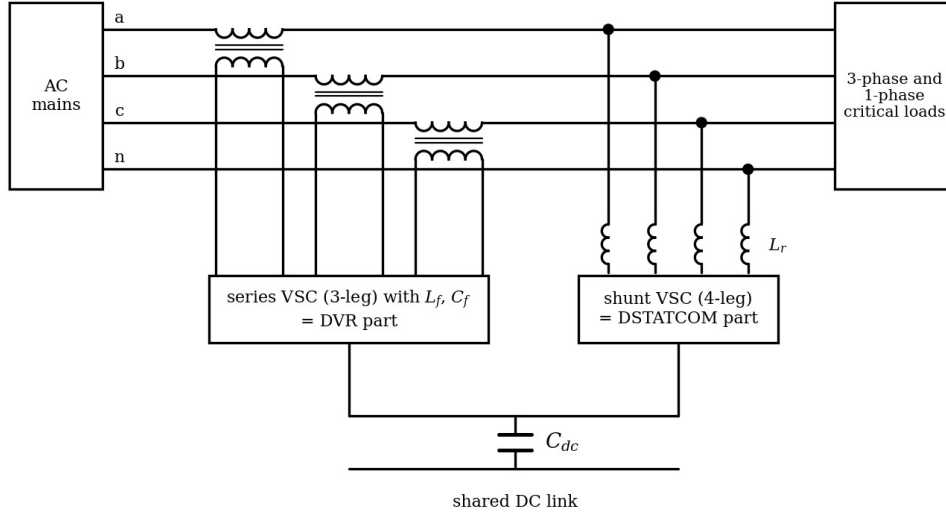


Figure 9.36. Three-phase four-wire right-shunt UPQC: three-leg series VSC (DVR part) and four-leg shunt VSC (DSTATCOM part) on a shared DC link.

UPQC-P, *UPQC-Q* and *UPQC-S*. These names describe operating strategies, not different hardware. They state how the series converter injects its voltage relative to the series current:

- *UPQC-P*: injection in phase with the supply voltage and the (unity-pf) source current, so the series converter handles only real power. Its injection voltage is the smallest possible, but the shunt converter must circulate that real power.
- *UPQC-Q*: injection in quadrature with the source current, so the series converter handles only reactive power. It needs a larger injection voltage, and it cannot correct swells, because adding a perpendicular voltage always increases the resultant magnitude.
- *UPQC-S*: injection at a chosen angle, so the series converter handles both real and reactive power. Khadkikar and Chandra [19] used this to let the series converter share the load reactive power during normal operation as well as during sags and swells; a minimum-VA angle can also be chosen.

The derivations below use the right-shunt topology, so the series current equals the source current, unless stated otherwise.

9.5.3 Phasor analysis and converter ratings

Take the sagged supply voltage $V_s' = V_{Lc}(1 - X)$ as real reference. With unity source power factor and a lossless UPQC, the source supplies exactly the load real power $P_L = V_{Lc} I_L \cos \phi$:

$$I_s = \frac{P_L}{V_s'} = \frac{I_L \cos \phi}{1 - X} \quad (9.45)$$

The source current rises as the supply sags. For a balanced three-phase system use per-phase quantities, or $I_s = P_{L,3\phi} / (\sqrt{3} V_{s,LL}')$. Figure 9.37 shows the three operating modes for a 30 % sag.

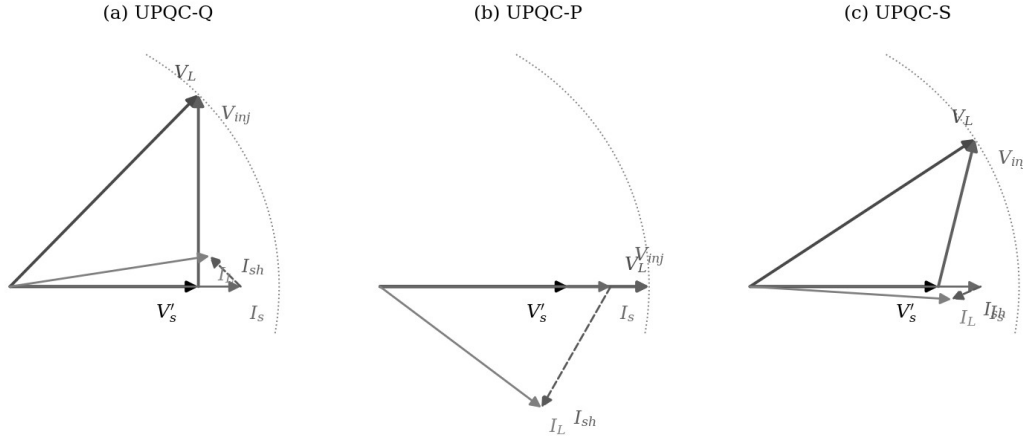


Figure 9.37. Phasor diagrams of a right-shunt UPQC during a 30 % sag: (a) UPQC-Q, (b) UPQC-P, (c) UPQC-S.

UPQC-Q. The injection is perpendicular to I_s , hence to V'_s , so $V_{Lc}^2 = |V'_s|^2 + U^2$:

$$U = |V_{inj}| = \sqrt{V_{Lc}^2 - |V'_s|^2}, \delta = \cos^{-1}(1 - X) \quad (9.46)$$

where δ is the angle by which the load voltage leads the supply. The series converter's duty is entirely reactive:

$$S_{se} = U I_s = V_{Lc} I_L \cos \phi \tan \delta, P_{se} = 0 \quad (9.47)$$

Writing V'_s in place of V_{Lc} in the last form is a common slip that underestimates the rating by the factor $1 - X$. The load current lags the rotated load voltage by ϕ , so it lies at $\delta - \phi$ from V'_s , and the shunt converter supplies the difference $I_{sh} = I_L - I_s$:

$$I_{sh} = I_L \frac{\sqrt{(1 - X)^2 + \cos^2 \phi - 2 \cos \phi \cos(\phi - \delta)(1 - X)}}{1 - X} \quad (9.48)$$

with $S_{sh} = V_{Lc} I_{sh}$. When $X = 1 - \cos \phi$, $\delta = \phi$ and $I_{sh} = 0$: the rotation of the load voltage has brought the load current into phase with the supply, so the series converter alone supplies all the load's reactive power. At other sag depths the shunt current is nonzero, and its reactive direction reverses across that point.

UPQC-P. The injection is in phase with V'_s , $V_{inj} = V_{Lc} X$ (signed, negative for a swell). The series duty is entirely real:

$$S_{se} = |P_{se}| = V_{Lc} |X| I_s = V_{Lc} |X| \frac{I_L \cos \phi}{1 - X} \quad (9.49)$$

The factor $1/(1 - X)$ is easily forgotten but essential: the series converter carries the increased source current, not the load's active current. The load voltage stays in phase with the supply, so the shunt current has a reactive part (the load's own) and an active part that returns the series converter's real power:

$$|I_{sh}| = I_L \sqrt{\sin^2 \phi + \cos^2 \phi \left(\frac{X}{1 - X} \right)^2} \quad (9.50)$$

UPQC-S. For a specified injection magnitude U , the angle measured from V'_s follows from Equation (9.35), and

$$P_{se} = U I_s \cos \psi, Q_{se} = U I_s \sin \psi, \delta = \sin^{-1} \left(\frac{U \sin \psi}{V_{Lc}} \right) \quad (9.51)$$

The shunt current is again $I_{sh} = I_L \angle (\delta - \phi) - I_s$. A minimum-combined-VA design varies the restored load angle δ and minimizes

$$J(\delta) = I_s \left| V_{Lc} e^{j\delta} - |V_s'| \right| + V_{Lc} \left| I_L e^{j(\delta - \phi)} - I_s \right| \quad (9.52)$$

subject to each converter's voltage, current and VA limits, DC-energy balance and the permissible phase shift. Figure 9.38 plots J for a 20 % sag on a 230 V, 25 A load: at 0.8 pf the minimum is the quadrature point ($\delta = 36.87^\circ$, 3450 VA), while at 0.95 pf it lies at $\delta = 10.5^\circ$ with 3132 VA.

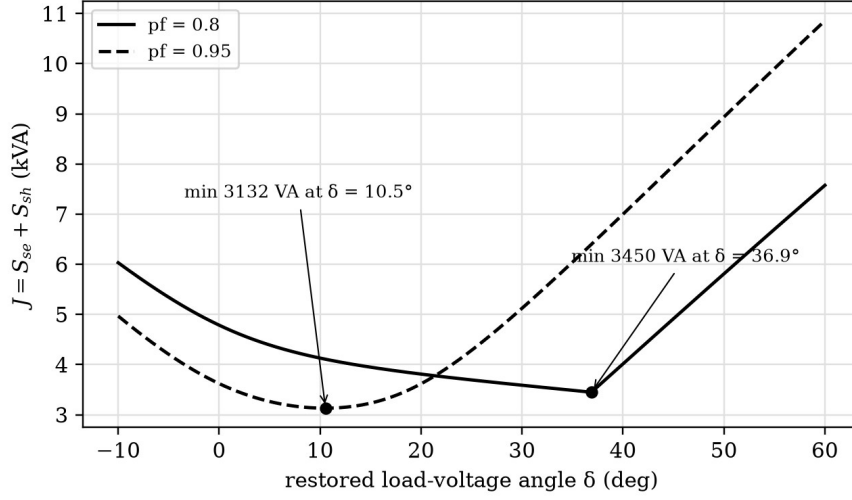


Figure 9.38. Combined operating VA of a UPQC versus restored load-voltage angle for a 20 % sag (230 V, 25 A).

Operating duty and installed rating. The numbers above are operating duties at one sag depth. The installed rating of each converter must cover its own maximum voltage and current over every specified condition, and those maxima do not occur at the same point. In particular, at nominal supply voltage the series converter is idle and the shunt converter supplies the whole load reactive current $I_L \sin \phi$. Figure 9.39 shows the series, shunt and total duties versus sag depth for a 230 V, 25 A, 0.8 pf load: in UPQC-Q the shunt duty falls from 3450 VA at nominal voltage to zero at a 20 % sag and rises again, while the series duty grows steadily; in UPQC-P the series duty is small but the shunt duty never falls below the nominal 3450 VA. A UPQC-Q designed for a 20 % sag therefore needs about 3450 VA in each converter, 6900 VA in total, although the operating total during the sag is only 3450 VA (Example 9.20). Figure 9.40 gives the series-converter VA of both strategies over a wider range of sags. Harmonic and unbalance compensation add RMS current and voltage that must also enter the thermal ratings, so fundamental kVA alone understates the installed requirement.

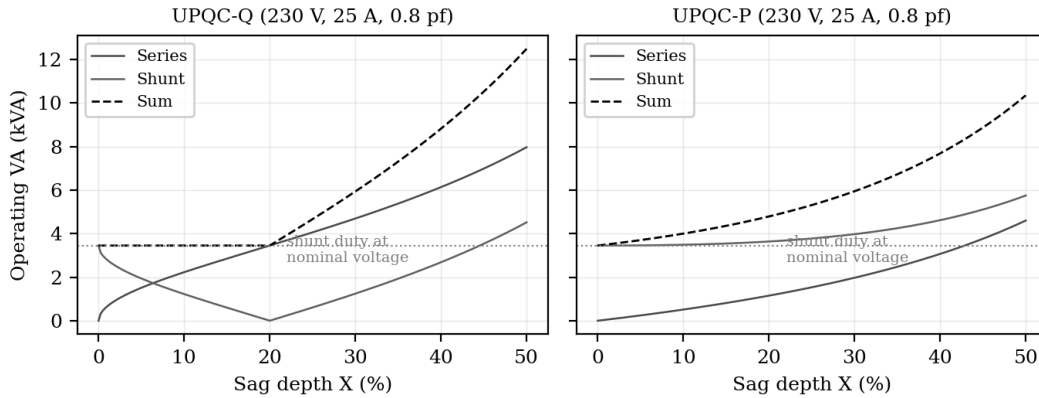


Figure 9.39. Series, shunt and total operating VA of a right-shunt UPQC versus sag depth (230 V, 25 A, 0.8 pf).

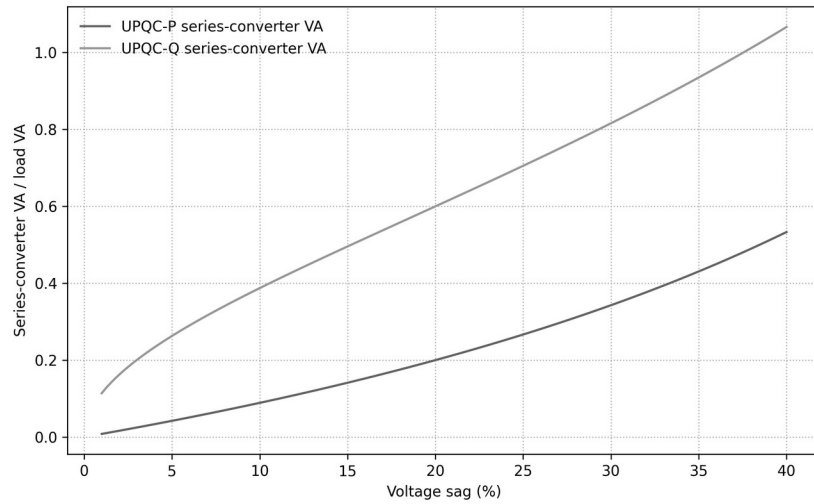


Figure 9.40. Normalized series-converter VA versus balanced sag depth for UPQC-P and UPQC-Q at 0.8 load power factor. These are screening results, not complete converter ratings.

Table 9.9. Injection strategies: current reference and design limits.

Strategy	Right-shunt UPQC ($I_{se} = I_s$)	Standalone DVR ($I_{se} = I_L$)
Quadrature	Sag only; zero series real power; $U = \sqrt{V_{Lc}^2 - V_s'^2}$	Must be perpendicular to I_L ; feasible only if $X \leq 1 - \cos \phi$; check both roots
In-phase	Minimum injection; shunt circulates the series real power	Real power $X P_L$ from storage; swell needs an absorption path
General angle (UPQC-S)	Shares P and Q; optimize a stated objective	Choose phase shift, voltage and energy together
Pre-sag	Restore pre-sag phasor; check both converters	Larger voltage and energy than in-phase

9.5.4 Left-shunt UPQC analysis

In a left-shunt UPQC the series converter carries the load current, so zero-real-power series operation needs injection perpendicular to I_L , exactly as for the standalone DVR, and is governed by Equation (9.32) and Equation (9.34). The shunt converter works at the disturbed PCC voltage V_s' and supplies the load's reactive current referred to the supply: if $\beta = \phi - \theta_L$ is the angle between V_s' and I_L , then $I_{sh} = I_L \sin \beta$ and $S_{sh} = V_s' I_L \sin \beta$. Unlike the right-shunt UPQC-Q, this arrangement can correct a swell with zero series real power, because the root of Equation (9.32) with $\theta_L < 0$ rotates the load voltage behind the supply and the injection then has a component opposing V_s' (Example 9.26).

9.5.5 Control of the UPQC

Figure 9.41 shows the usual coordinated structure. The series converter uses the DVR controller of Section 9.4.4: an SRF voltage reference built from the PLL angle, $v_{inj}^i = v_L^i - v_s$, with load-voltage feedback and PWM voltage control. The target may preserve the pre-sag phase or deliberately rotate the load voltage (UPQC-Q, UPQC-S), and the synchronization and reference extraction must reflect that choice. Unbalanced regulation needs phase or sequence controllers rather than one positive-sequence dq magnitude.

The shunt converter uses the DSTATCOM controller of Section 9.3.3: $\dot{i}_d^* = \dot{i}_{Ld} + \dot{i}_{loss}$, $\dot{i}_q^* = 0$ and $\dot{i}_{sh}^* = \dot{i}_L - \dot{i}_s$, with hysteresis or PWM current control. The DC-voltage PI output $\dot{i}_{loss}$ now absorbs not only losses but also the series converter's real-power exchange. A feedforward term, scaling $\dot{i}_{Ld}$ by the ratio of rated to measured supply voltage, makes the source current rise immediately when a sag starts instead of waiting for the DC voltage to fall, and greatly reduces the DC-link dip. Both converters share the PLL angle.

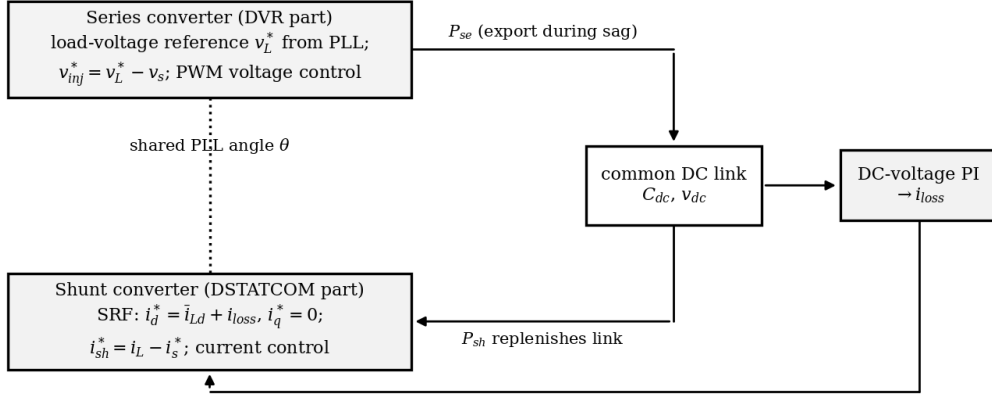


Figure 9.41. Coordinated UPQC control: series voltage loop, shunt current loop and common DC-link regulation.

The two controllers act on different variables but share the DC link, and poorly separated bandwidths make them fight: a fast shunt DC-voltage loop may react to the very transient power that the series loop is deliberately producing. A robust hierarchy (Figure 9.42) uses fast series-voltage and shunt-current loops, slower DC-link energy control, and a supervisor that selects UPQC-P, UPQC-Q or an optimized operating point. Current and voltage saturation must be coordinated as well. When the source voltage becomes very low, holding rated load voltage can demand both excessive series voltage and excessive source current, so the supervisor should reduce the restoration target or shed noncritical load before semiconductor limits are reached. This matters most in microgrids, where source strength can be low and PLL dynamics significant.

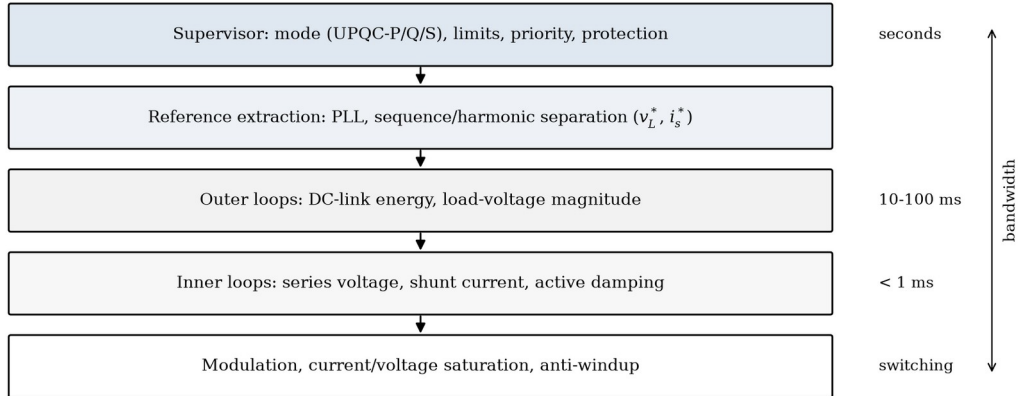


Figure 9.42. Layered control structure of a practical UPQC, from switching-level limits to the supervisory mode selection.

9.5.6 Design of a three-phase four-wire UPQC

The design combines the DSTATCOM design of Section 9.3.4 with the DVR design of Section 9.4.6, using one DC bus for both [2].

Shunt converter: V_{dc} from Equation (9.21); C_{dc} from Equation (9.22) (and Equation (9.23) if a split DC link carries the neutral current); L_r from Equation (9.24); switch ratings from Equation (9.26). *Series converter:* the injection voltage covers the specified range, for example $\pm 30\%$ in-phase, $V_{DVR} = 0.3 \times 239.6 = 71.9$ V per phase. Because both converters share the DC bus, the series converter's switches have the same voltage rating

as the shunt converter's, and a step-up injection transformer is used, $K_{DVR} = V_{VSC}/V_{DVR}$. The series current is largest during the deepest sag, $I_s = P_L/[3(1-X)V_{ph}]$, and smallest during the largest swell; the inductor is sized at the smallest current because ripple is specified as a percentage of it:

$$L_{DVR} = \frac{(\sqrt{3}/2)mV_{dc}K_{DVR}}{6af_s\Delta I_{DVR}} \quad (9.53)$$

9.5.7 Simulated performance

Figure 9.43 shows an average-value simulation of a 415 V right-shunt UPQC-P supplying a six-pulse rectifier (30 A DC) in parallel with a 0.8 pf linear load, about 27.5 kW in total. The supply sags to 0.75 pu from 0.10 to 0.20 s and swells to 1.2 pu from 0.28 to 0.36 s. The load voltage stays at rated value. The source current is sinusoidal throughout and grows by a factor $1/(1-X)=1.33$ during the sag, as Equation (9.45) predicts, because the supply must deliver the same power at lower voltage. The shunt current carries the load harmonics and reactive current plus the real power that feeds the series converter. The DC-link voltage dips by about 15 V at the start of the sag and swells similarly at its end before the PI loop restores it.

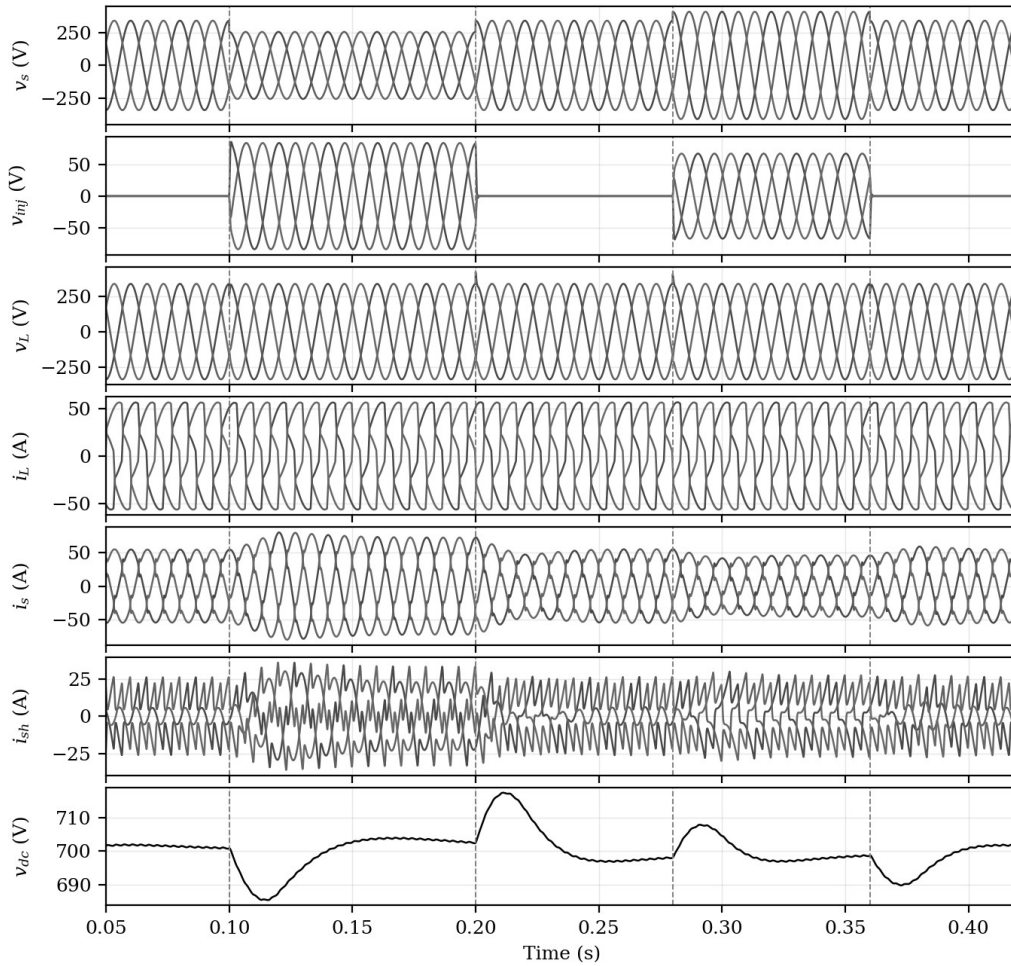


Figure 9.43. Simulated right-shunt UPQC-P during a 25 % sag and a 20 % swell with a nonlinear load: supply voltage, series injection, load voltage, load current, source current, shunt current and DC-link voltage.

Figure 9.44 confirms the power circulation of Equation (9.44). During the sag the series converter exports about 9.2 kW, which is $P_L X/(1-X) = 27.5 \times 0.25/0.75$ kW, and the shunt converter draws nearly the same from the load bus; their sum, the power drawn from the DC link, stays within about ± 2 kW and settles toward zero after each transition. During the swell the directions reverse.

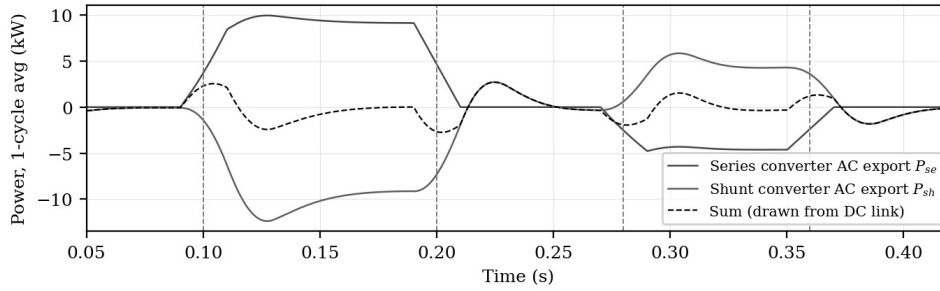


Figure 9.44. One-cycle-average real power of the series and shunt converters (AC export positive).

9.5.8 Worked examples: UPQC

Example 9.20. Right-shunt UPQC-Q: ratings during sag and at nominal voltage

A single-phase right-shunt UPQC-Q supplies a 230 V, 50 Hz, 25 A, 0.8 lagging pf load and keeps the source at unity power factor. The supply sags by 20 %. Find the voltage, current and VA of each converter during the sag, and the installed ratings when nominal operation is included.

Solution. $P_L = 230 \times 25 \times 0.8 = 4600$ W. Nominal: $V_{inj} = 0$; the shunt converter supplies $I_L \sin \phi = 15$ A at 230 V, 3450 VA. Sag: $V'_s = 184$ V and $I_s = 4600/184 = 25$ A. From Equation (9.46), $U = \sqrt{230^2 - 184^2} = 138$ V and $\delta = 36.87^\circ$; $S_{se} = 138 \times 25 = 3450$ VA, all reactive. Since $\delta = \phi$, Equation (9.48) gives $I_{sh} = 0$. Installed: series 138 V, 25 A, 3450 VA; shunt 230 V, 15 A, 3450 VA; total 6900 VA. The operating total during the sag is only 3450 VA, but each converter's rating is set by a different condition.

Example 9.21. UPQC-Q at a shallower sag

A lossless right-shunt UPQC-Q protects a 230 V, 20 A, 0.8 pf load during a 10 % sag. Find the series and shunt duties.

Solution. $V'_s = 207$ V, $P_L = 3680$ W and $I_s = 17.78$ A. $U = \sqrt{230^2 - 207^2} = 100.25$ V, so $S_{se} = 1782.3$ VA. $\delta = 25.84^\circ$, $I_L = 20 \angle -11.03^\circ$ A relative to V'_s and $|I_{sh}| = |I_L - I_s| = 4.251$ A, so $S_{sh} = 977.7$ VA. Total 2760.0 VA. For comparison, a standalone DVR protecting the same load at the same sag (Figure 9.26) needs only 863 VA at zero real power, because it carries a different current with a different geometry. Neither result supports a general claim that one arrangement always needs the larger rating.

Example 9.22. Right-shunt UPQC-P

For the load and sag of Example 9.20, find the converter duties in UPQC-P mode.

Solution. $V_{inj} = 230 \times 0.2 = 46$ V and $I_s = 25$ A, so $S_{se} = P_{se} = 1150$ VA. With $I_L = 20 - j15$ A, $I_{sh} = I_L - I_s = -5 - j15$ A and $|I_{sh}| = 15.81$ A, matching Equation (9.50): $25 \sqrt{0.36 + 0.64 \times 0.0625}$. $S_{sh} = 230 \times 15.81 = 3636.6$ VA and $P_{sh} = -1150$ W, returning the series converter's export. Total 4786.6 VA. The series duty is a third of the UPQC-Q value, but the total is larger, because the shunt converter carries both the load's reactive current and the circulated real power.

Example 9.23. UPQC-Q at a 25 % sag

A right-shunt UPQC-Q supplies 230 V, 20 A, 0.8 pf. The supply sags by 25 %. Find all ratings.

Solution. $P_L = 3680$ W, $V'_s = 172.5$ V and $I_s = 21.33$ A. $U = 152.13$ V and $S_{se} = 3245.5$ VA. $\delta = 41.41^\circ$, $\cos(\phi - \delta) = 0.99686$, and Equation (9.48) gives $I_{sh} = 2.111$ A and $S_{sh} = 485.5$ VA during the sag. At nominal voltage the shunt converter needs $20 \times 0.6 = 12$ A, 2760 VA, which governs. Installed total: $3245.5 + 2760 = 6005.5$ VA.

Example 9.24. UPQC-S with a limited series voltage

A single-phase right-shunt UPQC protects a 230 V, 25 A, 0.8 pf load against a 30 % sag. The series converter is rated for 130 V. Find the injection angle, its real and reactive powers, the load-voltage rotation and the shunt duty, and compare with UPQC-Q and UPQC-P.

Solution. $V_s' = 161$ V and $I_s = 4600/161 = 28.57$ A. From Equation (9.35), $\cos \psi = (230^2 - 161^2 - 130^2)/(2 \times 161 \times 130) = 0.2408$ and $\psi = 76.07^\circ$. $P_{se} = 130 \times 28.57 \times 0.2408 = 894.3$ W, $Q_{se} = 3605.0$ var and $S_{se} = 3714.3$ VA. $\delta = \sin^{-1}(130 \sin 76.07^\circ / 230) = 33.27^\circ$. $I_{sh} = |25 \angle -3.60^\circ - 28.57| = 3.95$ A, so $S_{sh} = 907.7$ VA and the total is 4621.9 VA. UPQC-Q at the same sag: $U = 164.25$ V, $S_{se} = 4692.9$ VA, $S_{sh} = 1242.9$ VA, total 5935.9 VA. UPQC-P: $U = 69$ V, $S_{se} = 1971.4$ VA, $S_{sh} = 3973.5$ VA, total 5945.0 VA. The UPQC-S angle reduces both the series voltage (130 V instead of 164 V) and the total operating VA, by about 22 %.

Example 9.25. Left-shunt UPQC-Q during a sag

A left-shunt UPQC-Q supplies 230 V, 25 A, 0.8 pf with a 10 % sag. The series converter injects perpendicular to the load current, and the shunt converter keeps unity pf at the source. Find both duties during the sag and the installed ratings.

Solution. $V_s' = 207$ V. From Equation (9.34), $\cos(\theta_L - \phi) = 184/207 = 0.8889$, so $\theta_L = 9.60^\circ$ (smaller root) and $V_{inj} = 43.17 \angle 62.73^\circ$ V. The series converter carries $I_L = 25$ A: $S_{se} = 1079.2$ VA, purely reactive. $\beta = \phi - \theta_L = 27.27^\circ$, $I_s = 4600/207 = 22.22$ A and $I_{sh} = 25 \sin 27.27^\circ = 11.45$ A at 207 V, $S_{sh} = 2370.8$ VA. At nominal voltage the shunt converter needs 15 A at 230 V, 3450 VA. Installed: $1079.2 + 3450 = 4529.2$ VA.

Example 9.26. Left-shunt UPQC-Q during a swell

For the left-shunt UPQC of Example 9.25, the supply swells by 10 %. Find the duties.

Solution. $V_s' = 253$ V and $\cos(\theta_L - \phi) = 184/253 = 0.7273$, so $\theta_L - \phi = \pm 43.34^\circ$. The root $\theta_L = -6.47^\circ$ (load voltage lagging the supply) gives $V_{inj} = 35.65 \angle -133.34^\circ$ V; the other root needs 311.6 V. $S_{se} = 35.65 \times 25 = 891.2$ VA. $\beta = 43.34^\circ$, so $I_{sh} = 25 \sin 43.34^\circ = 17.16$ A at 253 V and $S_{sh} = 4341.2$ VA. Installed: shunt 253 V and 17.16 A (4341 VA), series 891 VA, total 5232 VA.

9.5.9 Advanced and multi-converter UPQC structures

The two-converter UPQC is the reference structure, but modern distribution systems motivate extensions (Table 9.10). UPQC-DG connects distributed generation or storage to the common DC link so that the converters perform power-quality compensation and controlled active-power exchange together. An interline or multi-converter UPQC coordinates more than one feeder through a common DC link. Open and distributed UPQCs separate the converter locations while keeping coordinated objectives. Modular multilevel and cascaded converters extend the voltage rating and improve waveform quality at medium voltage. None of these is automatically superior: adding a PV array, battery or extra feeder changes the DC-link power balance, protection, islanding behaviour and converter rating, so a design must state which services are simultaneous, which are mutually exclusive under current limit, and how energy management interacts with fast power-quality control [20].

Table 9.10. Extensions of the UPQC.

Extension	Added capability	New design issue
UPQC-DG / PV-UPQC	PQ compensation plus generation interface	DC-link energy management and curtailment

Extension	Added capability	New design issue
UPQC-BESS	Sag support, peak shaving, ride-through	State of charge, battery power limit and protection
Interline / MC-UPQC	Coordination of several feeders	Cross-feeder real-power balance and fault isolation
Open / distributed UPQC	Flexible placement, less common hardware	Communication, synchronization and protection
MMC / cascaded UPQC	Medium-voltage scalability and lower dv/dt	Cell balancing, stored energy and fault handling

9.6 Series and Hybrid Active Power Filters

Three ideas are often mixed together and should be kept apart. A shunt active filter creates a compensating current; a series active filter creates a compensating voltage; a hybrid filter deliberately assigns part of the harmonic spectrum to passive components so that the converter handles only a residual or only a small voltage. The engineering question is therefore not merely whether harmonics exist, but what kind of source produces them, where they originate and which impedance path the compensator must create. A DSTATCOM operated purely for harmonic cancellation is a shunt active power filter, and its pq and SRF control were developed in Section 9.3.3. This section builds the network-level theory that decides which filter arrangement suits a given load and supply [21], [22], [23], [24], [25].

9.6.1 Harmonic-source models

Peng's classification of harmonic sources is the starting point [23] (Figure 9.45). A thyristor rectifier with a large DC inductor draws a nearly rectangular current set by the DC-side inductance; seen from the AC side it is a harmonic *current source* I_{Lh} with a large parallel impedance Z_L (Norton form). A diode rectifier with a large DC capacitor, typical of drives, switch-mode supplies and EV chargers, draws a pulsed current whose shape is set by the AC impedance in front of it; seen from the AC side it is a harmonic *voltage source* V_{Lh} behind a small series impedance Z_L (Thevenin form). The supply is represented at each harmonic by a background voltage V_{sh} behind the source impedance $Z_s = R_s + jhX_s$.

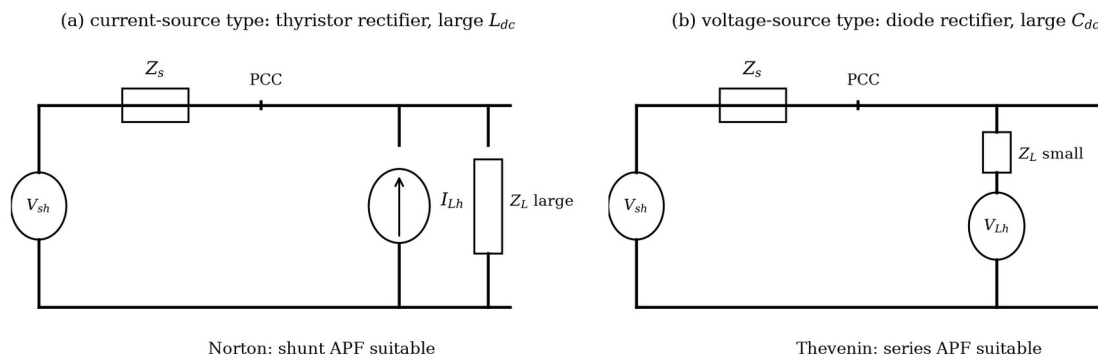


Figure 9.45. Harmonic-source models: (a) current-source type load (Norton), for which a shunt filter is natural; (b) voltage-source type load (Thevenin), for which a series filter is natural.

The distinction matters because the harmonic current that reaches the source depends on the ratio of the load's own impedance to the impedance the filter presents. A shunt path in parallel with a current-source load diverts the current; the same shunt path in parallel with a voltage-source load lowers the PCC impedance seen by the load and can actually increase the harmonic current it draws. A series impedance in front of a voltage-source load limits that current directly. The rest of the section quantifies these statements.

9.6.2 Passive and shunt active filtering of a current-source load

For a current-source load with a shunt passive filter of harmonic impedance Z_F at the PCC, superposition at harmonic h gives the source current

$$I_{sh} = \frac{Z_F}{Z_s + Z_F} I_{Lh} + \frac{1}{Z_s + Z_F} V_{sh} \quad (9.54)$$

The filtering characteristic is set by the ratio $Z_F/(Z_s + Z_F)$, so it depends on the source impedance, which the designer does not control and which changes with network configuration. Two failure modes follow directly from Equation (9.54). Below its tuning frequency the filter is capacitive, and it forms a parallel resonance with the inductive source at which $|Z_s + Z_F|$ becomes small: both terms of Equation (9.54) are then amplified. And because Z_F is low at its tuning frequency, the filter also sinks background distortion V_{sh} from the rest of the network and can be overloaded by harmonics it was never intended to absorb. For a plain capacitor bank of reactive power Q_c on a bus of short-circuit power S_{sc} , the parallel resonance occurs near

$$h_r \approx \sqrt{\frac{S_{sc}}{Q_c}} \quad (9.55)$$

and a series detuning reactor of p percent of the capacitor reactance moves the series resonance of the branch to $h_t = 1/\sqrt{p}$, below the lowest characteristic harmonic.

An ideal shunt active filter with load-current detection injects $i_c = G i_{Lh}$, where G is the harmonic gain of the reference generator ($G=1$ for complete cancellation). In parallel with a current-source load,

$$I_{sh} = (1 - G) \frac{Z_L}{Z_s + Z_L} I_{Lh} + \frac{1}{Z_s + Z_L} V_{sh} \quad (9.56)$$

so with $G \rightarrow 1$ the load harmonics vanish from the source independently of Z_s , which is the main advantage over a passive filter. The background term is not compensated in this mode; it is small for a current-source load because Z_L is large. The cost is a converter rated for the full harmonic (and any reactive or unbalanced) current, $S_{APF} = 3 V_{ph} \sqrt{\sum_h I_{ch}^2}$, which is economical at low voltage and becomes expensive at medium voltage.

9.6.3 Series active filter: voltage cancellation and harmonic isolation

A series APF (Figure 9.46(a)) can serve two distinct objectives. For *voltage cancellation* it isolates the load from measured supply-voltage harmonics by commanding $v_{inj,h} = -v_{s,h}$ within its voltage, current and energy limits. The desired fundamental is estimated first: a PLL with positive-sequence extraction and filtering in a synchronous frame gives $v_{s,1}^{+}$, and the reference becomes $v_{inj}^{\dot{}} = v_{s,1}^{+} - v_s^{\dot{}}$. With a balanced fundamental supply the residue contains only harmonics; with an unbalanced supply it also contains unwanted sequence components and must not be called harmonics alone. More generally, any desired load waveform $v_L^{\dot{}}$ gives $v_{inj}^{\dot{}} = v_L^{\dot{}} - v_s^{\dot{}}$, consistent with Equation (9.27), which is why the DVR and the series APF are the same hardware with different references.

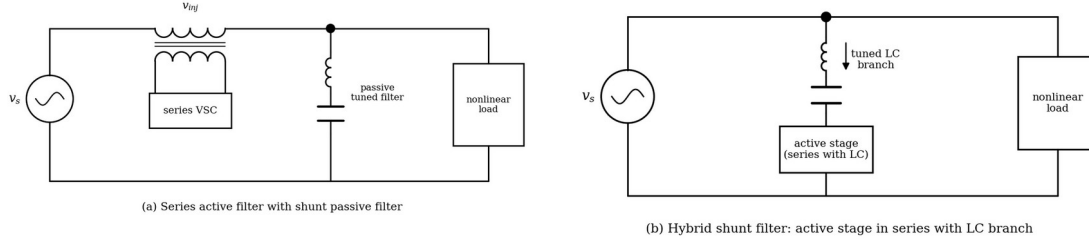


Figure 9.46. (a) Series active filter with a shunt passive filter; (b) hybrid shunt filter with the active stage in series with a tuned LC branch.

For *harmonic isolation* the series converter synthesizes a resistance at harmonic frequencies only. Its reference is derived from the detected harmonic source current, $v_c = K i_{sh}$ in the voltage-drop sense, with K in ohms (or per unit) and zero gain at the fundamental, so the converter presents a high series impedance to harmonics and none to the fundamental. Combined with a shunt passive filter, as proposed by Peng, Akagi and Nabae [22], the source harmonic current becomes

$$I_{sh} = \frac{Z_F}{Z_s + Z_F + K} I_{Lh} + \frac{1}{Z_s + Z_F + K} V_{sh} \quad (9.57)$$

When $K \gg |Z_s + Z_F|$, both terms are suppressed: the filtering no longer depends on Z_s , the passive filter no longer sinks background distortion from the network, and the parallel resonance between Z_F and Z_s is damped because K acts as a resistor inside the resonant loop. Figure 9.47 shows the effect for a 5th-harmonic tuned branch. The series converter carries the full line current, but its voltage is only $V_{ch} = K I_{sh}$, which is small once the harmonics have been isolated, so its kVA rating is typically a few percent of the load rating. A series filter alone cannot remove a nonlinear load's harmonic current, which needs a local shunt path; the passive branch provides it.

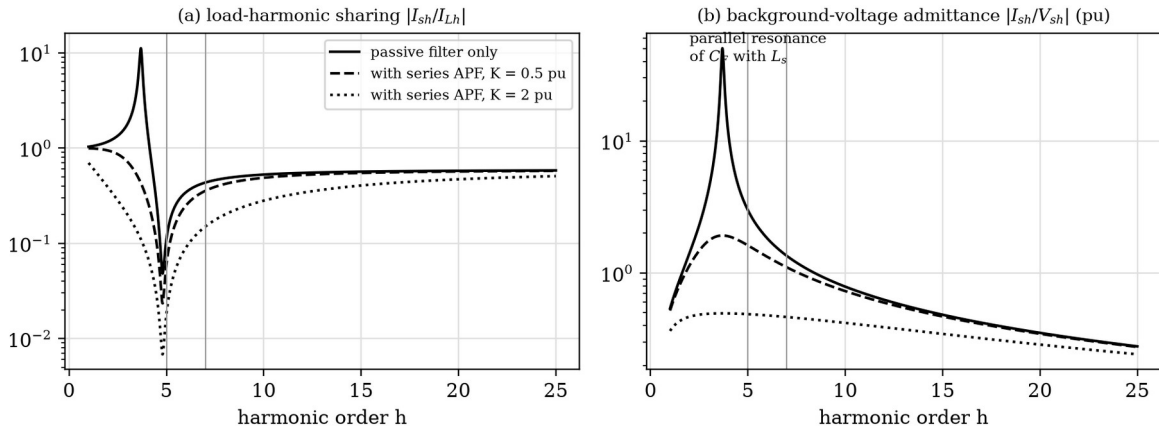


Figure 9.47. Frequency response of a shunt passive filter (tuned near the 5th, $Q=30$) on a source of $X_s=0.06$ pu, with and without a series active filter of harmonic gain K : (a) fraction of load harmonic current reaching the source; (b) source harmonic current per unit background voltage.

For a voltage-source type load the series filter plays a different role. The load harmonic current is $I_{Lh} = (V_{Lh} - V_{pcc,h})/Z_L$ with small Z_L , so a series impedance K in front of the load gives $I_{sh} = (V_{Lh} - V_{sh})/(Z_s + Z_L + K)$, which falls with K regardless of the source. A shunt active filter on such a load works against a low impedance and must inject a larger current than the load originally drew; this is the physical reason Peng recommends series compensation for capacitive-smoothed rectifiers and shunt compensation for inductive-smoothed ones [23].

9.6.4 Hybrid shunt filter with the active stage in series with an LC branch

A full shunt APF must carry its whole compensating current, and a series APF must carry the line current even when its voltage is small. In the hybrid arrangement of Figure 9.46(b), due to Fujita and Akagi [24], the active stage is placed in series with a tuned LC branch. Most of the fundamental voltage appears across the branch capacitor, so the converter needs only a small voltage to sharpen the branch's harmonic impedance and damp its resonance with the supply. If the converter is controlled as $v_{AF} = K i_{sh}$, the branch behaves as $Z_F + K$ in the harmonic loop and Equation (9.57) applies with the same benefits. The converter carries the branch current, including its fundamental capacitive component $I_{F1} \approx V_{ph}/X_{F1}$ and all its harmonic components, not the residual source current:

$$S_{AF} = V_{AF,rms} I_{F,rms}, I_{F,rms} = \sqrt{I_{F1}^2 + \sum_h I_{Fh}^2}, V_{AF,rms} \approx K \sqrt{\sum_h I_{sh}^2} \quad (9.58)$$

Because V_{AF} is a few percent of the supply voltage, the converter rating is a few percent of the load kVA even though its current is large. Other hybrids place a small shunt active filter in parallel with a passive filter; then the active stage carries only the residual. The two are easily confused, and Example 9.42 shows the consequence of doing so.

Example 9.27. Capacitor-bank resonance and detuning

An 11 kV bus has a short-circuit level of 100 MVA. A 3 Mvar power-factor-correction bank is to be added near a large six-pulse drive. Find the parallel resonant order and the order to which a 7 % detuning reactor tunes the branch.

Solution. From Equation (9.55), $h_r = \sqrt{100/3} = 5.77$, uncomfortably close to the 5th harmonic produced by the drive; the bank would amplify it. A 7 % series reactor tunes the branch to $h_t = 1/\sqrt{0.07} = 3.78$, below the 5th, so the branch is inductive at all characteristic harmonics and the resonance with the source moves below $h = 3.8$. A frequency scan over all credible network configurations is still required, because S_{sc} changes with switching state.

Example 9.28. Series active filter with a shunt passive filter

At the 5th harmonic a supply has $Z_s = 0.02 + j0.30$ pu and a shunt passive filter, slightly detuned by ageing, has $Z_F = 0.02 + j0.05$ pu. The load injects $I_{L5} = 0.20$ pu and the background supply distortion is $V_{s5} = 0.01$ pu. Find the source 5th-harmonic current with the passive filter alone and with a series active filter of $K = 1$ pu, and estimate the series-filter voltage and kVA on the load base.

Solution. Passive only, from Equation (9.54): $|Z_F/(Z_s + Z_F)| = 0.0539/0.3506 = 0.153$, so the load contributes 0.031 pu, and the background contributes $0.01/0.3506 = 0.028$ pu. Depending on phase, the source 5th-harmonic current is up to 0.059 pu, and the filter is also absorbing distortion from the rest of the network. With $K = 1$ pu, Equation (9.57) gives $0.0539/1.078 \times 0.20 = 0.0098$ pu and $0.01/1.078 = 0.0093$ pu, at most 0.019 pu in total, a reduction of about two thirds even with the filter detuned. The series-filter voltage is $V_{c5} = K I_{s5} \leq 0.019$ pu. Since it carries the full 1 pu line current, its rating is about $0.019 \times 1.0 = 1.9$ % of the load kVA, compared with roughly 20 % for a shunt APF that would have to carry the full 0.20 pu harmonic current.

9.6.5 Choosing among filter arrangements

A shunt APF and a DSTATCOM share the same hardware; the difference is the control objective. An APF emphasizes harmonic and unbalanced-current cancellation, while a DSTATCOM often includes fundamental reactive power and voltage regulation, and one converter can do both within its RMS current and bandwidth. Table 9.11 collects the trade-offs derived above.

Table 9.11. Filter arrangements compared.

Arrangement	Best-suited harmonic source	Dependence on Z_s	Resonance risk	Converter rating
Passive shunt filter	Current-source type, fixed spectrum	Strong (Equation (9.54))	High; sinks background distortion	None
Shunt APF	Current-source type, variable spectrum	Weak (Equation (9.56))	Low	Full harmonic + reactive current
Series APF + passive shunt	Either type; weak or variable grids	Removed when $K \gg \sqrt{Z_s + Z}$	Damped by K	Few % of load kVA, full line current
Active stage in series with LC	Current-source type at medium voltage	Removed for large K	Damped by K	Few % of load kVA, branch current
Small shunt APF parallel to passive	Current-source type	Moderate	Must be checked	Residual harmonic current only

Solid-state transfer switches, static breakers and fault-current limiters also belong to the custom-power family, but their function is source transfer, interruption or current limitation rather than continuous waveform conditioning; they were treated in Chapter 8. A UPS differs from a DVR because it can supply the entire load during source loss, provided storage and isolation are adequate. A solid-state transformer can integrate voltage transformation, isolation, DC ports and power-quality control, but its architecture and protection go well beyond a conventional compensator (Table 9.12).

Table 9.12. Custom-power devices and what each does not replace.

Device	Primary action	Best-suited disturbance	What it does not replace
DSTATCOM / shunt APF	Inject current	Reactive, harmonic, unbalanced and neutral current	Series restoration of a missing source voltage
DVR / series APF	Inject voltage	Sag, swell, voltage unbalance and harmonics	Long interruption without an energy source
UPQC	Series voltage + shunt current	Combined voltage and current problems	UPS function unless storage and isolation are designed
SSTS	Transfer load between sources	Source disturbance with a healthy alternate feeder	Continuous harmonic compensation
FCL	Increase fault-path impedance	Excessive fault current	Voltage restoration
UPS	Supply load from stored or alternate energy	Interruption and ride-through	Low-cost feeder-wide PQ correction

9.7 IEEE Std 519-2022 Compliance at the Distribution Level

IEEE Std 519-2022 [26] is a system-level standard applied at the point of common coupling (PCC), not a statement that a particular converter has “good THD”. Its philosophy is a shared responsibility: the user limits the harmonic current injected at the PCC, and the system owner limits the voltage distortion delivered there. Current compliance is judged by total demand distortion and by individual harmonics referred to the maximum

demand current, with limits that depend on the short-circuit ratio; voltage limits are checked separately. A device-level current THD measured at one operating point therefore cannot by itself establish compliance.

9.7.1 Definitions and the TDD-THD relation

The PCC is the point on the public system electrically nearest to the user's installation where other customers are, or could be, connected. For harmonic currents I_h at the PCC,

$$TDD = \frac{\sqrt{\sum_{h=2}^{50} I_h^2}}{I_L} \times 100\% \quad (9.59)$$

where I_L is the maximum demand load current, established as the average of the maximum demand currents (15- or 30-minute demand) over the preceding twelve months. It is not the instantaneous load current used elsewhere in this chapter. The voltage THD at the PCC is

$$THD_V = \frac{\sqrt{\sum_{h=2}^{50} V_h^2}}{V_1} \times 100\% \quad (9.60)$$

THD uses the contemporaneous fundamental I_1 , whereas TDD uses the fixed I_L , so

$$TDD = THD_I \frac{I_1}{I_L} \quad (9.61)$$

A lightly loaded drive can show a very high current THD while its TDD is small, because its harmonic currents are small in amperes (Figure 9.48). This is exactly why the standard uses TDD: it penalizes harmonic amperes, which create voltage distortion, rather than distortion percentages at light load.

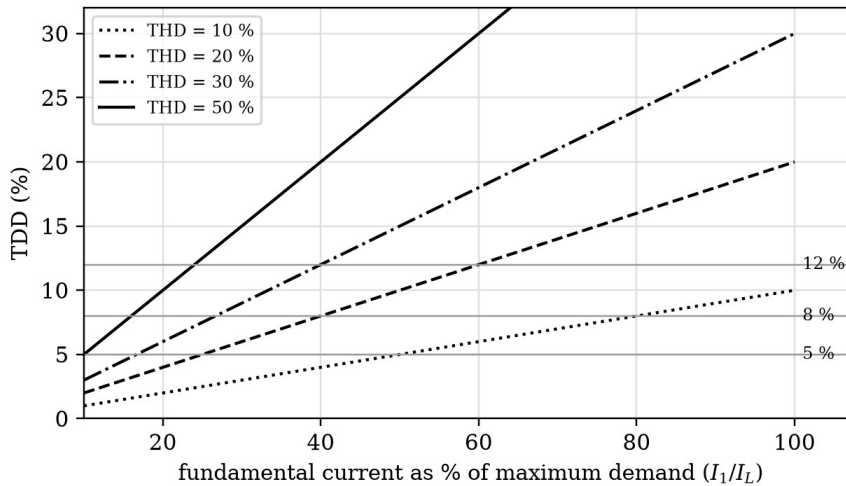


Figure 9.48. TDD versus loading for several values of current THD, with the 5 %, 8 % and 12 % TDD limits for $I_{sc}/I_L < 20$, 20 to 50 and 50 to 100.

9.7.2 Current and voltage limits

Table 9.13 gives the current limits for systems rated 120 V to 69 kV; the standard has separate tables for 69 to 161 kV and above 161 kV. The 2022 edition changed the treatment of even harmonics: in the 2014 edition all even harmonics were limited to 25 % of the odd-harmonic limits, whereas in 2022 only even harmonics of order 2, 4 and 6 are limited, to 50 % of the limit for their range, and even harmonics above the 6th share the limits of the odd harmonics in the same range.

Table 9.13. IEEE Std 519-2022 current distortion limits for systems from 120 V to 69 kV (percent of I_L ; even harmonics $h \leq 6$ are limited to 50 % of the values shown).

I_{sc}/I_L	$2 \leq h < 11$	$11 \leq h < 17$	$17 \leq h < 23$	$23 \leq h < 35$	$35 \leq h \leq 50$	TDD
below 20	4.0	2.0	1.5	0.6	0.3	5.0
20 to below 50	7.0	3.5	2.5	1.0	0.5	8.0
50 to below 100	10.0	4.5	4.0	1.5	0.7	12.0
100 to below 1000	12.0	5.5	5.0	2.0	1.0	15.0
1000 and above	15.0	7.0	6.0	2.5	1.4	20.0

Table 9.14. IEEE Std 519-2022 voltage distortion limits at the PCC.

Bus voltage V	Individual harmonic (%)	THD (%)
$V \leq 1$ kV	5.0	8.0
$1 < V \leq 69$ kV	3.0	5.0
$69 < V \leq 161$ kV	1.5	2.5
$V > 161$ kV	1.0	1.5

9.7.3 Why current limits scale with the short-circuit ratio

The structure of Table 9.13 follows from the voltage drop that a harmonic current produces. With the source treated as inductive at the PCC, the fundamental short-circuit reactance is $X_1 = V_1/I_{sc}$ per phase, so the harmonic reactance is hX_1 and

$$\frac{V_h}{V_1} = \frac{hX_1 I_h}{V_1} = h \frac{I_h}{I_{sc}} = \frac{h}{I_{sc}/I_L} \cdot \frac{I_h}{I_L} \quad (9.62)$$

An installation on a stiff bus (large I_{sc}/I_L) can inject more current for the same voltage distortion, which is why the permitted percentages rise with the short-circuit ratio. For example, at $I_{sc}/I_L = 20$ the 4 % limit on the 5th harmonic produces $V_5/V_1 = 5 \times 0.04/20 = 1$ %, comfortably inside the 3 % individual voltage limit at medium voltage, so that several users can share the voltage headroom. Equation (9.62) ignores network resonance, background distortion and capacitor banks, all of which a real study must include.

9.7.4 Statistical assessment

The limits are applied statistically, using the aggregation intervals of IEC 61000-4-30 [27]: very-short-time values over 3 s and short-time values over 10 min, each obtained as the RMS of the shorter-interval values,

$$X_{10min} = \sqrt{\frac{1}{N} \sum_{k=1}^N X_{3s,k}^2} \quad (9.63)$$

For current, the daily 99th-percentile very-short-time values should be below twice the table values, the weekly 99th-percentile short-time values below 1.5 times the table values, and the weekly 95th-percentile short-time values below the table values. For voltage, the daily 99th-percentile very-short-time values should be below 1.5 times the table values and the weekly 95th-percentile short-time values below them. Figure 9.49 illustrates the procedure for a synthetic week of 10-min TDD values: occasional values exceed 5 %, but the 95th percentile

(4.73 %) is below the limit and the 99th percentile (5.19 %) is below 1.5 times the limit, so this record passes the short-time tests.

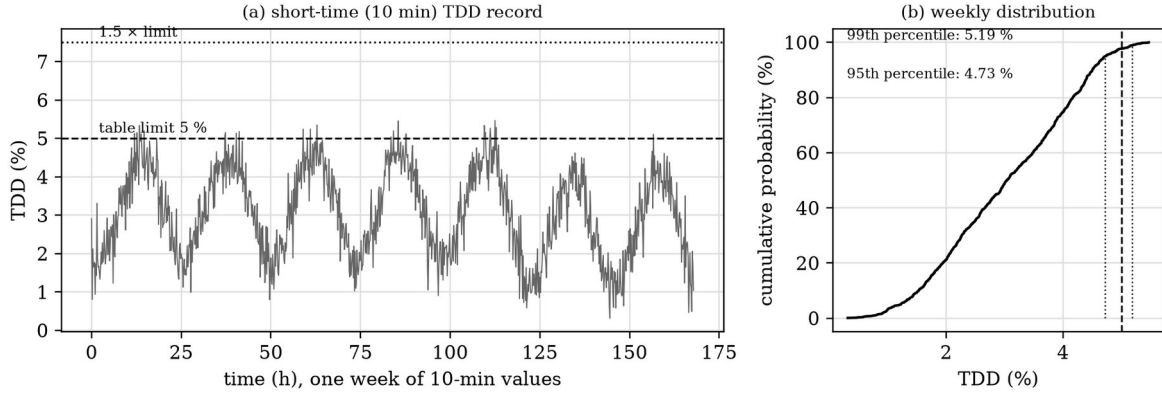


Figure 9.49. Illustrative statistical assessment of one week of 10-min TDD values ($I_{sc}/I_L < 20$): (a) time record; (b) cumulative distribution with the 95th and 99th percentiles. Synthetic data for explanation only.

9.7.5 Several harmonic sources at one PCC

When several nonlinear loads share a PCC, their harmonic currents add as phasors, and their phase angles become increasingly random at higher orders. IEC TR 61000-3-6 [28] uses a general summation law,

$$I_h = \left(\sum_i I_{h,i}^\alpha \right)^{1/\alpha}, \alpha = \begin{cases} 1, & h < 5 \\ 1.4, & 5 \leq h \leq 10 \\ 2, & h > 10 \end{cases} \quad (9.64)$$

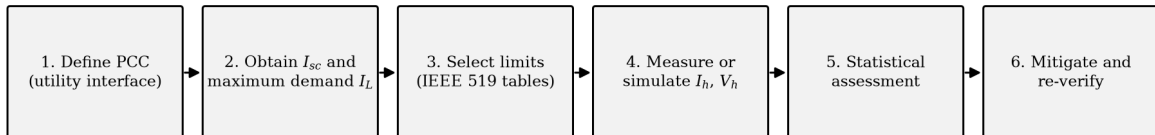
which lies between arithmetic addition ($\alpha=1$, worst case, appropriate for low orders where loads are in phase) and RMS addition ($\alpha=2$, random phase). Using arithmetic addition at all orders overstates the higher-order harmonics and can lead to an oversized filter.

9.7.6 Sizing mitigation for compliance

A compensator is sized from the residual spectrum it must leave. If a shunt APF removes a fraction $(1-k)$ of every harmonic uniformly, the residual is $k I_h$ and the required harmonic current rating is

$$I_{APF} = (1-k) \sqrt{\sum_h I_h^2}, k = \min \left(\frac{TDD_{target}}{TDD}, \min_h \frac{I_{h,lim}}{I_h} \right) \quad (9.65)$$

A selective-harmonic design can do better by attenuating only the offending orders, at the price of a more complex controller. Figure 9.50 summarizes the compliance workflow. Interharmonics, rapidly varying distortion and transients need separate interpretation, because the standard is framed around steady-state limits.



Voltage limits (Table of voltage distortion) are checked separately from current limits at every step.

Figure 9.50. IEEE Std 519 compliance workflow at a distribution PCC.

Example 9.29. THD versus TDD at light load

A drive operating at 40 % of the plant's maximum demand current draws a current with 20 % THD. Find its TDD.

Solution. From Equation (9.61), $TDD = 20 \times 0.40 = 8\%$. The same harmonic amperes at full load would give 20 % THD and 20 % TDD, while at 40 % load they read as 20 % THD but only 8 % TDD. Quoting THD at light load overstates the network impact.

Example 9.30. Voltage distortion produced by a harmonic current spectrum

The installation of Example 9.43 ($I_{sc} = 14$ kA; $I_5 = 14$, $I_7 = 9$, $I_{11} = 5$, $I_{13} = 3.5$, $I_{17} = 2$ A) is connected at a medium-voltage PCC. Estimate the voltage distortion it causes, ignoring resonance and background distortion.

Solution. From Equation (9.62), $V_5/V_1 = 5 \times 14/14\,000 = 0.50\%$, $V_7/V_1 = 0.45\%$, $V_{11}/V_1 = 0.39\%$, $V_{13}/V_1 = 0.33\%$ and $V_{17}/V_1 = 0.24\%$, so $THD_V = 0.88\%$. Every value is well inside the 3 % individual and 5 % total limits for $1 < V \leq 69$ kV. The installation's own contribution is small; any compliance problem at this PCC would come from background distortion or resonance, which a frequency scan must check.

Example 9.31. Sizing a shunt APF for compliance

A 415 V installation has $I_L = 500$ A and $I_{sc} = 8$ kA. Its spectrum is $I_5 = 40$, $I_7 = 25$, $I_{11} = 12$, $I_{13} = 8$, $I_{17} = 5$ and $I_{19} = 4$ A. Check compliance and size a shunt APF that brings the TDD to 4 % with uniform attenuation.

Solution. $I_{sc}/I_L = 16 < 20$, so the limits are 4 % (20 A) for $h < 11$, 2 % (10 A) for $11 \leq h < 17$, 1.5 % (7.5 A) for $17 \leq h < 23$, and 5 % TDD. The spectrum is 8.0, 5.0, 2.4, 1.6, 1.0 and 0.8 % of I_L : the 5th, 7th and 11th exceed their limits, and $\sqrt{\sum I_h^2} = 49.74$ A gives TDD = 9.95 %. Cutting each harmonic only to its individual limit would leave $\sqrt{20^2 + 20^2 + 10^2 + 8^2 + 5^2 + 4^2} = 31.7$ A, or 6.34 % TDD, still failing, so the TDD governs. From Equation (9.65), $k = 4/9.95 = 0.402$; the residual spectrum (16.1, 10.1, 4.8, 3.2, 2.0 and 1.6 A) meets every limit (Figure 9.51). The APF must supply $I_{APF} = (1 - 0.402) \times 49.74 = 29.7$ A RMS of harmonic current, or $3 \times 239.6 \times 29.7 = 21.4$ kVA, before allowances for reactive power, ripple and overload.

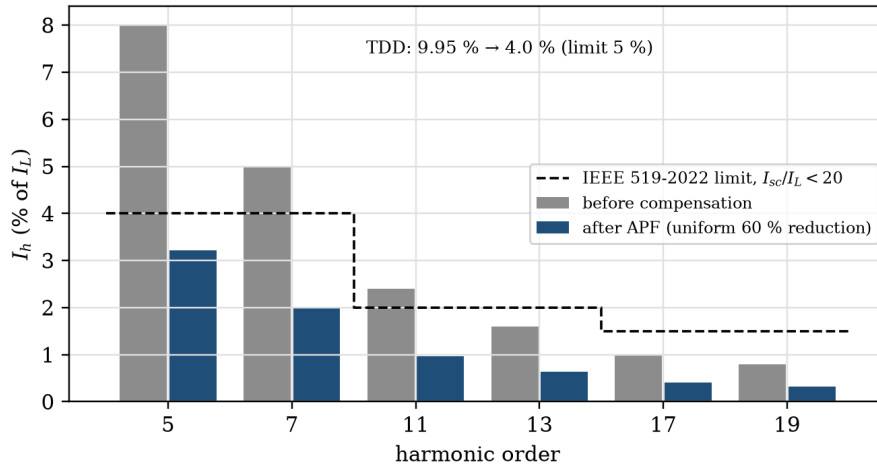


Figure 9.51. Harmonic spectrum of Example 9.31 before and after uniform APF attenuation, against the IEEE Std 519-2022 limits for $I_{sc}/I_L < 20$.

Example 9.32. Summation of two drives

Two drives at one PCC produce 5th-harmonic currents of 30 A and 20 A and 11th-harmonic currents of 8 A and 6 A. Estimate the combined values with Equation (9.64).

Solution. For $h=5$, $\alpha=1.4$: $(30^{1.4} + 20^{1.4})^{1/1.4} = 41.3$ A, between the arithmetic sum of 50 A and the RMS sum of 36.1 A. For $h=11$, $\alpha=2$: $\sqrt{8^2 + 6^2} = 10.0$ A, against an arithmetic 14 A. Arithmetic addition would oversize a filter for the 11th harmonic by 40 %.

9.8 Comparing and Selecting Custom Power Devices

Device selection should begin with the disturbed state variable and the required continuity of service. A voltage-quality problem upstream of a sensitive load points first toward a series compensator; a nonlinear or unbalanced load that pollutes the source points toward a shunt compensator; simultaneous voltage and current problems motivate a UPQC; a true interruption needs an energy-supported island or UPS function rather than a conventional conditioner. Converter rating, energy duration, fault duty and bypass philosophy are checked next, and only then is a control algorithm chosen. This section turns those statements into quantitative screening tools.

9.8.1 Functional comparison

Table 9.15 compares the three main devices by function, and Figure 9.52 arranges the candidate devices by the kind of disturbance and the side on which it originates. A DSTATCOM can move voltage through the feeder impedance but cannot guarantee restoration of a deep, externally imposed sag or an interruption. A DVR can compensate supply-voltage harmonics if designed for the bandwidth. A UPQC combines both services, subject to both converters' limits, the DC balance and the topology. The map is a first screen only: a DVR may suffice for a semiconductor process dominated by short sags, whereas the same plant may need a UPQC if its own drives also cause unacceptable source-current distortion; conversely, a UPQC where only displacement power factor is poor is unjustified if a capacitor bank or DSTATCOM meets the dynamic requirement.

Table 9.15. Comparison of DSTATCOM, DVR and UPQC.

Feature	DSTATCOM	DVR	UPQC
Connection	Shunt	Series (injection transformer)	Series and shunt, common DC link
Controlled quantity	Source current (and PCC voltage in ZVR)	Load voltage	Load voltage and source current
Main problems solved	Reactive power, unbalance, neutral current, current harmonics, flicker	Sag, swell, voltage unbalance, voltage harmonics, notches	All of these
Real-power source	None (losses only)	Storage, rectifier or none (quadrature, limited)	Grid via shunt converter
Typical rating	Load reactive and distortion power	Fraction of load ($X \times$ load for in-phase)	Series plus shunt; both converter maxima
Interruption ride-through	No	Only with storage and an island path	Only with storage, isolation and grid-forming control
Fault exposure	Shunt; current-limited	Carries downstream fault current; needs bypass	As DVR for the series part
Relative cost	Lowest per function	Moderate	Highest

interruption / severe event	UPS, island-capable storage (not a DVR)	SSCL / breaker (fault current)	UPS or storage-backed UPQC with isolation
harmonic / unbalance	DVR / series APF	shunt APF / DSTATCOM hybrid filter	UPQC or hybrid + DVR
fundamental (sag, swell, pf)	DVR (sag/swell) DSTATCOM (ZVR, weak feeder)	DSTATCOM (UPF, balancing)	UPQC
	supply-voltage disturbance	load-current disturbance	both

Figure 9.52. First-stage selection map for custom-power devices. Final selection still requires rating, protection, energy and economic studies.

9.8.2 Interruptions are an energy problem

A sag leaves some source voltage, so a grid-fed UPQC or a rectifier-supported DVR can draw the missing power from the remaining supply. An interruption leaves none. The protected load must then be fed entirely from storage, through a topology that isolates it from the dead network and forms its voltage and frequency. With usable stored energy E_{us} and load power P_L , the ideal ride-through time is

$$t_{rt} = \frac{\eta E_{us}}{P_L} \quad (9.66)$$

which for DC-link capacitors of practical size is a few milliseconds to a few hundred milliseconds. Problem 9.13 shows the scale. Longer interruptions require batteries, flywheels, a standby generator or a UPS, and once isolation and grid-forming control are added, the device is functionally a UPS rather than a power-quality conditioner.

9.8.3 Normalized converter ratings

The derivations of Sections 9.3 to 9.5 give closed-form ratings that make devices comparable on one axis. For a load of rated kVA S_L , power factor $\cos \phi$, current THD THD_I and negative-sequence current fraction u , full current compensation needs

$$\frac{S_{DST}}{S_L} = \sqrt{\sin^2 \phi + u^2 + THD_I^2} \quad (9.67)$$

An in-phase DVR designed for sag depth X needs $S_{DVR}/S_L = X$ plus allowances for unbalance and phase jump, and a stored energy $E = X P_L t_{sag}/\eta$. A right-shunt UPQC-P needs, from Equation (9.49) and Equation (9.50) with the harmonic duty added,

$$\frac{S_{UPQC}}{S_L} = \frac{X \cos \phi}{1 - X} + \sqrt{\sin^2 \phi + THD_I^2 + \cos^2 \phi \left(\frac{X}{1 - X} \right)^2} \quad (9.68)$$

Figure 9.53 plots these ratings for a 0.8 pf load with 25 % current THD. Two conclusions are useful for selection. First, a UPQC-P has roughly the same total converter rating as a separate DVR and DSTATCOM up to sags of about 25 %, but it needs no energy store, because the sag power circulates through the shunt converter. Second, beyond about 30 % the $1/(1 - X)$ term makes the UPQC rating rise steeply, and a storage-supported DVR plus a DSTATCOM becomes the leaner combination.

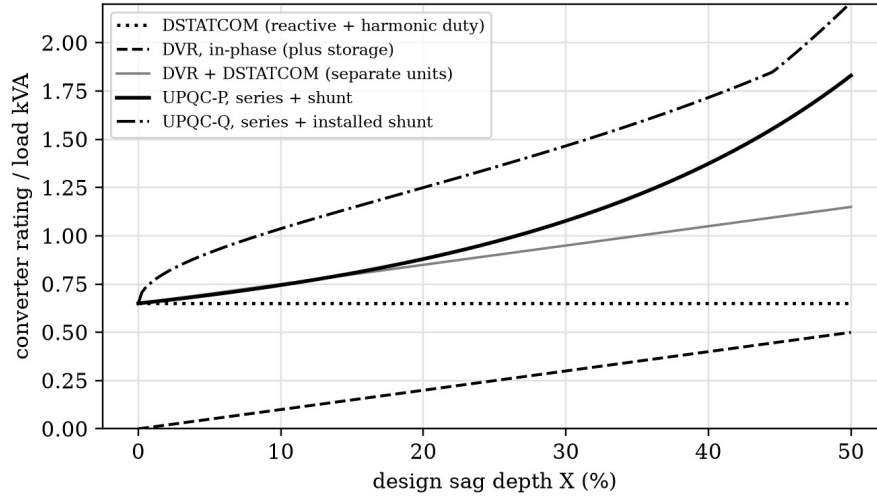


Figure 9.53. Normalized converter rating versus design sag depth for a 0.8 pf load with 25 % current THD.

9.8.4 Energy storage and ride-through

For a DVR the energy per event is $E = X P_L t_{sag} / \eta$ (Equation (9.40)). Figure 9.54 maps contours of stored energy per megawatt of load onto the sag depth-duration plane, the same plane as the voltage-tolerance curve of Figure 9.4. Short sags of a few cycles need only kilojoules per megawatt, which a DC capacitor or supercapacitor can supply; sags of a second or more at significant depth need hundreds of kilojoules per megawatt, which points to batteries or flywheels. Overlaying a measured sag scatter plot on this map shows immediately which storage class covers which fraction of events, and it is the most direct link between a power-quality audit and the energy rating of the device.

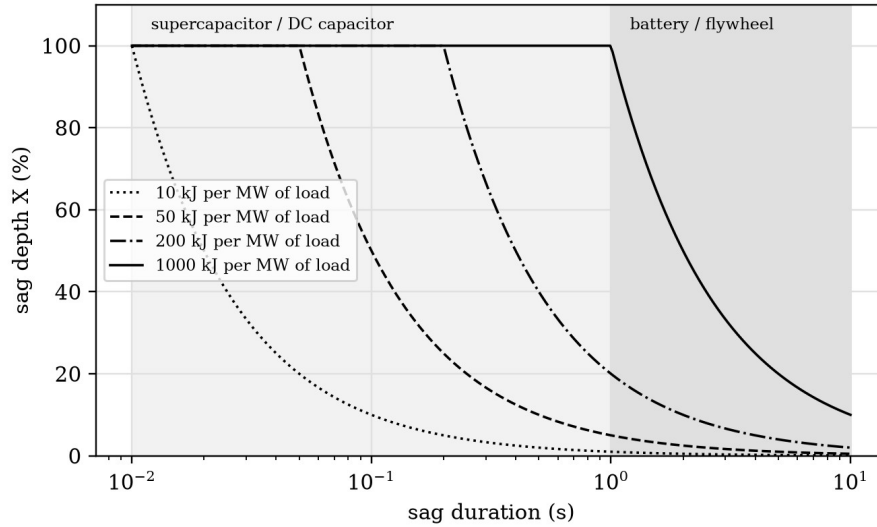


Figure 9.54. Stored energy required per MW of load for an in-phase DVR, as contours in the sag depth-duration plane ($\eta = 1$), with the typical domains of capacitor-type and battery-type storage.

9.8.5 Economic evaluation

A custom-power device is justified when the losses it avoids exceed its life-cycle cost. The expected annual cost of power-quality events without mitigation is

$$C_{PQ} = \sum_i N_i p_{trip,i} C_{trip,i} \quad (9.69)$$

where N_i is the annual number of events of class i (from the audit), $p_{trip,i}$ the probability that such an event trips the process (from equipment tolerance, as in IEEE Std 1346 [29]) and $C_{trip,i}$ the cost of one trip, including scrap, restart and lost production. The device's annualized cost is

$$C_{ann} = CRF \cdot C_{cap} + C_{OM} + C_{loss}, \quad CRF = \frac{r(1+r)^n}{(1+r)^n - 1} \quad (9.70)$$

with capital cost C_{cap} , discount rate r , life n years, operation and maintenance cost C_{OM} and the cost of converter losses C_{loss} . A DVR with bypass in standby has small losses; a DSTATCOM or UPQC that runs continuously can lose 1 to 3 % of its rating, and over a year this term can rival the capital charge. The device pays when $C_{PQ} - C_{PQ,res} > C_{ann}$, where $C_{PQ,res}$ is the residual cost of events the device cannot cover (Figure 9.55).

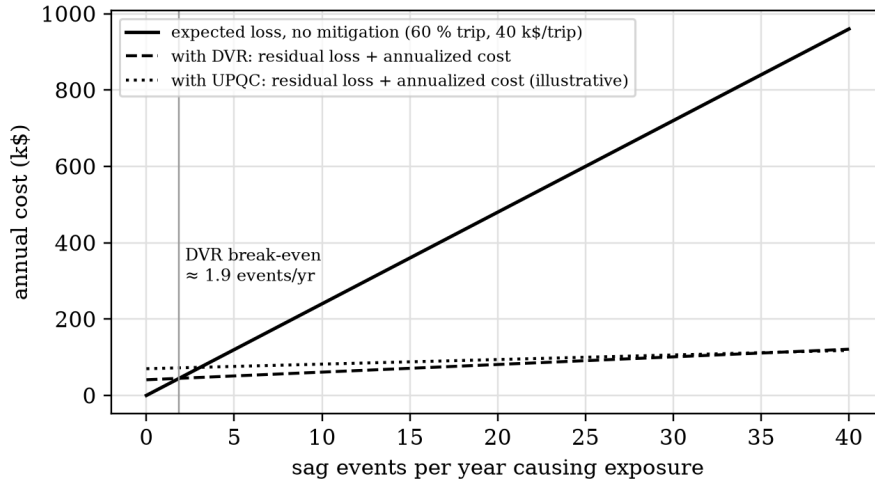


Figure 9.55. Annual cost versus number of exposing sag events for no mitigation and for mitigation with a DVR or a UPQC (illustrative values of Example 9.34).

9.8.6 Screening procedure and multi-criteria scoring

Figure 9.56 gives a first screening procedure; it is not a procurement rule. The essential first step is a power-quality audit that measures sag depth, duration and phase jumps, separates supply-voltage harmonics from load-current harmonics, checks TDD with the full method of Section 9.7 and identifies interruption exposure, which needs a separate energy path. When more than one device survives the screen, a weighted score $S_j = \sum_k w_k r_{jk}$, with weights w_k agreed in advance and ratings r_{jk} for each candidate j on criteria such as technical coverage, converter rating, energy, losses, footprint, redundancy and cost, makes the trade-off explicit and auditable (Example 9.35).

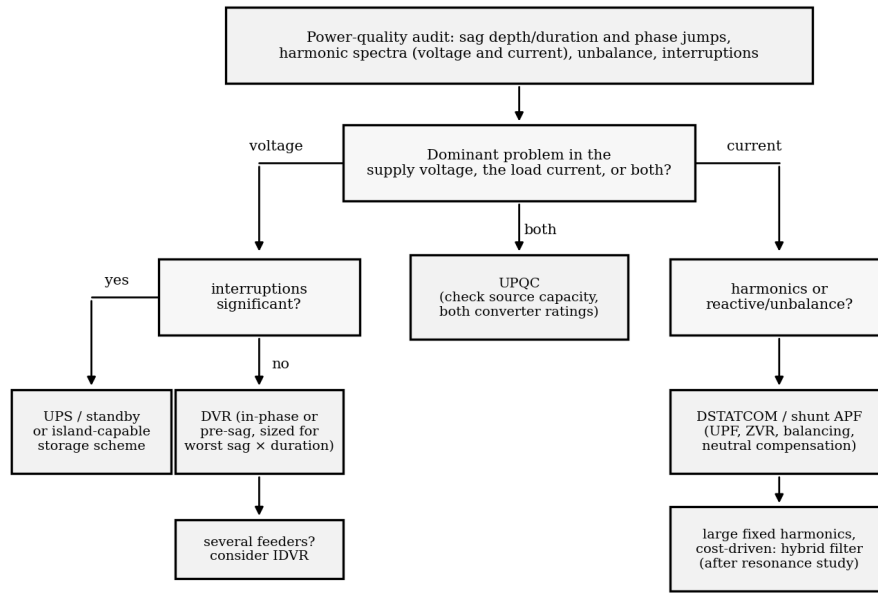


Figure 9.56. Screening procedure for selecting a custom power device.

Example 9.33. Normalized ratings of competing solutions

A 500 kVA, 0.8 pf plant has 25 % current THD and must ride through 30 % balanced sags lasting 300 ms. Compare the converter ratings and storage of (a) a DSTATCOM plus an in-phase DVR and (b) a right-shunt UPQC-P.

Solution. (a) From Equation (9.67) with $u=0$: $S_{DST}/S_L = \sqrt{0.36+0.0625} = 0.65$, so 325 kVA. The DVR needs $0.30 \times 500 = 150$ kVA plus allowances, and stores $E = 0.3 \times 400 \times 0.3 = 36$ kJ per event, so about 475 kVA of converters plus 36 kJ. (b) From Equation (9.68): series $0.3 \times 0.8/0.7 = 0.343$ and shunt $\sqrt{0.36+0.0625+0.1176} = 0.735$, total 1.078, or 539 kVA with no energy store. The UPQC needs about 13 % more converter rating but no storage, while the separate solution needs storage that must be maintained and replaced. Footprint, losses, redundancy and the cost of the store decide between them.

Example 9.34. Economic justification of a DVR

A plant experiences 20 exposing sags per year. Without mitigation 60 % of them trip the process at 40 k\$ per trip. A DVR costing 300 k\$ (life 15 years, discount rate 8 %, O&M 6 k\$ per year, negligible standby losses) reduces the trip probability to 5 %. Is it justified?

Solution. $C_{PQ} = 20 \times 0.6 \times 40 = 480$ k\$ per year; the residual is $20 \times 0.05 \times 40 = 40$ k\$. From Equation (9.70), $CRF = 0.08 \times 1.08^{15} / (1.08^{15} - 1) = 0.1168$, so $C_{ann} = 0.1168 \times 300 + 6 = 41.0$ k\$ per year. The net annual benefit is $480 - 40 - 41 = 399$ k\$, and the simple payback is $300 / (480 - 40 - 6) = 0.69$ year. The break-even exposure is $41.0 / (0.55 \times 40) = 1.9$ events per year (Figure 9.55). The result is driven by the trip cost and the audit's event count, which is why both must be measured rather than assumed.

Example 9.35. Multi-criteria screening

A semiconductor plant suffers frequent utility-side sags and significant harmonic distortion from its own drives, with comparable production losses from each. A second plant has rare, shallow sags but continuous harmonic distortion well above its TDD limit. Screen the technology for each, and set up a weighted score for the first plant.

Solution. For the first plant both problems are present and significant, so a UPQC deserves serious consideration: its shared DC link lets the series converter's sag energy come from the grid through the shunt converter rather than from a separately sized store. With weights of 0.35 (technical coverage), 0.20 (converter

rating), 0.15 (storage), 0.15 (losses and footprint) and 0.15 (cost), and ratings from 1 to 5 of 5, 3, 5, 3, 3 for the UPQC and 5, 4, 2, 3, 3 for a DVR plus a shunt APF, the scores are 4.0 and 3.75. The UPQC leads narrowly, so the decision is sensitive to the weights and should be confirmed with the plant's sag statistics, harmonic spectrum, ratings, losses, redundancy and life-cycle cost. For the second plant, a shunt APF or DSTATCOM, or a hybrid filter if a resonance study supports it, is the natural candidate; a DVR is hard to justify on the sag evidence.

9.9 Recent Developments and Research Directions

Recent work should be read as an extension of the physical models developed in this chapter. Higher switching frequency changes filter size and loss; grid-forming control changes the converter's synchronization and voltage-source behaviour; predictive and data-assisted control changes how references are computed; digital twins and hardware-in-the-loop change how claims are validated. None of these removes the semiconductor current, DC-energy, transformer, thermal or protection constraints. A recent scoping review of UPQCs in microgrids found that most studies still rely on PI and SRF control and on offline simulation, with comparatively few experimental or long-term stability assessments [20].

9.9.1 Wide-bandgap devices and higher switching frequency

SiC and GaN devices allow higher switching frequency and lower switching loss than silicon devices in suitable voltage classes. For custom-power equipment this shrinks the ripple filter (Equation (9.25) scales C_f with $1/f_s$) and the interfacing inductor (Equation (9.24) scales L_r with $1/f_s$), and it widens the current-loop bandwidth available for harmonic tracking. The gain is not free: dv/dt , common-mode current, insulation stress, electromagnetic compatibility, gate-drive design and thermal cycling become more demanding, and at medium voltage series connection or multilevel structures remain necessary. A meaningful comparison uses total converter loss, filter volume, junction-temperature swing and conducted emissions, not switching frequency alone.

9.9.2 Grid-forming and weak-grid operation

Conventional custom-power controllers assume a grid stiff enough for a PLL to extract the angle. In converter-dominated or island-capable microgrids this can fail, and the converter can no longer be treated as an ideal current or voltage source behind a fixed impedance. Its closed-loop output impedance $Z_{out}(s)$ interacts with the network impedance $Z_g(s)$, and a screening condition is based on the minor-loop ratio

$$L(s) = \frac{Z_g(s)}{Z_{out}(s)} \quad (9.71)$$

with a generalized Nyquist assessment required for coupled dq or sequence-domain models. Current limiting is critical because a grid-forming voltage controller may demand currents far beyond semiconductor capability during faults or deep sags. Grid codes now specify explicit tests of grid-forming functions, including voltage-source behaviour, fast fault-current injection and acting as a sink for harmonics and unbalance, rather than relying on nominal small-signal performance [30], [31]. For a DSTATCOM or UPQC, the research question is how voltage-source behaviour coexists with current limits and with the fast harmonic-compensation loops. Interconnection requirements for distributed energy resources, such as those of IEEE Std 1547-2018 [32], constrain what a converter sharing a DC link with generation may do during disturbances.

9.9.3 Predictive, repetitive, adaptive and data-assisted control

Model-predictive control can handle multivariable converter constraints explicitly. A generic constrained formulation for a UPQC is

$$J = \sum_{k=0}^{N-1} \left(w_v \|v_L - v_L^i\|^2 + w_i \|i_s - i_s^i\|^2 + w_{dc} (V_{dc} - V_{dc}^i)^2 + w_u \|\Delta u\|^2 \right) \quad (9.72)$$

subject to converter voltage, current, DC-energy and switching constraints. Repetitive and resonant controllers target periodic distortion, and adaptive observers or data-driven classifiers can improve disturbance identification or tune references. These methods should be evaluated against conventional SRF or PR control using the same converter limits, sampling delay, sensor noise and grid-impedance uncertainty, and the report should include execution time and robustness to parameter error. A lower simulation THD alone is not evidence of a better practical controller.

9.9.4 Digital twins, HIL and protection-aware validation

A digital twin of a UPQC or DVR is more than a simulation model. At minimum it synchronizes measured converter states with an electrical model, estimates slowly varying parameters such as filter inductance or DC-capacitor condition, compares predicted and measured residuals, and supports control and protection decisions without bypassing hard safety limits. Physics-constrained estimation is preferable to an unconstrained black-box predictor whenever extrapolation could command unsafe converter states. The twin should include saturation, current limiting, bypass logic and protection events, because these govern performance during the disturbances that matter most.

9.9.5 Renewable energy, EV charging and storage integration

PV inverters, battery systems and EV chargers are both sources of power-quality stress and controllable converter resources. A UPQC sharing a DC link with PV or a battery can use spare converter headroom for reactive and harmonic compensation while also managing real power. The scheduling problem is multi-timescale: energy management acts over seconds to hours, harmonic and sag compensation over cycles or milliseconds. The converter current limit couples these services and requires explicit priority logic.

9.9.6 What constitutes research-level validation

A research contribution becomes more convincing as it moves up the validation ladder of Figure 9.57. An analytical model should state network strength and R/X , load type, converter limits and sign conventions. EMT simulation should include switching or a justified averaged model, saturation, delay, DC-link dynamics and disturbance sequences. Sensitivity studies should vary grid impedance, load level and parameter error. Controller-HIL tests validate execution and timing; power-HIL adds power-interface dynamics; a prototype or field benchmark finally exposes sensing, dead time, thermal limits, protection and interference that ideal simulations hide. Comparisons should use identical hardware ratings and report the exact standard metric (for example TDD rather than THD), the detection delay and the transient peak current (Table 9.16).

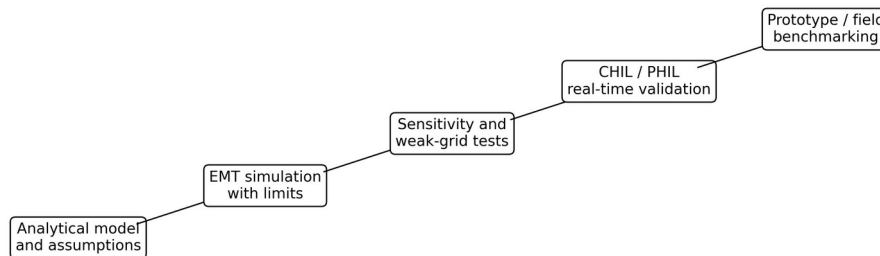


Figure 9.57. Research-validation ladder for advanced custom-power controllers.

Table 9.16. Validation levels and the evidence each provides.

Validation level	Question answered	Minimum evidence
Analytical / phasor	Is the compensation principle feasible?	Sign conventions, power balance, operating region
Averaged dynamic	Do control loops interact acceptably?	Bandwidths, saturation, DC-energy dynamics
Switching EMT	Are ripple and fast transients acceptable?	PWM, filters, delays, device limits
Controller HIL	Does the real controller execute correctly?	Sampling, computation delay, I/O scaling, protection logic
Power HIL / laboratory	Does hardware behave under realistic stress?	Interface stability, thermal and current limits, fault-safe operation

9.9.7 Industry perspective

Loads whose interruption is expensive, such as semiconductor fabrication, paper machines, plastics extrusion, data centres and hospitals, drive most custom-power installations. Field experience with inverter-based DVRs protecting industrial plants has been reported since the late 1990s [13], and medium-voltage DSTATCOMs of a few MVA are offered commercially, often coordinating switched capacitor and reactor banks. In each case the engineering argument rested on a measured disturbance profile compared with the equipment's own tolerance. A DVR or UPQC addresses sags; a UPS, standby generation or another topology is needed where outages dominate. The sub-millisecond responses simulated in this chapter do not by themselves prove that every protected process will be unaffected.

Key Equations

Table 9.17. Key equations and their assumptions.

Equation	Relationship	Meaning and scope
(9.2)	$P F = \cos \phi_1 / \sqrt{1 + T H D_I^2}$	True power factor with sinusoidal voltage
(9.4)	$i_s = i_L - i_c$	DSTATCOM KCL, i_c positive into the bus
(9.7)	Quadratic in I_q	Reactive source current for ZVR
(9.12)	$\dot{i}_d = \dot{i}_{Ld} + i_{loss}$	SRF reference; $\dot{i}_q = 0$ for UPF
(9.14)	$p = v_\alpha i_\alpha + v_\beta i_\beta, q = v_\alpha i_\beta - v_\beta i_\alpha$	Amplitude-invariant; $P = 3 p / 2$
(9.21)	$V_{dc} = 2 \sqrt{2} V_{LL} / (\sqrt{3} m)$	Three-leg VSC DC voltage
(9.24)	$L_r = \sqrt{3} m V_{dc} / (12 a f_s \Delta i)$	Interfacing inductor
(9.27)	$V_L = V_s + V_{inj}$	Series KVL, voltage-rise convention
(9.30)	$P_{DVR} = (V_{Lc} - V_s) I_L \cos \phi$	Standalone in-phase DVR

Equation	Relationship	Meaning and scope
(9.33)	$X_{max} = 1 - \cos \phi$	Zero-real-power limit, standalone DVR
(9.35)	$\cos \psi = (V_{Lc}^2 - V_s^2 - U^2) / (2 V_s U)$	Feasible angle for specified U
(9.42)	$C = 2 E / (V_{max}^2 - V_{min}^2)$	Energy-based capacitance
(9.45)	$I_s = I_L \cos \phi / (1 - X)$	Right-shunt UPQC source current
(9.47)	$S_{se} = V_{Lc} I_L \cos \phi \tan \delta$	UPQC-Q series duty
(9.49)	$S_{se} = V_{Lc} X I_L \cos \phi / (1 - X)$	UPQC-P series duty
(9.44)	$\dot{W}_{dc} = \sum P_{into} - P_{loss}$	Shared DC-link balance
(9.57)	$I_{sh} = \frac{Z_F I_{Lh} + V_{sh}}{Z_s + Z_F + K}$	Series APF with shunt passive filter
(9.55)	$h_r \approx \sqrt{S_{sc} / Q_c}$	Capacitor-bank parallel resonance
(9.59)	$TDD = 100 \sqrt{\sum I_h^2} / I_L$	Maximum-demand denominator
(9.62)	$V_h / V_1 = h I_h / I_{sc}$	Voltage distortion from a harmonic current
(9.64)	$I_h = (\sum I_{h,i}^\alpha)^{1/\alpha}$	Summation of several sources
(9.68)	Normalized UPQC-P rating	Device comparison
(9.70)	$C_{ann} = C R F C_{cap} + C_{OM} + C_{loss}$	Life-cycle cost

Common Misconceptions

- “A capacitor-supported DVR can correct any sag by injecting in quadrature.” For a standalone DVR the injection must be perpendicular to the load current, and the sag must satisfy $X \leq 1 - \cos \phi$. A unity-pf load cannot be helped at all without real power.
- “Quadrature to the supply voltage and quadrature to the current are the same thing.” They coincide only when the series current is in phase with the supply, as in a right-shunt UPQC with unity source pf.
- “A zero-power series converter needs no DC energy.” It still has conduction, switching and magnetic losses.
- “UPQC-Q can correct swells.” A perpendicular injection always increases the magnitude.
- “The UPQC-P series rating is $V_{Lc} X I_L \cos \phi$.” The series converter carries the increased source current, so the factor $1 / (1 - X)$ is required.
- “The smallest series VA means the smallest installation.” Check the shunt duty, the nominal-voltage duty and each converter’s own maxima.
- “A three-leg DSTATCOM can remove neutral current.” It needs a fourth leg, a split DC link or a zero-sequence transformer.

- “A DSTATCOM’s kVA rating includes the neutral-leg current times phase voltage.” The neutral leg’s AC terminal is at neutral potential.
- “A hybrid converter carries only residual source current.” In series with a passive branch it carries the branch current.
- “A shunt filter is always the right answer for harmonics.” For voltage-source type loads a series impedance is more effective, and a passive shunt filter can resonate with the source.
- “Passing current THD proves IEEE 519 compliance.” TDD, individual limits, voltage limits and the statistical tests are all required, and the low-voltage voltage-THD limit is 8 %, not 5 %.

Chapter Summary

Custom power devices apply the converter principles of transmission FACTS to distribution power quality. Problems divide into voltage-quality problems, which a series device corrects, and current-quality problems, which a shunt device corrects.

The DSTATCOM is a shunt VSC with only a capacitor on its DC side. It supplies the reactive, unbalanced, zero-sequence and harmonic parts of the load current so that the source supplies only balanced active current, or it adds reactive current to regulate the PCC voltage. Its topologies range from the three-leg VSC for three-wire systems to four-leg, split-capacitor and transformer-assisted converters for four-wire systems. Its references come from unit-template, SRF, pq and related algorithms, and its design follows from the DC-bus voltage, capacitor energy, inductor ripple and switch-rating relations.

The DVR is a series VSC with an injection transformer, filter, bypass switch and a DC link supplied by storage, a rectifier or only a capacitor. In-phase injection minimizes voltage rating and draws $X P_L$ from storage; pre-sag injection preserves phase; phase-advance injection saves energy; quadrature injection needs no real power but is feasible for a standalone DVR only when $X \leq 1 - \cos \phi$. Storage is sized from usable energy, real power and losses, and protection must bypass downstream faults without bypassing the upstream sags the DVR exists to correct.

The UPQC combines both on a common DC link, circulating the series converter’s real power through the shunt converter. UPQC-P minimizes series voltage, UPQC-Q makes the series duty reactive but cannot correct swells, and UPQC-S chooses the angle to share power or minimize combined VA. Each converter’s installed rating must cover its own worst condition, including nominal operation.

Harmonic filtering depends on the type of harmonic source and on the source impedance: shunt filters suit current-source loads, series filters suit voltage-source loads, and a series active filter with a passive shunt branch removes the dependence on source impedance at a small converter rating. Compliance with IEEE Std 519-2022 is judged by TDD, individual harmonics, voltage limits and statistical tests at the PCC. Device selection follows the measured disturbance profile, equipment tolerance, normalized converter ratings, storage requirements and a life-cycle cost assessment.

Solved Numerical Practice Problems

The following problems are chapter-level synthesis exercises. Unless stated otherwise, three-phase quantities use 415 V line-to-line (239.6 V phase), and each solution connects the calculation with converter physics, rating limits and model assumptions.

Example 9.36. Injection transformer rating

A DVR for a 415 V, 20 kVA system must inject up to 171 V per phase on the line side, with a converter-side voltage of 50 V. Find the turns ratio, the transformer kVA and the converter-side current.

Solution. $n=50/171=0.292$. $I_s=27.82$ A, so $S=3 \times 171 \times 27.82=14.27$ kVA. The converter-side current is $27.82 \times 171/50=95.1$ A.

Example 9.37. DVR with source impedance: why zero-power injection fails

A 415 V supply with $Z_s = 0.3 + j1.8 \Omega$ per phase feeds a star load $Z_L = 18 + j12 \Omega$ per phase. The source EMF sags by 25 %. A DVR at the load must keep the load voltage at 239.6 V. (a) Find the load current and voltage without the DVR. (b) Can a capacitor-supported DVR restore the voltage? (c) Find the minimum-magnitude injection and its real power.

Solution. (a) $V_M = 0.75 \times 239.6 = 179.7$ V, $I = 179.7 / |Z_s + Z_L| = 7.84$ A and $|V_L| = 7.84 \times 21.63 = 169.6$ V (293.8 V line). (b) With the load restored, $|I_L| = 239.6 / 21.63 = 11.08$ A and the circuit resistance absorbs $11.08^2 \times 18.3 = 2245$ W per phase. The sagged EMF can deliver at most $179.7 \times 11.08 = 1990$ W at that current, so zero-real-power restoration is impossible. (c) $V_{inj} = V_L(1 + Z_s/Z_L) - V_M$ is smallest when $V_L(1 + Z_s/Z_L)$ is aligned with V_M : $|V_{inj}| = 239.6 \times 1.0595 - 179.7 = 74.15$ V per phase. The DVR then supplies $P_{DVR} = 1.97$ kW (three-phase) and its duty is 2.46 kVA. A quadrature injection of $\sqrt{239.6^2 - 179.7^2} = 158.5$ V is not a valid zero-power answer here; the power-balance test in (b) settles feasibility before any injection angle is chosen.

Example 9.38. Unbalanced sag compensation

At the DVR terminals $V_p = 0.85 \angle 0^\circ$ pu, $V_n = 0.15 \angle 0^\circ$ pu and $V_0 = 0$. Restore the balanced target $[1, a^2, a]$ pu.

Solution. From Equation (9.36), $V_a = 1$, $V_b = -0.5 - j0.6062 = 0.7858 \angle -129.52^\circ$ and $V_c = 0.7858 \angle 129.52^\circ$ pu. Equation (9.37) gives $V_{inj,a} = 0$, $V_{inj,b} = -j0.2598$ and $V_{inj,c} = +j0.2598$ pu. Phase a already meets its target; b and c need equal magnitudes with opposite angles. Restoring phase magnitudes alone does not ensure balanced phase relationships.

Example 9.39. Real and reactive power of an angled injection

A single-phase DVR injects 60 V at 45° ahead of a 15 A load current. Find its P, Q and VA.

Solution. $P = 60 \times 15 \cos 45^\circ = 636.4$ W, $Q = 636.4$ var and $S = 900$ VA.

Example 9.40. IDVR real-power sharing

Three series compensators share a DC bus. Feeder 2 needs 12 kW for 150 ms; feeders 1 and 3 can each absorb up to 8 kW while keeping acceptable voltage. Check sufficiency, neglecting losses.

Solution. 16 kW is available against 12 kW needed. Commanding 6 kW absorption on each supporting converter gives $P_1 = -6$, $P_2 = 12$ and $P_3 = -6$ kW, and the mean DC energy is constant. A standalone storage-only DVR would need $12 \times 0.15 = 1.8$ kJ. With total loss P_{loss} the absorption must sum to $12 + P_{loss}$ kW, so up to 4 kW of loss can be covered. This establishes steady power sufficiency, not zero DC capacitance: transient delay and ripple still need buffer energy.

Example 9.41. Design of a 415 V, 30 kW three-phase four-wire UPQC

Design the series part of a UPQC for a 415 V, 30 kW load with $\pm 30\%$ supply variation, a 700 V DC bus, $f_s = 10$ kHz, $m = 1$, $a = 1.2$ and 10 % current ripple. Size the DC-link capacitor for a 5 % droop over 10 ms at the shunt converter's overload current.

Solution. $V_{DVR} = 0.3 \times 239.6 = 71.9$ V and $K_{DVR} = 239.6 / 71.9 = 3.33$; a ratio of 3 is chosen. During the deepest sag $I_s = 30\,000 / (3 \times 0.7 \times 239.6) = 59.6$ A, so the transformer is rated $3 \times 71.9 \times 59.6 = 12.86$ kVA and the series VSC device current is $59.6 / 3 = 19.9$ A (32 A devices with margin). During the largest swell $I_s = 30\,000 / (3 \times 1.3 \times 239.6) = 32.1$ A, and Equation (9.53) with $\Delta I = 3.21$ A gives $L_{DVR} = 0.866 \times 700 \times 3 / (6 \times 1.2 \times 10^4 \times 3.21) = 7.87$ mH. The ripple filter uses $R_r = 5 \Omega$ and $C_r = 10 \mu\text{F}$.

The shunt converter is designed as in Section 9.3.4 at $V_{dc}=677.7 \rightarrow 700$ V. With $I=69.56$ A and $a=1.2$, Equation (9.22) gives $\Delta E=3 \times 239.6 \times 1.2 \times 69.56 \times 0.01=600$ J and $C_{dc}=2 \times 600/(700^2 - 665^2)=25.1$ mF. The four-leg shunt converter does not pass neutral current through the capacitor, so Equation (9.23) does not apply.

Example 9.42. Hybrid branch current and residual source harmonic

For the topology of Figure 9.46(b), a load draws 40 A RMS of fifth harmonic. Assume, for this simplified same-phase exercise, that the passive branch diverts 92 % under nominal tuning and 85 % when detuned. Find the residual source harmonic and the fifth-harmonic current through the active stage. Is a 10 A stage enough?

Solution. Nominal: $I_{s,5}=40 \times 0.08=3.2$ A and the branch carries 36.8 A. Detuned: 6.0 A residual and 34.0 A in the branch. The active stage in series with the branch carries 36.8 A or 34.0 A, so a 10 A stage is insufficient; a 45 A stage gives 22 % margin over the nominal case, before the branch's fundamental capacitive current is added (Equation (9.58)). Assigning the 6 A residual to this converter would confuse it with a separate shunt residual-current stage. In a real network the current division is complex-valued and depends on the source and filter impedances (Equation (9.54)).

Example 9.43. PCC TDD and individual-harmonic screening

An installation at a 120 V to 69 kV PCC has $I_L=400$ A and $I_{sc}=14$ kA. Its RMS spectrum is $I_5=14$, $I_7=9$, $I_{11}=5$, $I_{13}=3.5$ and $I_{17}=2$ A, with $I_1=380$ A. Screen it.

Solution. $I_{sc}/I_L=35$, the 20 to 50 band: TDD limit 8 %, individual limits 7 % (5th, 7th), 3.5 % (11th, 13th) and 2.5 % (17th). $I_H=\sqrt{318.25}=17.84$ A, so TDD = 4.46 % and THD = 4.69 %. The individual values are 3.5, 2.25, 1.25, 0.875 and 0.5 %, all within limits. A shunt APF that reduced every harmonic by 80 % would bring TDD to 0.89 %, but the spectrum already passes this screen, so such a filter would need a separate justification. Full compliance also needs the statistical assessment of Section 9.7.4.

Unsolved Practice Problems

Unless stated otherwise, UPQC problems use the lossless right-shunt topology with unity source pf and RMS single-phase quantities; standalone DVR problems use the actual load current.

Problem 9.1 (UPQC-Q) A 230 V load draws 18 A at 0.75 pf; the supply sags by 18 %. Find V_s' , I_s , V_{inj} and S_{se} .

Problem 9.2 For the load of Problem 9.1, find V_{inj} and S_{se} in UPQC-P mode.

Problem 9.3 For Problem 9.1, find δ and the shunt current.

Problem 9.4 For Problems 9.1 and 9.3, find the total UPQC-Q operating VA.

Problem 9.5 (Standalone DVR, in-phase) 400 V (line), 15 kW, 0.85 pf, 25 % sag, 180 ms. Find V_{ph} , V_{inj} , I_{DVR} , P_{DVR} and the ideal E .

Problem 9.6 For Problem 9.5, the DC bus may fall from 650 V to 600 V. Find C_{dc} .

Problem 9.7 (IDVR) Feeder 2 needs 10 kW; feeders 1 and 3 can supply 5 and 4 kW. Is this sufficient?

Problem 9.8 A load has 5th = 7 A, 7th = 3.5 A and fundamental 45 A. Find the THD and the APF compensating current.

Problem 9.9 A facility has $I_L=350$ A with 5th = 12 A, 7th = 8 A, 11th = 4 A and 13th = 3 A. Find the TDD.

Problem 9.10 (Hybrid filter) A load draws 35 A of harmonic current; the passive branch diverts 88 %. Find the residual source harmonic and the harmonic current in an active stage in series with the branch.

Problem 9.11 (Unbalanced sag) $V_p=0.80$ pu and $V_n=0.12$ pu in phase with V_p , $V_0=0$. Find the DVR injection magnitude in each phase to restore a balanced 1.0 pu set.

Problem 9.12 (TDD compliance) The facility of Problem 9.9 has $I_{sc}/I_L=28$. Check the TDD and each individual harmonic.

Problem 9.13 (Interruption bound) A UPQC DC bus stores 10 kJ and must retain 75 % of it. With an island-capable path and a 12 kW load, find the ideal ride-through time.

Problem 9.14 (Strategies) A right-shunt UPQC protects 230 V, 18 A, 0.75 pf against a 15 % sag. Find the series voltage and VA in UPQC-Q and UPQC-P modes.

Problem 9.15 (Loss-inclusive energy) A standalone DVR supplies 6000 W for 180 ms at 95 % efficiency. Find the energy and its excess over the ideal value.

Problem 9.16 (DSTATCOM, UPF) A 415 V, 40 kW, 0.75 pf load is compensated to UPF. Find the source current before and after, the DSTATCOM current and its kVA.

Problem 9.17 (DSTATCOM design) For a 415 V DSTATCOM rated 50 A, find V_{dc} ($m=1$), L_r ($f_s=10$ kHz, $a=1.2$, 5 % ripple) using a 700 V bus, and C_{dc} for a 5 % droop with $\Delta E=3 V_{ph} a I_c t$, $t=10$ ms.

Problem 9.18 (Load balancing) A 30 kW UPF single-phase load is connected between lines b and c of a 415 V system. Find the load current, the balanced source current, the DSTATCOM currents and its kVA.

Problem 9.19 (Harmonic compensation) A 415 V load draws 60 A fundamental at unity displacement with 5th, 7th and 11th harmonics of 18, 12 and 6 %. Find the current THD, the DSTATCOM current and its kVA.

Problem 9.20 (Capacitor-supported DVR limit) A standalone capacitor-supported DVR protects a 230 V, 0.75 pf load. Find the maximum sag correctable without real power, the injection magnitude at that limit and the load-voltage phase shift.

Problem 9.21 (BESS energy) A 400 V, 50 kVA, 0.9 pf load is protected by an in-phase DVR against a 40 % sag for 300 ms with 95 % efficiency. Find P_{DVR} and the energy in kJ and Wh.

Problem 9.22 (Injection transformer) A 415 V, 30 kVA DVR must inject 0.35 pu with a converter-side voltage of 100 V. Find the line-side voltage, ratio, kVA and converter-side current.

Problem 9.23 (UPQC-S) A right-shunt UPQC with a 110 V series limit protects 230 V, 20 A, 0.85 pf against a 25 % sag. Find ψ , δ , P_{se} , S_{se} , S_{sh} and the total.

Problem 9.24 (Left-shunt UPQC-Q) A left-shunt UPQC-Q protects 230 V, 20 A, 0.9 pf during an 8 % sag. Using the smaller root, find θ_L , $|V_{inj}|$, S_{se} and the shunt operating VA.

Problem 9.25 (Neutral current) A four-wire load draws $I_a=30 \angle 0^\circ$, $I_b=20 \angle -150^\circ$ and $I_c=25 \angle 100^\circ$ A. Find the neutral current the DSTATCOM must supply.

Problem 9.26 (Resonance) A 400 V bus has a short-circuit level of 15 MVA. Find the parallel resonant order with a 250 kvar capacitor bank, and the tuning order with a 5.67 % detuning reactor.

Problem 9.27 (Series hybrid) Repeat Example 9.28 with $K=2$ pu and find the maximum source 5th-harmonic current.

Problem 9.28 (Economics) For Example 9.34, find the number of exposing events per year below which the DVR is not justified if the trip cost falls to 15 k\$.

Answer Key (Unsolved Problems)

9.1 $V'_s=188.60$ V, $I_s=16.463$ A, $V_{inj}=131.64$ V, $S_{se}=2167.3$ VA.

9.2 $V_{inj}=41.40$ V, $S_{se}=681.6$ VA.

9.3 $\delta=34.92^\circ$, $I_{sh}=2.483$ A.

9.4 2738.4 VA.

9.5 $V_{ph}=230.94$ V, $V_{inj}=57.74$ V, $I_{DVR}=25.47$ A, $P_{DVR}=3750$ W, $E=675$ J.

9.6 $C_{dc}=21.6$ mF.

9.7 9 kW available: insufficient by 1 kW.

9.8 THD = 17.39 %, $I_{comp}=7.826$ A.

9.9 TDD = 4.36 %.

9.10 Residual 4.20 A; active-stage branch harmonic current 30.80 A.

9.11 0.08, 0.28 and 0.28 pu.

9.12 TDD 4.36 % < 8 %; 5th 3.43 % and 7th 2.29 % (< 7 %); 11th 1.14 % and 13th 0.86 % (< 3.5 %). Passes the spectrum screen.

9.13 Usable energy 2500 J; $t_{max}=0.208$ s.

9.14 UPQC-Q: 121.16 V, 1924.3 VA; UPQC-P: 34.5 V, 547.9 VA.

9.15 $E=1136.8$ J, 5.3 % above 1080 J.

9.16 74.20 A before, 55.65 A after; $I_{DST}=49.08$ A; 35.28 kVA.

9.17 $V_{dc}=677.7$ V (700 V used); $L_r=3.37$ mH; $C_{dc}=18.05$ mF.

9.18 $I_{Lb}=-I_{Lc}=72.29 \angle -90^\circ$ A; $I_s=41.74$ A; $I_{ca}=41.74 \angle 180^\circ$, $I_{cb}=41.74 \angle -60^\circ$, $I_{cc}=41.74 \angle 60^\circ$ A; 30 kVA.

9.19 THD = 22.45 %; 13.47 A; 9.68 kVA.

9.20 25 %; 152.1 V; the load voltage leads by 41.41° .

9.21 18 kW; 5.684 kJ = 1.579 Wh.

9.22 83.86 V; $n=100/83.86=1.192$; 10.5 kVA; 35.0 A.

9.23 $\psi=73.08^\circ$, $\delta=27.23^\circ$, $P_{se}=725.6$ W, $S_{se}=2493.3$ VA, $S_{sh}=726.6$ VA, total 3219.9 VA.

9.24 $\theta_L=13.87^\circ$, $|V_{inj}|=56.37$ V, $S_{se}=1127.5$ VA, $I_{sh}=4.15$ A at 211.6 V, 877.6 VA.

9.25 $16.83 \angle 60.30^\circ$ A.

9.26 $h_r=\sqrt{15/0.25}=7.75$; $h_t=1/\sqrt{0.0567}=4.20$.

9.27 $0.0052+0.0048=0.0100$ pu, half the value with $K=1$ pu.

9.28 $41.0/(0.55 \times 15)=4.97$, so about 5 events per year.

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